

PSSE Contingency Analysis - Year 5, Topology 6

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Chapter 1

Introduction

1.1 Study Description

The Year 5 Topology 6 contingency analysis of the hypothetical SAVNW system is carried out to report the branches reporting loading greater than 100% and buses reporting upper and lower voltage limit violations. Topology 6 is the system in Year 6 with the same topology as Year 4 but with increased in load in all areas. Hence the firm and nonfirm capacity of the system remains the same. For Year 5, Topology 6, similar to Year 4, Topology 5 the variability of hydro, wind and solar resources are tested. The systems single line diagram can be seen in Figure 1.1 for Year 5 Topology 6.

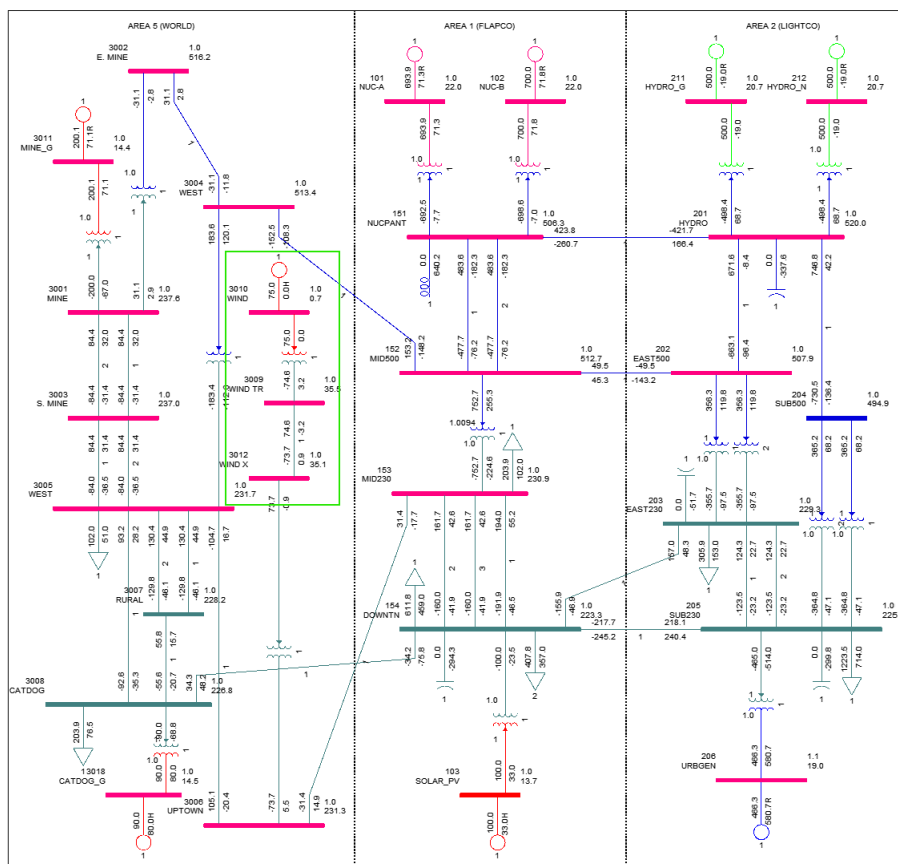


Figure 1.1: Single Line Diagram Year 5, Topology 6

Detailed analysis of Year 5 Topology 6 Base Case was carried out. Analysis of system totals by area, generator contributions to each scenario, and the considered load scenarios can be found here. The overload and voltage violations can be found here.

1.2 Contingency Analysis

AC contingency calculation was conducted on the hypothetical SAVNW system for 81 different scenarios and hence on 81 different case files corresponding to Topology 6. The same configuration files are used for the 81 scenarios and are:

- Subsystem file savnw.sub - Studied subsystems of the studied scenario/ case are defined via the Subsystem Definition data file (Figure 1.2)
- Monitor file savnw.mon - Monitored Element Data File identifies the branches that are to be monitored for flow violations and the buses that are to be monitored for voltage violations (Figure 1.3)
- Contingency file savnw.con - Contingency cases that are to be tested are defined in the Contingency Definition data file (Figure 1.4)

```
SUBSYSTEM CON
AREA 1
AREA 2
AREA 5
END

SUBSYSTEM MON
AREA 1
AREA 2
AREA 5
END

END
```

Figure 1.2: The subsystem file corresponding to Year 5 Topology 6

```
MONITOR VOLTAGE RANGE ALL BUSES 0.9 1.1
MONITOR ALL BRANCHES
MONITOR TIES FROM AREA 1 TO AREAS 2 5
MONITOR TIES FROM AREA 2 TO AREA 5
END
END
```

Figure 1.3: The monitored file corresponding to Year 5 Topology 6

```
SINGLE BRANCH IN AREA 1
SINGLE BRANCH IN AREA 2
SINGLE BRANCH IN AREA 5
SINGLE BUS IN AREA 1
SINGLE BUS IN AREA 2
SINGLE BUS IN AREA 5
SINGLE MACHINE IN AREA 1
SINGLE MACHINE IN AREA 2
SINGLE MACHINE IN AREA 5
END
```

Figure 1.4: The contingency file corresponding to Year 5 Topology 6

For each of the 81 studied scenarios, the API DFAX_2 is used to construct 81 different distribution factor data files corresponding to each .sav file, and the above defined .sub, .mon, .con configuration files. For each of the 81 scenarios, by running the AC contingency calculation function ACCC_WITH_DSP_3, the contingency solution output .acc files are obtained.

For each of the 81 scenarios, using the contingency solution output files, the results are exported as excel files for further analysis. The results exported are ACCC Analysis Summary, Monitored Branch Flows (MVA), Monitored Bus Voltages.

Python code to conduct AC contingency calculation is:

```
import psspy
list_gens = [0,50,100]
list_lsc = ['LLS','RLS','HLS']
list_gen_hydro = [400,500,600] #dry, average, wet hydrological scenarios
list_gen_wind = [0,40,75] # for 3 wind gust scenarios
for gen in list_gens:
    for gen_hy in list_gen_hydro:
        for lsc in list_lsc:
            for gen_wi in list_gen_wind:
                file_in = 'sav\savnw_sol_' + str(gen) + '_hy_' + str(gen_hy) + '_wi_' + str(gen_wi) + '_' + str(lsc) + '.sav'
                file_dist = 'savnw_sol_' + str(gen) + '_hy_' + str(gen_hy) + '_wi_' + str(gen_wi) + '_' + str(lsc) + '.dfx'
                file_out = 'savnw_sol_' + str(gen) + '_hy_' + str(gen_hy) + '_wi_' + str(gen_wi) + '_' + str(lsc) + '.acc'
                file_sav = r"{}".format(file_in)
                file_dfx = r"{}".format(file_dist)
                file_acc = r"{}".format(file_out)
                psspy.case(file_in)
                psspy.fdns([0,1,0,0,0,0,0,0])
                psspy.dfax_2([1,1,0],r""savnw.sub"",r""savnw.mon"",r""savnw.con"",file_dfx)
                psspy.accc_with_dsp_3(0.1,[0,1,0,0,0,0,0,0,0,0,0,0],"",file_dfx,file_acc,"","")
```

Python code to export AC contingency solution output file as excel is:

```
import pssexcel
list_gens = [0,50,100]
list_lsc = ['LLS','RLS','HLS']
list_gen_hydro = [400,500,600] #dry, average, wet hydrological scenarios
list_gen_wind = [0,40,75] # for 3 wind gust scenarios
for gen in list_gens:
    for gen_hy in list_gen_hydro:
        for lsc in list_lsc:
            for gen_wi in list_gen_wind:
                file_in = 'acc\savnw_sol_' + str(gen) + '_hy_' + str(gen_hy) + '_wi_' + str(gen_wi) + '_' + str(lsc) + '.acc'
                file_out = 'savnw_sol_' + str(gen) + '_hy_' + str(gen_hy) + '_wi_' + str(gen_wi) + '_' + str(lsc) + '.xlsx'
                file_acc = r"{}".format(file_in)
                file_xlsx = r"{}".format(file_out)
                pssexcel.accc(file_acc, ['s','v','b'], colabel='', stype='contingency', busmsm=0.5, sysmsm=5.0, rating='a', namesplit=False, xlsfile=file_out, sheet='', overwritesheet=True, show=False, ratecon='b', baseflowvio=True, basevoltvio=True, flowlimit=100.0, flowchange=0.0, voltchange=0.0, swdrating='a', swdratecon='b', baseswdfio=False, basenodevoltvio=False, overloadreport=False)
```

Contingency analysis was carried out in PSSE for each of the studied scenario for Year 5, Topology 6. It was seen that the power flow solution did not converge for some of the tested contingencies for the studied scenario. For each of the scenario, the contingencies for which power flow solution did not converge are given in Tables 5.2, 5.3, and 5.4 Appendix 5 of the document.

For the converged contingencies for each of the studied scenario, the results of the contingency analysis were analysed to check for branch overload ($> 100\%$) and out of range bus voltage violations (lower emergency limit $< 0.9PU$ and upper emergency limit $> 1.1PU$). It was seen that for some of the contingencies there were no branch overload or bus voltage violations reported. Rest of the contingencies violating branch overload and bus voltage emergency ranges are reported in the subsequent chapters. Chapter 2 of this document gives the observed branch flow violations for each of the studied scenario. Chapter 3 gives the upper voltage violations reported for each of the studied scenarios. Chapter 4 gives the lower voltage violations reported for each of the studied scenarios. To conclude, Chapter 5 summarises the result of the contingency analysis carried out on Year 5, Topology 6 of the hypothetical SAVNW system.

Chapter 2

Branch Overload Violation

2.1 Introduction

In this chapter, for each of the studied scenario, the branches that are loaded more than 100% of their rating is tabulated. Branches that are loaded more than 130% of the rating are said to be severely/ critically loaded and are noted down for reporting as operating these branches for prolonged duration is not recommended for a safe and reliable power system.

2.2 Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.1. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 205 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.1: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	386.80	379.16	300.00	126.39
3001-3003(2)	SING OPN LIN 1 101-151(1)	386.80	379.16	300.00	126.39
3003-3005(1)	SING OPN LIN 1 101-151(1)	-375.04	379.38	350.00	108.39
3003-3005(2)	SING OPN LIN 1 101-151(1)	-375.04	379.38	350.00	108.39
101-151(1)	SING OPN LIN 2 102-151(1)	1516.41	1516.41	1350.00	112.33
154-203(1)	SING OPN LIN 6 152-153(1)	298.16	328.73	250.00	131.49
154-205(1)	SING OPN LIN 6 152-153(1)	761.43	839.50	660.00	127.20
154-203(1)	BUS 153	280.58	301.66	250.00	120.66
154-205(1)	BUS 153	684.33	735.74	660.00	111.48
153-154(1)	BUS 205	-276.44	364.06	350.00	104.02
154-203(1)	BUS 205	380.42	501.01	250.00	200.40
3001-3003(1)	UNIT 101(1)	386.87	379.23	300.00	126.41
3001-3003(2)	UNIT 101(1)	386.87	379.23	300.00	126.41
3003-3005(1)	UNIT 101(1)	-375.11	379.45	350.00	108.41
3003-3005(2)	UNIT 101(1)	-375.11	379.45	350.00	108.41
101-151(1)	UNIT 102(1)	1516.41	1516.41	1350.00	112.33

2.3 Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.2. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 205 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.2: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	372.04	364.19	300.00	121.40
3001-3003(2)	SING OPN LIN 1 101-151(1)	372.04	364.19	300.00	121.40
3003-3005(1)	SING OPN LIN 1 101-151(1)	-361.05	364.44	350.00	104.13
3003-3005(2)	SING OPN LIN 1 101-151(1)	-361.05	364.44	350.00	104.13
101-151(1)	SING OPN LIN 2 102-151(1)	1516.48	1516.48	1350.00	112.33
154-203(1)	SING OPN LIN 6 152-153(1)	296.13	325.81	250.00	130.32
154-205(1)	SING OPN LIN 6 152-153(1)	755.65	831.37	660.00	125.97
154-203(1)	BUS 153	280.16	301.04	250.00	120.41
154-205(1)	BUS 153	683.04	733.93	660.00	111.20
153-154(1)	BUS 203	-324.87	350.28	350.00	100.08
153-154(1)	BUS 205	-278.20	369.32	350.00	105.52
154-203(1)	BUS 205	381.33	506.23	250.00	202.49
3001-3003(1)	UNIT 101(1)	372.11	364.26	300.00	121.42
3001-3003(2)	UNIT 101(1)	372.11	364.26	300.00	121.42
3003-3005(1)	UNIT 101(1)	-361.11	364.50	350.00	104.14
3003-3005(2)	UNIT 101(1)	-361.11	364.50	350.00	104.14
101-151(1)	UNIT 102(1)	1516.48	1516.48	1350.00	112.33

2.4 Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.3. No branches reported a loading greater than 130% of the branch rating.

Table 2.3: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	359.49	351.54	300.00	117.18
3001-3003(2)	SING OPN LIN 1 101-151(1)	359.49	351.54	300.00	117.18
3003-3005(1)	SING OPN LIN 1 101-151(1)	-349.05	351.82	350.00	100.52
3003-3005(2)	SING OPN LIN 1 101-151(1)	-349.05	351.82	350.00	100.52
101-151(1)	SING OPN LIN 2 102-151(1)	1516.60	1516.60	1350.00	112.34
154-203(1)	SING OPN LIN 6 152-153(1)	294.57	324.01	250.00	129.60
154-205(1)	SING OPN LIN 6 152-153(1)	751.86	827.00	660.00	125.30
154-203(1)	BUS 153	279.92	300.86	250.00	120.34
154-205(1)	BUS 153	682.60	733.65	660.00	111.16
153-154(1)	BUS 203	-326.44	352.21	350.00	100.63
3001-3003(1)	UNIT 101(1)	359.55	351.61	300.00	117.20
3001-3003(2)	UNIT 101(1)	359.55	351.61	300.00	117.20
3003-3005(1)	UNIT 101(1)	-349.11	351.89	350.00	100.54
3003-3005(2)	UNIT 101(1)	-349.11	351.89	350.00	100.54
101-151(1)	UNIT 102(1)	1516.60	1516.60	1350.00	112.34

2.5 Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.4. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.4: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	428.13	424.23	300.00	141.41
3001-3003(2)	SING OPN LIN 1 101-151(1)	428.13	424.23	300.00	141.41
3003-3005(1)	SING OPN LIN 1 101-151(1)	-408.26	424.87	350.00	121.39
3003-3005(2)	SING OPN LIN 1 101-151(1)	-408.26	424.87	350.00	121.39
101-151(1)	SING OPN LIN 2 102-151(1)	1594.72	1594.72	1350.00	118.13
154-203(1)	SING OPN LIN 6 152-153(1)	310.70	364.11	250.00	145.64
154-205(1)	SING OPN LIN 6 152-153(1)	812.73	952.43	660.00	144.31
203-205(1)	SING OPN LIN 6 152-153(1)	-220.00	253.11	250.00	101.24
203-205(2)	SING OPN LIN 6 152-153(1)	-220.00	253.11	250.00	101.24
205-206(1)	SING OPN LIN 12 201-211(1)	-1291.26	1291.26	1250.00	103.30
205-206(1)	SING OPN LIN 13 201-212(1)	-1291.26	1291.26	1250.00	103.30
154-203(1)	BUS 153	293.58	331.95	250.00	132.78
154-205(1)	BUS 153	731.49	827.08	660.00	125.31
153-154(1)	BUS 203	-336.33	380.69	350.00	108.77
154-203(1)	BUS 3005	229.20	273.34	250.00	109.33
154-205(1)	BUS 3005	641.30	764.78	660.00	115.88
154-3008(1)	BUS 3005	-363.01	458.49	440.00	104.20
3001-3003(1)	UNIT 101(1)	428.30	424.41	300.00	141.47
3001-3003(2)	UNIT 101(1)	428.30	424.41	300.00	141.47
3003-3005(1)	UNIT 101(1)	-408.40	425.04	350.00	121.44
3003-3005(2)	UNIT 101(1)	-408.40	425.04	350.00	121.44
101-151(1)	UNIT 102(1)	1594.72	1594.72	1350.00	118.13
205-206(1)	UNIT 211(1)	-1289.97	1289.97	1250.00	103.20
205-206(1)	UNIT 212(1)	-1289.97	1289.97	1250.00	103.20

2.6 Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.5. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.5: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	412.73	408.08	300.00	136.03
3001-3003(2)	SING OPN LIN 1 101-151(1)	412.73	408.08	300.00	136.03
3003-3005(1)	SING OPN LIN 1 101-151(1)	-394.42	408.72	350.00	116.78
3003-3005(2)	SING OPN LIN 1 101-151(1)	-394.42	408.72	350.00	116.78
101-151(1)	SING OPN LIN 2 102-151(1)	1594.66	1594.66	1350.00	118.12
154-203(1)	SING OPN LIN 6 152-153(1)	308.70	360.63	250.00	144.25
154-205(1)	SING OPN LIN 6 152-153(1)	806.76	942.47	660.00	142.80
203-205(1)	SING OPN LIN 6 152-153(1)	-218.60	250.76	250.00	100.31
203-205(2)	SING OPN LIN 6 152-153(1)	-218.60	250.76	250.00	100.31
205-206(1)	SING OPN LIN 12 201-211(1)	-1291.09	1291.09	1250.00	103.29
205-206(1)	SING OPN LIN 13 201-212(1)	-1291.09	1291.09	1250.00	103.29
154-203(1)	BUS 153	293.17	331.22	250.00	132.49
154-205(1)	BUS 153	730.17	824.93	660.00	124.99
153-154(1)	BUS 203	-338.16	382.66	350.00	109.33
154-203(1)	BUS 3005	228.23	271.14	250.00	108.46
154-205(1)	BUS 3005	637.39	757.22	660.00	114.73
154-3008(1)	BUS 3005	-362.85	456.22	440.00	103.69
3001-3003(1)	UNIT 101(1)	412.89	408.24	300.00	136.08
3001-3003(2)	UNIT 101(1)	412.89	408.24	300.00	136.08
3003-3005(1)	UNIT 101(1)	-394.56	408.88	350.00	116.82
3003-3005(2)	UNIT 101(1)	-394.56	408.88	350.00	116.82
101-151(1)	UNIT 102(1)	1594.66	1594.66	1350.00	118.12
205-206(1)	UNIT 211(1)	-1289.83	1289.83	1250.00	103.19
205-206(1)	UNIT 212(1)	-1289.83	1289.83	1250.00	103.19

2.7 Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.6. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.6: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	399.83	394.71	300.00	131.57
3001-3003(2)	SING OPN LIN 1 101-151(1)	399.83	394.71	300.00	131.57
3003-3005(1)	SING OPN LIN 1 101-151(1)	-382.59	395.37	350.00	112.96
3003-3005(2)	SING OPN LIN 1 101-151(1)	-382.59	395.37	350.00	112.96
101-151(1)	SING OPN LIN 2 102-151(1)	1594.83	1594.83	1350.00	118.14
154-203(1)	SING OPN LIN 6 152-153(1)	307.24	358.80	250.00	143.52
154-205(1)	SING OPN LIN 6 152-153(1)	802.93	937.68	660.00	142.07
205-206(1)	SING OPN LIN 12 201-211(1)	-1291.03	1291.03	1250.00	103.28
205-206(1)	SING OPN LIN 13 201-212(1)	-1291.03	1291.03	1250.00	103.28
154-203(1)	BUS 153	292.96	331.09	250.00	132.44
154-205(1)	BUS 153	729.74	824.72	660.00	124.96
153-154(1)	BUS 203	-339.61	384.65	350.00	109.90
154-203(1)	BUS 3005	227.64	270.28	250.00	108.11
154-205(1)	BUS 3005	635.70	754.79	660.00	114.36
154-3008(1)	BUS 3005	-362.83	455.89	440.00	103.61
3001-3003(1)	UNIT 101(1)	399.98	394.87	300.00	131.62
3001-3003(2)	UNIT 101(1)	399.98	394.87	300.00	131.62
3003-3005(1)	UNIT 101(1)	-382.73	395.53	350.00	113.01
3003-3005(2)	UNIT 101(1)	-382.73	395.53	350.00	113.01
101-151(1)	UNIT 102(1)	1594.83	1594.83	1350.00	118.14
205-206(1)	UNIT 211(1)	-1289.72	1289.72	1250.00	103.18
205-206(1)	UNIT 212(1)	-1289.72	1289.72	1250.00	103.18

2.8 Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.7. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2), 3003-3005(1), 3003-3005(2) for single line open contingency SING OPN LIN 1 101-151(1); 3001-3003(1), 3001-3003(2), 3003-3005(1), 3003-3005(2) for unit fault contingency UNIT 101(1); 154-203(1), 154-205(1) for bus fault contingency BUS 153.

Table 2.7: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	463.69	464.38	300.00	154.79
3001-3003(2)	SING OPN LIN 1 101-151(1)	463.69	464.38	300.00	154.79
3003-3005(1)	SING OPN LIN 1 101-151(1)	-434.60	465.34	350.00	132.95
3003-3005(2)	SING OPN LIN 1 101-151(1)	-434.60	465.34	350.00	132.95
101-151(1)	SING OPN LIN 2 102-151(1)	1660.01	1660.01	1350.00	122.96
205-206(1)	SING OPN LIN 12 201-211(1)	-1354.66	1354.66	1250.00	108.37
205-206(1)	SING OPN LIN 13 201-212(1)	-1354.66	1354.66	1250.00	108.37
154-203(1)	BUS 153	305.41	365.08	250.00	146.03
154-205(1)	BUS 153	768.20	918.27	660.00	139.13
203-205(1)	BUS 153	-218.75	256.38	250.00	102.55
203-205(2)	BUS 153	-218.75	256.38	250.00	102.55
153-154(1)	BUS 203	-347.32	413.98	350.00	118.28
3001-3003(1)	UNIT 101(1)	463.72	464.41	300.00	154.80
3001-3003(2)	UNIT 101(1)	463.72	464.41	300.00	154.80
3003-3005(1)	UNIT 101(1)	-434.63	465.37	350.00	132.96
3003-3005(2)	UNIT 101(1)	-434.63	465.37	350.00	132.96
101-151(1)	UNIT 102(1)	1660.00	1660.00	1350.00	122.96
205-206(1)	UNIT 211(1)	-1354.27	1354.27	1250.00	108.34
205-206(1)	UNIT 212(1)	-1354.27	1354.27	1250.00	108.34

2.9 Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.8. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1), 154-205(1) for bus fault contingency BUS 153; 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.8: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	447.71	447.18	300.00	149.06
3001-3003(2)	SING OPN LIN 1 101-151(1)	447.71	447.18	300.00	149.06
3003-3005(1)	SING OPN LIN 1 101-151(1)	-420.85	448.15	350.00	128.04
3003-3005(2)	SING OPN LIN 1 101-151(1)	-420.85	448.15	350.00	128.04
101-151(1)	SING OPN LIN 2 102-151(1)	1659.87	1659.87	1350.00	122.95
205-206(1)	SING OPN LIN 12 201-211(1)	-1354.44	1354.44	1250.00	108.35
205-206(1)	SING OPN LIN 13 201-212(1)	-1354.44	1354.44	1250.00	108.35
154-203(1)	BUS 153	305.01	364.18	250.00	145.67
154-205(1)	BUS 153	766.86	915.63	660.00	138.73
203-205(1)	BUS 153	-218.46	255.77	250.00	102.31
203-205(2)	BUS 153	-218.46	255.77	250.00	102.31
153-154(1)	BUS 203	-349.39	416.46	350.00	118.99
3001-3003(1)	UNIT 101(1)	447.74	447.21	300.00	149.07
3001-3003(2)	UNIT 101(1)	447.74	447.21	300.00	149.07
3003-3005(1)	UNIT 101(1)	-420.87	448.18	350.00	128.05
3003-3005(2)	UNIT 101(1)	-420.87	448.18	350.00	128.05
101-151(1)	UNIT 102(1)	1659.86	1659.86	1350.00	122.95
205-206(1)	UNIT 211(1)	-1354.06	1354.06	1250.00	108.32
205-206(1)	UNIT 212(1)	-1354.06	1354.06	1250.00	108.32

2.10 Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.9. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1), 154-205(1) for bus fault contingency BUS 153; 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.9: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	434.45	433.10	300.00	144.37
3001-3003(2)	SING OPN LIN 1 101-151(1)	434.45	433.10	300.00	144.37
3003-3005(1)	SING OPN LIN 1 101-151(1)	-409.09	434.10	350.00	124.03
3003-3005(2)	SING OPN LIN 1 101-151(1)	-409.09	434.10	350.00	124.03
101-151(1)	SING OPN LIN 2 102-151(1)	1660.05	1660.05	1350.00	122.97
205-206(1)	SING OPN LIN 12 201-211(1)	-1354.32	1354.32	1250.00	108.35
205-206(1)	SING OPN LIN 13 201-212(1)	-1354.32	1354.32	1250.00	108.35
154-203(1)	BUS 153	304.84	364.14	250.00	145.66
154-205(1)	BUS 153	766.44	915.54	660.00	138.72
203-205(1)	BUS 153	-218.35	255.75	250.00	102.30
203-205(2)	BUS 153	-218.35	255.75	250.00	102.30
153-154(1)	BUS 203	-350.54	418.56	350.00	119.59
3001-3003(1)	UNIT 101(1)	434.47	433.13	300.00	144.38
3001-3003(2)	UNIT 101(1)	434.47	433.13	300.00	144.38
3003-3005(1)	UNIT 101(1)	-409.12	434.13	350.00	124.04
3003-3005(2)	UNIT 101(1)	-409.12	434.13	350.00	124.04
101-151(1)	UNIT 102(1)	1660.04	1660.04	1350.00	122.97
205-206(1)	UNIT 211(1)	-1353.94	1353.94	1250.00	108.31
205-206(1)	UNIT 212(1)	-1353.94	1353.94	1250.00	108.31

2.11 Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.10. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.10: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	389.00	381.91	300.00	127.30
3001-3003(2)	SING OPN LIN 1 101-151(1)	389.00	381.91	300.00	127.30
3003-3005(1)	SING OPN LIN 1 101-151(1)	-376.07	382.24	350.00	109.21
3003-3005(2)	SING OPN LIN 1 101-151(1)	-376.07	382.24	350.00	109.21
101-151(1)	SING OPN LIN 2 102-151(1)	1520.99	1520.99	1350.00	112.67
154-203(1)	SING OPN LIN 6 152-153(1)	319.52	358.62	250.00	143.45
154-205(1)	SING OPN LIN 6 152-153(1)	717.75	805.57	660.00	122.06
203-205(1)	SING OPN LIN 6 152-153(1)	-235.67	260.33	250.00	104.13
203-205(2)	SING OPN LIN 6 152-153(1)	-235.67	260.33	250.00	104.13
154-203(1)	BUS 153	303.93	331.99	250.00	132.80
154-205(1)	BUS 153	644.21	703.68	660.00	106.62
153-154(1)	BUS 203	-349.70	387.51	350.00	110.72
154-203(1)	BUS 3005	229.34	257.53	250.00	103.01
3001-3003(1)	UNIT 101(1)	389.08	381.99	300.00	127.33
3001-3003(2)	UNIT 101(1)	389.08	381.99	300.00	127.33
3003-3005(1)	UNIT 101(1)	-376.14	382.32	350.00	109.23
3003-3005(2)	UNIT 101(1)	-376.14	382.32	350.00	109.23
101-151(1)	UNIT 102(1)	1520.99	1520.99	1350.00	112.67

2.12 Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.11. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.11: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	374.04	366.57	300.00	122.19
3001-3003(2)	SING OPN LIN 1 101-151(1)	374.04	366.57	300.00	122.19
3003-3005(1)	SING OPN LIN 1 101-151(1)	-362.20	366.90	350.00	104.83
3003-3005(2)	SING OPN LIN 1 101-151(1)	-362.20	366.90	350.00	104.83
101-151(1)	SING OPN LIN 2 102-151(1)	1521.01	1521.01	1350.00	112.67
154-203(1)	SING OPN LIN 6 152-153(1)	317.54	355.54	250.00	142.22
154-205(1)	SING OPN LIN 6 152-153(1)	711.88	797.09	660.00	120.77
203-205(1)	SING OPN LIN 6 152-153(1)	-234.30	258.23	250.00	103.29
203-205(2)	SING OPN LIN 6 152-153(1)	-234.30	258.23	250.00	103.29
154-203(1)	BUS 153	303.52	331.33	250.00	132.53
154-205(1)	BUS 153	642.92	701.83	660.00	106.34
153-154(1)	BUS 203	-351.62	389.56	350.00	111.30
154-203(1)	BUS 3005	228.39	255.87	250.00	102.35
3001-3003(1)	UNIT 101(1)	374.12	366.65	300.00	122.22
3001-3003(2)	UNIT 101(1)	374.12	366.65	300.00	122.22
3003-3005(1)	UNIT 101(1)	-362.27	366.98	350.00	104.85
3003-3005(2)	UNIT 101(1)	-362.27	366.98	350.00	104.85
101-151(1)	UNIT 102(1)	1521.01	1521.01	1350.00	112.67

2.13 Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.12. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.12: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	361.40	353.75	300.00	117.92
3001-3003(2)	SING OPN LIN 1 101-151(1)	361.40	353.75	300.00	117.92
3003-3005(1)	SING OPN LIN 1 101-151(1)	-350.30	354.10	350.00	101.17
3003-3005(2)	SING OPN LIN 1 101-151(1)	-350.30	354.10	350.00	101.17
101-151(1)	SING OPN LIN 2 102-151(1)	1520.90	1520.90	1350.00	112.66
154-203(1)	SING OPN LIN 6 152-153(1)	315.98	353.69	250.00	141.47
154-205(1)	SING OPN LIN 6 152-153(1)	708.17	792.69	660.00	120.10
203-205(1)	SING OPN LIN 6 152-153(1)	-233.19	256.93	250.00	102.77
203-205(2)	SING OPN LIN 6 152-153(1)	-233.19	256.93	250.00	102.77
154-203(1)	BUS 153	303.27	331.16	250.00	132.46
154-205(1)	BUS 153	642.55	701.64	660.00	106.31
153-154(1)	BUS 203	-353.15	391.60	350.00	111.88
154-203(1)	BUS 3005	227.70	255.04	250.00	102.01
3001-3003(1)	UNIT 101(1)	361.35	353.70	300.00	117.90
3001-3003(2)	UNIT 101(1)	361.35	353.70	300.00	117.90
3003-3005(1)	UNIT 101(1)	-350.26	354.05	350.00	101.16
3003-3005(2)	UNIT 101(1)	-350.26	354.05	350.00	101.16
101-151(1)	UNIT 102(1)	1520.90	1520.90	1350.00	112.66

2.14 Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.13. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.13: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	431.55	428.41	300.00	142.80
3001-3003(2)	SING OPN LIN 1 101-151(1)	431.55	428.41	300.00	142.80
3003-3005(1)	SING OPN LIN 1 101-151(1)	-410.03	429.14	350.00	122.61
3003-3005(2)	SING OPN LIN 1 101-151(1)	-410.03	429.14	350.00	122.61
101-151(1)	SING OPN LIN 2 102-151(1)	1600.53	1600.53	1350.00	118.56
154-203(1)	SING OPN LIN 6 152-153(1)	331.55	398.40	250.00	159.36
154-205(1)	SING OPN LIN 6 152-153(1)	766.93	921.57	660.00	139.63
203-205(1)	SING OPN LIN 6 152-153(1)	-243.02	286.40	250.00	114.56
203-205(2)	SING OPN LIN 6 152-153(1)	-243.02	286.40	250.00	114.56
154-203(1)	BUS 153	315.73	363.94	250.00	145.58
154-205(1)	BUS 153	692.20	797.91	660.00	120.90
203-205(1)	BUS 153	-234.24	265.31	250.00	106.12
203-205(2)	BUS 153	-234.24	265.31	250.00	106.12
153-154(1)	BUS 203	-362.59	425.60	350.00	121.60
153-154(2)	BUS 203	-302.80	355.42	350.00	101.55
153-154(3)	BUS 203	-302.80	355.42	350.00	101.55
3001-3003(1)	UNIT 101(1)	431.75	428.63	300.00	142.88
3001-3003(2)	UNIT 101(1)	431.75	428.63	300.00	142.88
3003-3005(1)	UNIT 101(1)	-410.21	429.36	350.00	122.67
3003-3005(2)	UNIT 101(1)	-410.21	429.36	350.00	122.67
101-151(1)	UNIT 102(1)	1600.52	1600.52	1350.00	118.56

2.15 Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.14. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.14: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	416.01	412.06	300.00	137.35
3001-3003(2)	SING OPN LIN 1 101-151(1)	416.01	412.06	300.00	137.35
3003-3005(1)	SING OPN LIN 1 101-151(1)	-396.17	412.80	350.00	117.94
3003-3005(2)	SING OPN LIN 1 101-151(1)	-396.17	412.80	350.00	117.94
101-151(1)	SING OPN LIN 2 102-151(1)	1600.48	1600.48	1350.00	118.55
154-203(1)	SING OPN LIN 6 152-153(1)	329.61	394.58	250.00	157.83
154-205(1)	SING OPN LIN 6 152-153(1)	760.86	910.82	660.00	138.00
203-205(1)	SING OPN LIN 6 152-153(1)	-241.68	283.82	250.00	113.53
203-205(2)	SING OPN LIN 6 152-153(1)	-241.68	283.82	250.00	113.53
154-203(1)	BUS 153	315.33	363.16	250.00	145.26
154-205(1)	BUS 153	690.89	795.69	660.00	120.56
203-205(1)	BUS 153	-233.97	264.78	250.00	105.91
203-205(2)	BUS 153	-233.97	264.78	250.00	105.91
153-154(1)	BUS 203	-364.37	427.56	350.00	122.16
153-154(2)	BUS 203	-304.27	357.04	350.00	102.01
153-154(3)	BUS 203	-304.27	357.04	350.00	102.01
3001-3003(1)	UNIT 101(1)	416.21	412.27	300.00	137.42
3001-3003(2)	UNIT 101(1)	416.21	412.27	300.00	137.42
3003-3005(1)	UNIT 101(1)	-396.35	413.01	350.00	118.00
3003-3005(2)	UNIT 101(1)	-396.35	413.01	350.00	118.00
101-151(1)	UNIT 102(1)	1600.48	1600.48	1350.00	118.55

2.16 Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.15. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.15: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	403.03	398.58	300.00	132.86
3001-3003(2)	SING OPN LIN 1 101-151(1)	403.03	398.58	300.00	132.86
3003-3005(1)	SING OPN LIN 1 101-151(1)	-384.34	399.34	350.00	114.10
3003-3005(2)	SING OPN LIN 1 101-151(1)	-384.34	399.34	350.00	114.10
101-151(1)	SING OPN LIN 2 102-151(1)	1600.69	1600.69	1350.00	118.57
154-203(1)	SING OPN LIN 6 152-153(1)	328.15	392.64	250.00	157.06
154-205(1)	SING OPN LIN 6 152-153(1)	757.23	906.07	660.00	137.28
203-205(1)	SING OPN LIN 6 152-153(1)	-240.64	282.46	250.00	112.99
203-205(2)	SING OPN LIN 6 152-153(1)	-240.64	282.46	250.00	112.99
154-203(1)	BUS 153	315.12	363.05	250.00	145.22
154-205(1)	BUS 153	690.52	795.57	660.00	120.54
203-205(1)	BUS 153	-233.82	264.70	250.00	105.88
203-205(2)	BUS 153	-233.82	264.70	250.00	105.88
153-154(1)	BUS 203	-365.76	429.72	350.00	122.78
153-154(2)	BUS 203	-305.43	358.84	350.00	102.53
153-154(3)	BUS 203	-305.43	358.84	350.00	102.53
3001-3003(1)	UNIT 101(1)	403.23	398.78	300.00	132.93
3001-3003(2)	UNIT 101(1)	403.23	398.78	300.00	132.93
3003-3005(1)	UNIT 101(1)	-384.51	399.55	350.00	114.16
3003-3005(2)	UNIT 101(1)	-384.51	399.55	350.00	114.16
101-151(1)	UNIT 102(1)	1600.68	1600.68	1350.00	118.57

2.17 Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.16. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2), 3003-3005(1), 3003-3005(2) for single line open contingency SING OPN LIN 1 101-151(1); 3001-3003(1), 3001-3003(2), 3003-3005(1), 3003-3005(2) for unit fault contingency UNIT 101(1).

Table 2.16: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	185.43	203.47	200.00	101.74
3001-3003(1)	SING OPN LIN 1 101-151(1)	467.44	469.09	300.00	156.36
3001-3003(2)	SING OPN LIN 1 101-151(1)	467.44	469.09	300.00	156.36
3003-3005(1)	SING OPN LIN 1 101-151(1)	-436.38	470.13	350.00	134.32
3003-3005(2)	SING OPN LIN 1 101-151(1)	-436.38	470.13	350.00	134.32
101-151(1)	SING OPN LIN 2 102-151(1)	1666.66	1666.66	1350.00	123.46
205-206(1)	SING OPN LIN 12 201-211(1)	-1269.49	1269.49	1250.00	101.56
205-206(1)	SING OPN LIN 13 201-212(1)	-1269.49	1269.49	1250.00	101.56
3001-3003(1)	UNIT 101(1)	467.49	469.14	300.00	156.38
3001-3003(2)	UNIT 101(1)	467.49	469.14	300.00	156.38
3003-3005(1)	UNIT 101(1)	-436.42	470.18	350.00	134.34
3003-3005(2)	UNIT 101(1)	-436.42	470.18	350.00	134.34
101-151(1)	UNIT 102(1)	1666.64	1666.64	1350.00	123.45
205-206(1)	UNIT 211(1)	-1268.61	1268.61	1250.00	101.49
205-206(1)	UNIT 212(1)	-1268.61	1268.61	1250.00	101.49

2.18 Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.17. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.17: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	185.71	203.72	200.00	101.86
3001-3003(1)	SING OPN LIN 1 101-151(1)	451.32	451.64	300.00	150.55
3001-3003(2)	SING OPN LIN 1 101-151(1)	451.32	451.64	300.00	150.55
3003-3005(1)	SING OPN LIN 1 101-151(1)	-422.61	452.70	350.00	129.34
3003-3005(2)	SING OPN LIN 1 101-151(1)	-422.61	452.70	350.00	129.34
101-151(1)	SING OPN LIN 2 102-151(1)	1666.54	1666.54	1350.00	123.45
205-206(1)	SING OPN LIN 12 201-211(1)	-1269.27	1269.27	1250.00	101.54
205-206(1)	SING OPN LIN 13 201-212(1)	-1269.27	1269.27	1250.00	101.54
3001-3003(1)	UNIT 101(1)	451.36	451.69	300.00	150.56
3001-3003(2)	UNIT 101(1)	451.36	451.69	300.00	150.56
3003-3005(1)	UNIT 101(1)	-422.65	452.74	350.00	129.36
3003-3005(2)	UNIT 101(1)	-422.65	452.74	350.00	129.36
101-151(1)	UNIT 102(1)	1666.52	1666.52	1350.00	123.45
205-206(1)	UNIT 211(1)	-1268.42	1268.42	1250.00	101.47
205-206(1)	UNIT 212(1)	-1268.42	1268.42	1250.00	101.47

2.19 Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.18. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.18: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	186.04	204.20	200.00	102.10
3001-3003(1)	SING OPN LIN 1 101-151(1)	437.97	437.43	300.00	145.81
3001-3003(2)	SING OPN LIN 1 101-151(1)	437.97	437.43	300.00	145.81
3003-3005(1)	SING OPN LIN 1 101-151(1)	-410.87	438.52	350.00	125.29
3003-3005(2)	SING OPN LIN 1 101-151(1)	-410.87	438.52	350.00	125.29
101-151(1)	SING OPN LIN 2 102-151(1)	1666.76	1666.76	1350.00	123.46
205-206(1)	SING OPN LIN 12 201-211(1)	-1269.17	1269.17	1250.00	101.53
205-206(1)	SING OPN LIN 13 201-212(1)	-1269.17	1269.17	1250.00	101.53
3001-3003(1)	UNIT 101(1)	438.01	437.47	300.00	145.82
3001-3003(2)	UNIT 101(1)	438.01	437.47	300.00	145.82
3003-3005(1)	UNIT 101(1)	-410.90	438.56	350.00	125.30
3003-3005(2)	UNIT 101(1)	-410.90	438.56	350.00	125.30
101-151(1)	UNIT 102(1)	1666.74	1666.74	1350.00	123.46
205-206(1)	UNIT 211(1)	-1268.32	1268.32	1250.00	101.47
205-206(1)	UNIT 212(1)	-1268.32	1268.32	1250.00	101.47

2.20 Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.19. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.19: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	392.25	386.00	300.00	128.67
3001-3003(2)	SING OPN LIN 1 101-151(1)	392.25	386.00	300.00	128.67
3003-3005(1)	SING OPN LIN 1 101-151(1)	-377.48	386.49	350.00	110.43
3003-3005(2)	SING OPN LIN 1 101-151(1)	-377.48	386.49	350.00	110.43
101-151(1)	SING OPN LIN 2 102-151(1)	1527.26	1527.26	1350.00	113.13
154-203(1)	SING OPN LIN 6 152-153(1)	341.24	393.63	250.00	157.45
154-205(1)	SING OPN LIN 6 152-153(1)	672.26	775.48	660.00	117.50
203-205(1)	SING OPN LIN 6 152-153(1)	-259.46	294.37	250.00	117.75
203-205(2)	SING OPN LIN 6 152-153(1)	-259.46	294.37	250.00	117.75
154-203(1)	BUS 153	327.50	366.30	250.00	146.52
154-205(1)	BUS 153	603.72	675.25	660.00	102.31
203-205(1)	BUS 153	-252.45	278.07	250.00	111.23
203-205(2)	BUS 153	-252.45	278.07	250.00	111.23
153-154(1)	BUS 203	-377.04	435.78	350.00	124.51
153-154(2)	BUS 203	-314.73	363.77	350.00	103.93
153-154(3)	BUS 203	-314.73	363.77	350.00	103.93
154-203(1)	BUS 3005	246.24	285.12	250.00	114.05
3001-3003(1)	UNIT 101(1)	392.37	386.12	300.00	128.71
3001-3003(2)	UNIT 101(1)	392.37	386.12	300.00	128.71
3003-3005(1)	UNIT 101(1)	-377.59	386.61	350.00	110.46
3003-3005(2)	UNIT 101(1)	-377.59	386.61	350.00	110.46
101-151(1)	UNIT 102(1)	1527.25	1527.25	1350.00	113.13

2.21 Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.20. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.20: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	376.93	370.24	300.00	123.41
3001-3003(2)	SING OPN LIN 1 101-151(1)	376.93	370.24	300.00	123.41
3003-3005(1)	SING OPN LIN 1 101-151(1)	-363.40	370.73	350.00	105.92
3003-3005(2)	SING OPN LIN 1 101-151(1)	-363.40	370.73	350.00	105.92
101-151(1)	SING OPN LIN 2 102-151(1)	1527.28	1527.28	1350.00	113.13
154-203(1)	SING OPN LIN 6 152-153(1)	339.30	390.26	250.00	156.11
154-205(1)	SING OPN LIN 6 152-153(1)	666.38	766.48	660.00	116.13
203-205(1)	SING OPN LIN 6 152-153(1)	-258.12	292.05	250.00	116.82
203-205(2)	SING OPN LIN 6 152-153(1)	-258.12	292.05	250.00	116.82
154-203(1)	BUS 153	327.10	365.60	250.00	146.24
154-205(1)	BUS 153	602.45	673.34	660.00	102.02
203-205(1)	BUS 153	-252.18	277.58	250.00	111.03
203-205(2)	BUS 153	-252.18	277.58	250.00	111.03
153-154(1)	BUS 203	-378.89	437.83	350.00	125.10
153-154(2)	BUS 203	-316.27	365.48	350.00	104.42
153-154(3)	BUS 203	-316.27	365.48	350.00	104.42
153-154(1)	BUS 3005	-303.56	350.51	350.00	100.15
154-203(1)	BUS 3005	245.30	283.24	250.00	113.30
3001-3003(1)	UNIT 101(1)	376.94	370.25	300.00	123.42
3001-3003(2)	UNIT 101(1)	376.94	370.25	300.00	123.42
3003-3005(1)	UNIT 101(1)	-363.41	370.74	350.00	105.93
3003-3005(2)	UNIT 101(1)	-363.41	370.74	350.00	105.93
101-151(1)	UNIT 102(1)	1527.28	1527.28	1350.00	113.13

2.22 Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.21. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.21: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	364.26	357.34	300.00	119.11
3001-3003(2)	SING OPN LIN 1 101-151(1)	364.26	357.34	300.00	119.11
3003-3005(1)	SING OPN LIN 1 101-151(1)	-351.55	357.86	350.00	102.25
3003-3005(2)	SING OPN LIN 1 101-151(1)	-351.55	357.86	350.00	102.25
101-151(1)	SING OPN LIN 2 102-151(1)	1527.50	1527.50	1350.00	113.15
154-203(1)	SING OPN LIN 6 152-153(1)	337.75	388.33	250.00	155.33
154-205(1)	SING OPN LIN 6 152-153(1)	662.84	762.10	660.00	115.47
203-205(1)	SING OPN LIN 6 152-153(1)	-257.03	290.70	250.00	116.28
203-205(2)	SING OPN LIN 6 152-153(1)	-257.03	290.70	250.00	116.28
154-203(1)	BUS 153	326.86	365.44	250.00	146.18
154-205(1)	BUS 153	602.16	673.25	660.00	102.01
203-205(1)	BUS 153	-252.00	277.47	250.00	110.99
203-205(2)	BUS 153	-252.00	277.47	250.00	110.99
153-154(1)	BUS 203	-380.35	440.07	350.00	125.73
153-154(2)	BUS 203	-317.49	367.33	350.00	104.95
153-154(3)	BUS 203	-317.49	367.33	350.00	104.95
153-154(1)	BUS 3005	-304.66	351.68	350.00	100.48
154-203(1)	BUS 3005	244.64	282.39	250.00	112.96
3001-3003(1)	UNIT 101(1)	364.27	357.35	300.00	119.12
3001-3003(2)	UNIT 101(1)	364.27	357.35	300.00	119.12
3003-3005(1)	UNIT 101(1)	-351.56	357.87	350.00	102.25
3003-3005(2)	UNIT 101(1)	-351.56	357.87	350.00	102.25
101-151(1)	UNIT 102(1)	1527.50	1527.50	1350.00	113.15

2.23 Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.22. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency

SING OPN LIN 1 101-151(1).

Table 2.22: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	194.24	209.85	200.00	104.93
3001-3003(1)	SING OPN LIN 1 101-151(1)	436.09	434.12	300.00	144.71
3001-3003(2)	SING OPN LIN 1 101-151(1)	436.09	434.12	300.00	144.71
3003-3005(1)	SING OPN LIN 1 101-151(1)	-412.08	435.00	350.00	124.28
3003-3005(2)	SING OPN LIN 1 101-151(1)	-412.08	435.00	350.00	124.28
101-151(1)	SING OPN LIN 2 102-151(1)	1608.09	1608.09	1350.00	119.12
3001-3003(1)	UNIT 101(1)	436.36	434.41	300.00	144.80
3001-3003(2)	UNIT 101(1)	436.36	434.41	300.00	144.80
3003-3005(1)	UNIT 101(1)	-412.31	435.29	350.00	124.37
3003-3005(2)	UNIT 101(1)	-412.31	435.29	350.00	124.37
101-151(1)	UNIT 102(1)	1608.08	1608.08	1350.00	119.12

2.24 Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.23. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.23: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	194.54	210.13	200.00	105.06
3001-3003(1)	SING OPN LIN 1 101-151(1)	420.37	417.49	300.00	139.16
3001-3003(2)	SING OPN LIN 1 101-151(1)	420.37	417.49	300.00	139.16
3003-3005(1)	SING OPN LIN 1 101-151(1)	-398.21	418.38	350.00	119.54
3003-3005(2)	SING OPN LIN 1 101-151(1)	-398.21	418.38	350.00	119.54
101-151(1)	SING OPN LIN 2 102-151(1)	1608.05	1608.05	1350.00	119.11
3001-3003(1)	UNIT 101(1)	420.63	417.77	300.00	139.26
3001-3003(2)	UNIT 101(1)	420.63	417.77	300.00	139.26
3003-3005(1)	UNIT 101(1)	-398.43	418.66	350.00	119.62
3003-3005(2)	UNIT 101(1)	-398.43	418.66	350.00	119.62
101-151(1)	UNIT 102(1)	1608.04	1608.04	1350.00	119.11

2.25 Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.24. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.24: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	194.87	210.61	200.00	105.30
3001-3003(1)	SING OPN LIN 1 101-151(1)	406.06	402.54	300.00	134.18
3001-3003(2)	SING OPN LIN 1 101-151(1)	406.06	402.54	300.00	134.18
3003-3005(1)	SING OPN LIN 1 101-151(1)	-385.31	403.46	350.00	115.27
3003-3005(2)	SING OPN LIN 1 101-151(1)	-385.31	403.46	350.00	115.27
101-151(1)	SING OPN LIN 2 102-151(1)	1608.30	1608.30	1350.00	119.13
3001-3003(1)	UNIT 101(1)	406.08	402.56	300.00	134.19
3001-3003(2)	UNIT 101(1)	406.08	402.56	300.00	134.19
3003-3005(1)	UNIT 101(1)	-385.33	403.48	350.00	115.28
3003-3005(2)	UNIT 101(1)	-385.33	403.48	350.00	115.28
101-151(1)	UNIT 102(1)	1608.30	1608.30	1350.00	119.13

2.26 Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.25. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2), 3003-3005(1), 3003-3005(2) for single line open contingency SING OPN LIN 1 101-151(1); 3001-3003(1), 3001-3003(2), 3003-3005(1), 3003-3005(2) for unit fault contingency UNIT 101(1).

Table 2.25: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	199.60	222.29	200.00	111.15
3001-3003(1)	SING OPN LIN 1 101-151(1)	472.72	475.88	300.00	158.63
3001-3003(2)	SING OPN LIN 1 101-151(1)	472.72	475.88	300.00	158.63
3003-3005(1)	SING OPN LIN 1 101-151(1)	-438.47	477.06	350.00	136.30
3003-3005(2)	SING OPN LIN 1 101-151(1)	-438.47	477.06	350.00	136.30
101-151(1)	SING OPN LIN 2 102-151(1)	1675.53	1675.53	1350.00	124.11
3001-3003(1)	UNIT 101(1)	472.80	475.97	300.00	158.66
3001-3003(2)	UNIT 101(1)	472.80	475.97	300.00	158.66
3003-3005(1)	UNIT 101(1)	-438.53	477.15	350.00	136.33
3003-3005(2)	UNIT 101(1)	-438.53	477.15	350.00	136.33
101-151(1)	UNIT 102(1)	1675.52	1675.52	1350.00	124.11

2.27 Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.26. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2), 3003-3005(1), 3003-3005(2) for single line open contingency SING OPN LIN 1 101-151(1); 3001-3003(1), 3001-3003(2), 3003-3005(1), 3003-3005(2) for unit fault contingency UNIT 101(1).

Table 2.26: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	199.89	222.55	200.00	111.27
3001-3003(1)	SING OPN LIN 1 101-151(1)	456.34	458.05	300.00	152.68
3001-3003(2)	SING OPN LIN 1 101-151(1)	456.34	458.05	300.00	152.68
3003-3005(1)	SING OPN LIN 1 101-151(1)	-424.68	459.25	350.00	131.21
3003-3005(2)	SING OPN LIN 1 101-151(1)	-424.68	459.25	350.00	131.21
101-151(1)	SING OPN LIN 2 102-151(1)	1675.42	1675.42	1350.00	124.11
3001-3003(1)	UNIT 101(1)	456.40	458.12	300.00	152.71
3001-3003(2)	UNIT 101(1)	456.40	458.12	300.00	152.71
3003-3005(1)	UNIT 101(1)	-424.74	459.32	350.00	131.23
3003-3005(2)	UNIT 101(1)	-424.74	459.32	350.00	131.23
101-151(1)	UNIT 102(1)	1675.41	1675.41	1350.00	124.10

2.28 Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.27. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.27: Contingencies causing branch flow greater than 100% for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	200.22	223.07	200.00	111.53
3001-3003(1)	SING OPN LIN 1 101-151(1)	445.77	446.79	300.00	148.93
3001-3003(2)	SING OPN LIN 1 101-151(1)	445.77	446.79	300.00	148.93
3003-3005(1)	SING OPN LIN 1 101-151(1)	-415.37	448.02	350.00	128.00
3003-3005(2)	SING OPN LIN 1 101-151(1)	-415.37	448.02	350.00	128.00
101-151(1)	SING OPN LIN 2 102-151(1)	1675.71	1675.71	1350.00	124.13
3001-3003(1)	UNIT 101(1)	442.91	443.68	300.00	147.89
3001-3003(2)	UNIT 101(1)	442.91	443.68	300.00	147.89
3003-3005(1)	UNIT 101(1)	-413.00	444.91	350.00	127.12
3003-3005(2)	UNIT 101(1)	-413.00	444.91	350.00	127.12
101-151(1)	UNIT 102(1)	1675.69	1675.69	1350.00	124.13

2.29 Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.28. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 205.

Table 2.28: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	368.04	360.11	300.00	120.04
3001-3003(2)	SING OPN LIN 1 101-151(1)	368.04	360.11	300.00	120.04
3003-3005(1)	SING OPN LIN 1 101-151(1)	-357.32	360.35	350.00	102.96
3003-3005(2)	SING OPN LIN 1 101-151(1)	-357.32	360.35	350.00	102.96
101-151(1)	SING OPN LIN 2 102-151(1)	1462.97	1462.97	1350.00	108.37
154-203(1)	SING OPN LIN 6 152-153(1)	289.96	317.52	250.00	127.01
154-205(1)	SING OPN LIN 6 152-153(1)	733.94	803.71	660.00	121.77
154-203(1)	BUS 153	271.63	290.30	250.00	116.12
154-205(1)	BUS 153	656.65	701.78	660.00	106.33
154-203(1)	BUS 205	368.83	472.12	250.00	188.85
3001-3003(1)	UNIT 101(1)	368.10	360.16	300.00	120.05
3001-3003(2)	UNIT 101(1)	368.10	360.16	300.00	120.05
3003-3005(1)	UNIT 101(1)	-357.37	360.41	350.00	102.97
3003-3005(2)	UNIT 101(1)	-357.37	360.41	350.00	102.97
101-151(1)	UNIT 102(1)	1462.97	1462.97	1350.00	108.37

2.30 Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.29. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 205.

Table 2.29: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	353.40	345.34	300.00	115.11
3001-3003(2)	SING OPN LIN 1 101-151(1)	353.40	345.34	300.00	115.11
101-151(1)	SING OPN LIN 2 102-151(1)	1463.04	1463.04	1350.00	108.37
154-203(1)	SING OPN LIN 6 152-153(1)	287.92	314.66	250.00	125.86
154-205(1)	SING OPN LIN 6 152-153(1)	728.16	795.78	660.00	120.57
154-203(1)	BUS 153	271.21	289.69	250.00	115.88
154-205(1)	BUS 153	655.37	700.02	660.00	106.06
154-203(1)	BUS 205	369.76	476.11	250.00	190.44
3001-3003(1)	UNIT 101(1)	353.45	345.39	300.00	115.13
3001-3003(2)	UNIT 101(1)	353.45	345.39	300.00	115.13
101-151(1)	UNIT 102(1)	1463.04	1463.04	1350.00	108.37

2.31 Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.30. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 205.

Table 2.30: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	340.92	332.84	300.00	110.95
3001-3003(2)	SING OPN LIN 1 101-151(1)	340.92	332.84	300.00	110.95
101-151(1)	SING OPN LIN 2 102-151(1)	1463.15	1463.15	1350.00	108.38
154-203(1)	SING OPN LIN 6 152-153(1)	286.36	312.88	250.00	125.15
154-205(1)	SING OPN LIN 6 152-153(1)	724.40	791.50	660.00	119.92
154-203(1)	BUS 153	270.97	289.50	250.00	115.80
154-205(1)	BUS 153	654.95	699.75	660.00	106.02
153-154(1)	BUS 205	-270.98	352.27	350.00	100.65
154-203(1)	BUS 205	370.83	482.08	250.00	192.83
3001-3003(1)	UNIT 101(1)	340.98	332.89	300.00	110.96
3001-3003(2)	UNIT 101(1)	340.98	332.89	300.00	110.96
101-151(1)	UNIT 102(1)	1463.15	1463.15	1350.00	108.38

2.32 Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.31. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.31: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	407.99	402.92	300.00	134.31
3001-3003(2)	SING OPN LIN 1 101-151(1)	407.99	402.92	300.00	134.31
3003-3005(1)	SING OPN LIN 1 101-151(1)	-390.57	403.53	350.00	115.29
3003-3005(2)	SING OPN LIN 1 101-151(1)	-390.57	403.53	350.00	115.29
101-151(1)	SING OPN LIN 2 102-151(1)	1539.94	1539.94	1350.00	114.07
154-203(1)	SING OPN LIN 6 152-153(1)	302.85	351.37	250.00	140.55
154-205(1)	SING OPN LIN 6 152-153(1)	785.11	910.88	660.00	138.01
205-206(1)	SING OPN LIN 12 201-211(1)	-1289.96	1289.96	1250.00	103.20
205-206(1)	SING OPN LIN 13 201-212(1)	-1289.96	1289.96	1250.00	103.20
154-203(1)	BUS 153	284.91	319.56	250.00	127.82
154-205(1)	BUS 153	703.91	789.51	660.00	119.62
153-154(1)	BUS 203	-328.33	368.43	350.00	105.27
154-203(1)	BUS 3005	222.70	262.04	250.00	104.82
154-205(1)	BUS 3005	618.65	727.95	660.00	110.30
154-3008(1)	BUS 3005	-362.47	450.61	440.00	102.41
3001-3003(1)	UNIT 101(1)	408.12	403.06	300.00	134.35
3001-3003(2)	UNIT 101(1)	408.12	403.06	300.00	134.35
3003-3005(1)	UNIT 101(1)	-390.69	403.66	350.00	115.33
3003-3005(2)	UNIT 101(1)	-390.69	403.66	350.00	115.33
101-151(1)	UNIT 102(1)	1539.94	1539.94	1350.00	114.07
205-206(1)	UNIT 211(1)	-1288.89	1288.89	1250.00	103.11
205-206(1)	UNIT 212(1)	-1288.89	1288.89	1250.00	103.11

2.33 Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.32. A loading of greater than 130% of the branch rating were reported for the branches 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.32: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	392.81	387.16	300.00	129.05
3001-3003(2)	SING OPN LIN 1 101-151(1)	392.81	387.16	300.00	129.05
3003-3005(1)	SING OPN LIN 1 101-151(1)	-376.77	387.77	350.00	110.79
3003-3005(2)	SING OPN LIN 1 101-151(1)	-376.77	387.77	350.00	110.79
101-151(1)	SING OPN LIN 2 102-151(1)	1539.90	1539.90	1350.00	114.07
154-203(1)	SING OPN LIN 6 152-153(1)	300.83	348.02	250.00	139.21
154-205(1)	SING OPN LIN 6 152-153(1)	779.19	901.40	660.00	136.58
205-206(1)	SING OPN LIN 12 201-211(1)	-1290.05	1290.05	1250.00	103.20
205-206(1)	SING OPN LIN 13 201-212(1)	-1290.05	1290.05	1250.00	103.20
154-203(1)	BUS 153	284.49	318.85	250.00	127.54
154-205(1)	BUS 153	702.60	787.44	660.00	119.31
153-154(1)	BUS 203	-330.16	370.38	350.00	105.82
154-203(1)	BUS 3005	221.74	260.03	250.00	104.01
154-205(1)	BUS 3005	614.87	721.05	660.00	109.25
154-3008(1)	BUS 3005	-362.34	448.66	440.00	101.97
3001-3003(1)	UNIT 101(1)	392.93	387.29	300.00	129.10
3001-3003(2)	UNIT 101(1)	392.93	387.29	300.00	129.10
3003-3005(1)	UNIT 101(1)	-376.88	387.90	350.00	110.83
3003-3005(2)	UNIT 101(1)	-376.88	387.90	350.00	110.83
101-151(1)	UNIT 102(1)	1539.89	1539.89	1350.00	114.07
205-206(1)	UNIT 211(1)	-1288.77	1288.77	1250.00	103.10
205-206(1)	UNIT 212(1)	-1288.77	1288.77	1250.00	103.10

2.34 Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.33. A loading of greater than 130% of the branch rating were reported for the branches 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.33: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	380.05	374.07	300.00	124.69
3001-3003(2)	SING OPN LIN 1 101-151(1)	380.05	374.07	300.00	124.69
3003-3005(1)	SING OPN LIN 1 101-151(1)	-364.96	374.70	350.00	107.06
3003-3005(2)	SING OPN LIN 1 101-151(1)	-364.96	374.70	350.00	107.06
101-151(1)	SING OPN LIN 2 102-151(1)	1540.07	1540.07	1350.00	114.08
154-203(1)	SING OPN LIN 6 152-153(1)	299.37	346.21	250.00	138.49
154-205(1)	SING OPN LIN 6 152-153(1)	775.43	896.77	660.00	135.87
205-206(1)	SING OPN LIN 12 201-211(1)	-1289.72	1289.72	1250.00	103.18
205-206(1)	SING OPN LIN 13 201-212(1)	-1289.72	1289.72	1250.00	103.18
154-203(1)	BUS 153	284.28	318.71	250.00	127.49
154-205(1)	BUS 153	702.18	787.22	660.00	119.28
153-154(1)	BUS 203	-331.63	372.34	350.00	106.38
154-203(1)	BUS 3005	221.14	259.20	250.00	103.68
154-205(1)	BUS 3005	613.22	718.75	660.00	108.90
154-3008(1)	BUS 3005	-362.32	448.36	440.00	101.90
3001-3003(1)	UNIT 101(1)	380.17	374.19	300.00	124.73
3001-3003(2)	UNIT 101(1)	380.17	374.19	300.00	124.73
3003-3005(1)	UNIT 101(1)	-365.07	374.83	350.00	107.09
3003-3005(2)	UNIT 101(1)	-365.07	374.83	350.00	107.09
101-151(1)	UNIT 102(1)	1540.06	1540.06	1350.00	114.08
205-206(1)	UNIT 211(1)	-1288.67	1288.67	1250.00	103.09
205-206(1)	UNIT 212(1)	-1288.67	1288.67	1250.00	103.09

2.35 Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.34. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1), 154-205(1) for bus fault contingency BUS 153; 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.34: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	442.62	441.45	300.00	147.15
3001-3003(2)	SING OPN LIN 1 101-151(1)	442.62	441.45	300.00	147.15
3003-3005(1)	SING OPN LIN 1 101-151(1)	-417.01	442.38	350.00	126.39
3003-3005(2)	SING OPN LIN 1 101-151(1)	-417.01	442.38	350.00	126.39
101-151(1)	SING OPN LIN 2 102-151(1)	1604.46	1604.46	1350.00	118.85
205-206(1)	SING OPN LIN 12 201-211(1)	-1353.11	1353.11	1250.00	108.25
205-206(1)	SING OPN LIN 13 201-212(1)	-1353.11	1353.11	1250.00	108.25
154-203(1)	BUS 153	296.85	350.70	250.00	140.28
154-205(1)	BUS 153	740.89	875.29	660.00	132.62
153-154(1)	BUS 203	-339.58	399.88	350.00	114.25
3001-3003(1)	UNIT 101(1)	442.64	441.47	300.00	147.16
3001-3003(2)	UNIT 101(1)	442.64	441.47	300.00	147.16
3003-3005(1)	UNIT 101(1)	-417.03	442.40	350.00	126.40
3003-3005(2)	UNIT 101(1)	-417.03	442.40	350.00	126.40
101-151(1)	UNIT 102(1)	1604.45	1604.45	1350.00	118.85
205-206(1)	UNIT 211(1)	-1352.84	1352.84	1250.00	108.23
205-206(1)	UNIT 212(1)	-1352.84	1352.84	1250.00	108.23

2.36 Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.35. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 154-203(1), 154-205(1) for bus fault contingency BUS 153; 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.35: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	428.25	426.17	300.00	142.06
3001-3003(2)	SING OPN LIN 1 101-151(1)	428.25	426.17	300.00	142.06
3003-3005(1)	SING OPN LIN 1 101-151(1)	-404.42	427.12	350.00	122.03
3003-3005(2)	SING OPN LIN 1 101-151(1)	-404.42	427.12	350.00	122.03
101-151(1)	SING OPN LIN 2 102-151(1)	1604.35	1604.35	1350.00	118.84
154-203(1)	SING OPN LIN 6 152-153(1)	312.06	386.56	250.00	154.62
154-205(1)	SING OPN LIN 6 152-153(1)	818.91	1014.43	660.00	153.70
203-205(1)	SING OPN LIN 6 152-153(1)	-221.24	268.30	250.00	107.32
203-205(2)	SING OPN LIN 6 152-153(1)	-221.24	268.30	250.00	107.32
205-206(1)	SING OPN LIN 12 201-211(1)	-1352.91	1352.91	1250.00	108.23
205-206(1)	SING OPN LIN 13 201-212(1)	-1352.91	1352.91	1250.00	108.23
154-203(1)	BUS 153	296.44	349.84	250.00	139.94
154-205(1)	BUS 153	739.55	872.79	660.00	132.24
153-154(1)	BUS 203	-341.29	401.74	350.00	114.78
3001-3003(1)	UNIT 101(1)	426.98	424.82	300.00	141.61
3001-3003(2)	UNIT 101(1)	426.98	424.82	300.00	141.61
3003-3005(1)	UNIT 101(1)	-403.31	425.76	350.00	121.65
3003-3005(2)	UNIT 101(1)	-403.31	425.76	350.00	121.65
101-151(1)	UNIT 102(1)	1604.34	1604.34	1350.00	118.84
205-206(1)	UNIT 211(1)	-1352.66	1352.66	1250.00	108.21
205-206(1)	UNIT 212(1)	-1352.66	1352.66	1250.00	108.21

2.37 Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.36. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 154-203(1), 154-205(1) for bus fault contingency BUS 153; 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.36: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	415.16	412.45	300.00	137.48
3001-3003(2)	SING OPN LIN 1 101-151(1)	415.16	412.45	300.00	137.48
3003-3005(1)	SING OPN LIN 1 101-151(1)	-392.67	413.42	350.00	118.12
3003-3005(2)	SING OPN LIN 1 101-151(1)	-392.67	413.42	350.00	118.12
101-151(1)	SING OPN LIN 2 102-151(1)	1604.52	1604.52	1350.00	118.85
154-203(1)	SING OPN LIN 6 152-153(1)	310.74	384.72	250.00	153.89
154-205(1)	SING OPN LIN 6 152-153(1)	815.35	1009.46	660.00	152.95
203-205(1)	SING OPN LIN 6 152-153(1)	-220.32	267.04	250.00	106.81
203-205(2)	SING OPN LIN 6 152-153(1)	-220.32	267.04	250.00	106.81
205-206(1)	SING OPN LIN 12 201-211(1)	-1352.80	1352.80	1250.00	108.22
205-206(1)	SING OPN LIN 13 201-212(1)	-1352.80	1352.80	1250.00	108.22
154-203(1)	BUS 153	296.26	349.78	250.00	139.91
154-205(1)	BUS 153	739.14	872.67	660.00	132.22
153-154(1)	BUS 203	-344.14	406.36	350.00	116.10
3001-3003(1)	UNIT 101(1)	415.42	412.71	300.00	137.57
3001-3003(2)	UNIT 101(1)	415.42	412.71	300.00	137.57
3003-3005(1)	UNIT 101(1)	-392.89	413.68	350.00	118.19
3003-3005(2)	UNIT 101(1)	-392.89	413.68	350.00	118.19
101-151(1)	UNIT 102(1)	1604.52	1604.52	1350.00	118.85
205-206(1)	UNIT 211(1)	-1352.54	1352.54	1250.00	108.20
205-206(1)	UNIT 212(1)	-1352.54	1352.54	1250.00	108.20

2.38 Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.37. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.37: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	369.78	362.05	300.00	120.68
3001-3003(2)	SING OPN LIN 1 101-151(1)	369.78	362.05	300.00	120.68
3003-3005(1)	SING OPN LIN 1 101-151(1)	-358.56	362.35	350.00	103.53
3003-3005(2)	SING OPN LIN 1 101-151(1)	-358.56	362.35	350.00	103.53
101-151(1)	SING OPN LIN 2 102-151(1)	1466.59	1466.59	1350.00	108.64
154-203(1)	SING OPN LIN 6 152-153(1)	311.45	346.80	250.00	138.72
154-205(1)	SING OPN LIN 6 152-153(1)	690.56	768.94	660.00	116.51
203-205(1)	SING OPN LIN 6 152-153(1)	-230.50	252.69	250.00	101.08
203-205(2)	SING OPN LIN 6 152-153(1)	-230.50	252.69	250.00	101.08
154-203(1)	BUS 153	295.03	320.04	250.00	128.02
154-205(1)	BUS 153	617.25	669.58	660.00	101.45
153-154(1)	BUS 203	-341.57	375.44	350.00	107.27
3001-3003(1)	UNIT 101(1)	369.84	362.12	300.00	120.71
3001-3003(2)	UNIT 101(1)	369.84	362.12	300.00	120.71
3003-3005(1)	UNIT 101(1)	-358.62	362.41	350.00	103.55
3003-3005(2)	UNIT 101(1)	-358.62	362.41	350.00	103.55
101-151(1)	UNIT 102(1)	1466.59	1466.59	1350.00	108.64

2.39 Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.38. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.38: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	354.94	346.99	300.00	115.66
3001-3003(2)	SING OPN LIN 1 101-151(1)	354.94	346.99	300.00	115.66
101-151(1)	SING OPN LIN 2 102-151(1)	1466.58	1466.58	1350.00	108.64
154-203(1)	SING OPN LIN 6 152-153(1)	309.45	343.80	250.00	137.52
154-205(1)	SING OPN LIN 6 152-153(1)	684.75	760.77	660.00	115.27
203-205(1)	SING OPN LIN 6 152-153(1)	-229.11	250.64	250.00	100.26
203-205(2)	SING OPN LIN 6 152-153(1)	-229.11	250.64	250.00	100.26
154-203(1)	BUS 153	294.62	319.40	250.00	127.76
154-205(1)	BUS 153	615.97	667.78	660.00	101.18
153-154(1)	BUS 203	-343.49	377.47	350.00	107.85
3001-3003(1)	UNIT 101(1)	355.00	347.05	300.00	115.68
3001-3003(2)	UNIT 101(1)	355.00	347.05	300.00	115.68
101-151(1)	UNIT 102(1)	1466.58	1466.58	1350.00	108.64

2.40 Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.39. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.39: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	342.47	334.48	300.00	111.49
3001-3003(2)	SING OPN LIN 1 101-151(1)	342.47	334.48	300.00	111.49
101-151(1)	SING OPN LIN 2 102-151(1)	1466.74	1466.74	1350.00	108.65
154-203(1)	SING OPN LIN 6 152-153(1)	307.88	341.96	250.00	136.78
154-205(1)	SING OPN LIN 6 152-153(1)	681.17	756.56	660.00	114.63
154-203(1)	BUS 153	294.37	319.22	250.00	127.69
154-205(1)	BUS 153	615.62	667.59	660.00	101.15
153-154(1)	BUS 203	-345.04	379.49	350.00	108.42
3001-3003(1)	UNIT 101(1)	342.52	334.54	300.00	111.51
3001-3003(2)	UNIT 101(1)	342.52	334.54	300.00	111.51
101-151(1)	UNIT 102(1)	1466.73	1466.73	1350.00	108.65

2.41 Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.40. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.40: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	411.11	406.69	300.00	135.56
3001-3003(2)	SING OPN LIN 1 101-151(1)	411.11	406.69	300.00	135.56
3003-3005(1)	SING OPN LIN 1 101-151(1)	-392.25	407.40	350.00	116.40
3003-3005(2)	SING OPN LIN 1 101-151(1)	-392.25	407.40	350.00	116.40
101-151(1)	SING OPN LIN 2 102-151(1)	1545.36	1545.36	1350.00	114.47
154-203(1)	SING OPN LIN 6 152-153(1)	322.89	382.73	250.00	153.09
154-205(1)	SING OPN LIN 6 152-153(1)	742.86	880.52	660.00	133.41
203-205(1)	SING OPN LIN 6 152-153(1)	-236.93	275.59	250.00	110.24
203-205(2)	SING OPN LIN 6 152-153(1)	-236.93	275.59	250.00	110.24
154-203(1)	BUS 153	307.00	350.46	250.00	140.19
154-205(1)	BUS 153	665.35	759.54	660.00	115.08
203-205(1)	BUS 153	-228.41	256.32	250.00	102.53
203-205(2)	BUS 153	-228.41	256.32	250.00	102.53
153-154(1)	BUS 203	-354.68	411.45	350.00	117.56
3001-3003(1)	UNIT 101(1)	411.27	406.87	300.00	135.62
3001-3003(2)	UNIT 101(1)	411.27	406.87	300.00	135.62
3003-3005(1)	UNIT 101(1)	-392.39	407.57	350.00	116.45
3003-3005(2)	UNIT 101(1)	-392.39	407.57	350.00	116.45
101-151(1)	UNIT 102(1)	1545.36	1545.36	1350.00	114.47

2.42 Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.41. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.41: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	395.81	390.77	300.00	130.26
3001-3003(2)	SING OPN LIN 1 101-151(1)	395.81	390.77	300.00	130.26
3003-3005(1)	SING OPN LIN 1 101-151(1)	-378.43	391.48	350.00	111.85
3003-3005(2)	SING OPN LIN 1 101-151(1)	-378.43	391.48	350.00	111.85
101-151(1)	SING OPN LIN 2 102-151(1)	1545.33	1545.33	1350.00	114.47
154-203(1)	SING OPN LIN 6 152-153(1)	320.93	379.11	250.00	151.65
154-205(1)	SING OPN LIN 6 152-153(1)	736.88	870.46	660.00	131.89
203-205(1)	SING OPN LIN 6 152-153(1)	-235.57	273.14	250.00	109.26
203-205(2)	SING OPN LIN 6 152-153(1)	-235.57	273.14	250.00	109.26
154-203(1)	BUS 153	306.60	349.72	250.00	139.89
154-205(1)	BUS 153	664.05	757.43	660.00	114.76
203-205(1)	BUS 153	-228.14	255.82	250.00	102.33
203-205(2)	BUS 153	-228.14	255.82	250.00	102.33
153-154(1)	BUS 203	-356.48	413.42	350.00	118.12
3001-3003(1)	UNIT 101(1)	395.97	390.93	300.00	130.31
3001-3003(2)	UNIT 101(1)	395.97	390.93	300.00	130.31
3003-3005(1)	UNIT 101(1)	-378.57	391.65	350.00	111.90
3003-3005(2)	UNIT 101(1)	-378.57	391.65	350.00	111.90
101-151(1)	UNIT 102(1)	1545.33	1545.33	1350.00	114.47

2.43 Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.42. A loading of greater than 130% of the branch rating were reported for the branches: 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 154-203(1) for bus fault contingency BUS 153.

Table 2.42: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	382.99	377.58	300.00	125.86
3001-3003(2)	SING OPN LIN 1 101-151(1)	382.99	377.58	300.00	125.86
3003-3005(1)	SING OPN LIN 1 101-151(1)	-366.62	378.32	350.00	108.09
3003-3005(2)	SING OPN LIN 1 101-151(1)	-366.62	378.32	350.00	108.09
101-151(1)	SING OPN LIN 2 102-151(1)	1545.54	1545.54	1350.00	114.48
154-203(1)	SING OPN LIN 6 152-153(1)	319.46	377.23	250.00	150.89
154-205(1)	SING OPN LIN 6 152-153(1)	733.26	865.85	660.00	131.19
203-205(1)	SING OPN LIN 6 152-153(1)	-234.53	271.83	250.00	108.73
203-205(2)	SING OPN LIN 6 152-153(1)	-234.53	271.83	250.00	108.73
154-203(1)	BUS 153	306.39	349.60	250.00	139.84
154-205(1)	BUS 153	663.71	757.31	660.00	114.74
203-205(1)	BUS 153	-227.99	255.73	250.00	102.29
203-205(2)	BUS 153	-227.99	255.73	250.00	102.29
153-154(1)	BUS 203	-357.90	415.52	350.00	118.72
3001-3003(1)	UNIT 101(1)	383.15	377.74	300.00	125.91
3001-3003(2)	UNIT 101(1)	383.15	377.74	300.00	125.91
3003-3005(1)	UNIT 101(1)	-366.76	378.48	350.00	108.14
3003-3005(2)	UNIT 101(1)	-366.76	378.48	350.00	108.14
101-151(1)	UNIT 102(1)	1545.54	1545.54	1350.00	114.48

2.44 Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.43. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.43: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	446.03	445.66	300.00	148.55
3001-3003(2)	SING OPN LIN 1 101-151(1)	446.03	445.66	300.00	148.55
3003-3005(1)	SING OPN LIN 1 101-151(1)	-418.71	446.68	350.00	127.62
3003-3005(2)	SING OPN LIN 1 101-151(1)	-418.71	446.68	350.00	127.62
101-151(1)	SING OPN LIN 2 102-151(1)	1610.66	1610.66	1350.00	119.31
205-206(1)	SING OPN LIN 12 201-211(1)	-1267.79	1267.79	1250.00	101.42
205-206(1)	SING OPN LIN 13 201-212(1)	-1267.79	1267.79	1250.00	101.42
153-154(1)	BUS 203	-365.72	451.47	350.00	128.99
153-154(2)	BUS 203	-305.46	377.07	350.00	107.74
153-154(3)	BUS 203	-305.46	377.07	350.00	107.74
3001-3003(1)	UNIT 101(1)	446.06	445.70	300.00	148.57
3001-3003(2)	UNIT 101(1)	446.06	445.70	300.00	148.57
3003-3005(1)	UNIT 101(1)	-418.74	446.71	350.00	127.63
3003-3005(2)	UNIT 101(1)	-418.74	446.71	350.00	127.63
101-151(1)	UNIT 102(1)	1610.65	1610.65	1350.00	119.31
205-206(1)	UNIT 211(1)	-1267.13	1267.13	1250.00	101.37
205-206(1)	UNIT 212(1)	-1267.13	1267.13	1250.00	101.37

2.45 Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.44. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.44: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	431.99	430.69	300.00	143.56
3001-3003(2)	SING OPN LIN 1 101-151(1)	431.99	430.69	300.00	143.56
3003-3005(1)	SING OPN LIN 1 101-151(1)	-406.47	431.72	350.00	123.35
3003-3005(2)	SING OPN LIN 1 101-151(1)	-406.47	431.72	350.00	123.35
101-151(1)	SING OPN LIN 2 102-151(1)	1610.57	1610.57	1350.00	119.30
205-206(1)	SING OPN LIN 12 201-211(1)	-1267.60	1267.60	1250.00	101.41
205-206(1)	SING OPN LIN 13 201-212(1)	-1267.60	1267.60	1250.00	101.41
153-154(1)	BUS 203	-367.38	453.30	350.00	129.51
153-154(2)	BUS 203	-306.83	378.59	350.00	108.17
153-154(3)	BUS 203	-306.83	378.59	350.00	108.17
3001-3003(1)	UNIT 101(1)	430.28	428.86	300.00	142.95
3001-3003(2)	UNIT 101(1)	430.28	428.86	300.00	142.95
3003-3005(1)	UNIT 101(1)	-405.00	429.89	350.00	122.83
3003-3005(2)	UNIT 101(1)	-405.00	429.89	350.00	122.83
101-151(1)	UNIT 102(1)	1610.56	1610.56	1350.00	119.30
205-206(1)	UNIT 211(1)	-1266.96	1266.96	1250.00	101.36
205-206(1)	UNIT 212(1)	-1266.96	1266.96	1250.00	101.36

2.46 Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.45. A loading of greater than 130% of the branch rating were reported for the branches: 153-154(1) for bus fault contingency BUS 203; 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.45: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	418.82	416.82	300.00	138.94
3001-3003(2)	SING OPN LIN 1 101-151(1)	418.82	416.82	300.00	138.94
3003-3005(1)	SING OPN LIN 1 101-151(1)	-394.72	417.89	350.00	119.40
3003-3005(2)	SING OPN LIN 1 101-151(1)	-394.72	417.89	350.00	119.40
101-151(1)	SING OPN LIN 2 102-151(1)	1610.79	1610.79	1350.00	119.32
205-206(1)	SING OPN LIN 12 201-211(1)	-1267.54	1267.54	1250.00	101.40
205-206(1)	SING OPN LIN 13 201-212(1)	-1267.54	1267.54	1250.00	101.40
153-154(1)	BUS 203	-368.65	455.66	350.00	130.19
153-154(2)	BUS 203	-307.88	380.55	350.00	108.73
153-154(3)	BUS 203	-307.88	380.55	350.00	108.73
3001-3003(1)	UNIT 101(1)	419.12	417.14	300.00	139.05
3001-3003(2)	UNIT 101(1)	419.12	417.14	300.00	139.05
3003-3005(1)	UNIT 101(1)	-394.98	418.21	350.00	119.49
3003-3005(2)	UNIT 101(1)	-394.98	418.21	350.00	119.49
101-151(1)	UNIT 102(1)	1610.78	1610.78	1350.00	119.32
205-206(1)	UNIT 211(1)	-1266.90	1266.90	1250.00	101.35
205-206(1)	UNIT 212(1)	-1266.90	1266.90	1250.00	101.35

2.47 Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.46. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.46: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	372.64	365.66	300.00	121.89
3001-3003(2)	SING OPN LIN 1 101-151(1)	372.64	365.66	300.00	121.89
3003-3005(1)	SING OPN LIN 1 101-151(1)	-359.81	366.12	350.00	104.61
3003-3005(2)	SING OPN LIN 1 101-151(1)	-359.81	366.12	350.00	104.61
101-151(1)	SING OPN LIN 2 102-151(1)	1472.52	1472.52	1350.00	109.08
154-203(1)	SING OPN LIN 6 152-153(1)	333.33	380.73	250.00	152.29
154-205(1)	SING OPN LIN 6 152-153(1)	645.63	737.43	660.00	111.73
203-205(1)	SING OPN LIN 6 152-153(1)	-254.43	285.94	250.00	114.38
203-205(2)	SING OPN LIN 6 152-153(1)	-254.43	285.94	250.00	114.38
154-203(1)	BUS 153	318.69	353.43	250.00	141.37
203-205(1)	BUS 153	-246.64	269.47	250.00	107.79
203-205(2)	BUS 153	-246.64	269.47	250.00	107.79
153-154(1)	BUS 203	-368.99	421.59	350.00	120.45
153-154(2)	BUS 203	-308.05	351.96	350.00	100.56
153-154(3)	BUS 203	-308.05	351.96	350.00	100.56
154-203(1)	BUS 3005	239.92	274.68	250.00	109.87
3001-3003(1)	UNIT 101(1)	372.73	365.75	300.00	121.92
3001-3003(2)	UNIT 101(1)	372.73	365.75	300.00	121.92
3003-3005(1)	UNIT 101(1)	-359.89	366.21	350.00	104.63
3003-3005(2)	UNIT 101(1)	-359.89	366.21	350.00	104.63
101-151(1)	UNIT 102(1)	1472.52	1472.52	1350.00	109.08

2.48 Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.47. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.47: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	357.61	350.33	300.00	116.78
3001-3003(2)	SING OPN LIN 1 101-151(1)	357.61	350.33	300.00	116.78
3003-3005(1)	SING OPN LIN 1 101-151(1)	-345.86	350.80	350.00	100.23
3003-3005(2)	SING OPN LIN 1 101-151(1)	-345.86	350.80	350.00	100.23
101-151(1)	SING OPN LIN 2 102-151(1)	1472.56	1472.56	1350.00	109.08
154-203(1)	SING OPN LIN 6 152-153(1)	330.93	376.75	250.00	150.70
154-205(1)	SING OPN LIN 6 152-153(1)	640.71	729.42	660.00	110.52
203-205(1)	SING OPN LIN 6 152-153(1)	-252.59	282.99	250.00	113.20
203-205(2)	SING OPN LIN 6 152-153(1)	-252.59	282.99	250.00	113.20
154-203(1)	BUS 153	318.29	352.75	250.00	141.10
203-205(1)	BUS 153	-246.37	269.00	250.00	107.60
203-205(2)	BUS 153	-246.37	269.00	250.00	107.60
153-154(1)	BUS 203	-370.86	423.64	350.00	121.04
153-154(2)	BUS 203	-309.60	353.67	350.00	101.05
153-154(3)	BUS 203	-309.60	353.67	350.00	101.05
154-203(1)	BUS 3005	238.98	272.90	250.00	109.16
3001-3003(1)	UNIT 101(1)	357.62	350.33	300.00	116.78
3001-3003(2)	UNIT 101(1)	357.62	350.33	300.00	116.78
3003-3005(1)	UNIT 101(1)	-345.87	350.80	350.00	100.23
3003-3005(2)	UNIT 101(1)	-345.87	350.80	350.00	100.23
101-151(1)	UNIT 102(1)	1472.56	1472.56	1350.00	109.08

2.49 Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.48. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.48: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	345.06	337.64	300.00	112.55
3001-3003(2)	SING OPN LIN 1 101-151(1)	345.06	337.64	300.00	112.55
101-151(1)	SING OPN LIN 2 102-151(1)	1472.77	1472.77	1350.00	109.09
154-203(1)	SING OPN LIN 6 152-153(1)	329.37	374.83	250.00	149.93
154-205(1)	SING OPN LIN 6 152-153(1)	637.11	725.04	660.00	109.85
203-205(1)	SING OPN LIN 6 152-153(1)	-251.50	281.66	250.00	112.66
203-205(2)	SING OPN LIN 6 152-153(1)	-251.50	281.66	250.00	112.66
154-203(1)	BUS 153	318.04	352.58	250.00	141.03
203-205(1)	BUS 153	-246.19	268.88	250.00	107.55
203-205(2)	BUS 153	-246.19	268.88	250.00	107.55
153-154(1)	BUS 203	-372.34	425.81	350.00	121.66
153-154(2)	BUS 203	-310.83	355.47	350.00	101.56
153-154(3)	BUS 203	-310.83	355.47	350.00	101.56
154-203(1)	BUS 3005	238.31	272.04	250.00	108.82
3001-3003(1)	UNIT 101(1)	345.06	337.65	300.00	112.55
3001-3003(2)	UNIT 101(1)	345.06	337.65	300.00	112.55
101-151(1)	UNIT 102(1)	1472.77	1472.77	1350.00	109.09

2.50 Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.49. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.49: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	188.70	202.85	200.00	101.43
3001-3003(1)	SING OPN LIN 1 101-151(1)	415.21	411.80	300.00	137.27
3001-3003(2)	SING OPN LIN 1 101-151(1)	415.21	411.80	300.00	137.27
3003-3005(1)	SING OPN LIN 1 101-151(1)	-394.16	412.65	350.00	117.90
3003-3005(2)	SING OPN LIN 1 101-151(1)	-394.16	412.65	350.00	117.90
101-151(1)	SING OPN LIN 2 102-151(1)	1552.42	1552.42	1350.00	114.99
154-205(1)	BUS 153	626.38	735.67	660.00	111.46
203-205(1)	BUS 153	-252.85	291.65	250.00	116.66
203-205(2)	BUS 153	-252.85	291.65	250.00	116.66
3001-3003(1)	UNIT 101(1)	415.42	412.03	300.00	137.34
3001-3003(2)	UNIT 101(1)	415.42	412.03	300.00	137.34
3003-3005(1)	UNIT 101(1)	-394.35	412.88	350.00	117.97
3003-3005(2)	UNIT 101(1)	-394.35	412.88	350.00	117.97
101-151(1)	UNIT 102(1)	1552.41	1552.41	1350.00	114.99

2.51 Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.50. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.50: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	189.01	203.12	200.00	101.56
3001-3003(1)	SING OPN LIN 1 101-151(1)	399.76	395.64	300.00	131.88
3001-3003(2)	SING OPN LIN 1 101-151(1)	399.76	395.64	300.00	131.88
3003-3005(1)	SING OPN LIN 1 101-151(1)	-380.34	396.51	350.00	113.29
3003-3005(2)	SING OPN LIN 1 101-151(1)	-380.34	396.51	350.00	113.29
101-151(1)	SING OPN LIN 2 102-151(1)	1552.40	1552.40	1350.00	114.99
154-205(1)	BUS 153	625.06	733.39	660.00	111.12
203-205(1)	BUS 153	-252.59	291.08	250.00	116.43
203-205(2)	BUS 153	-252.59	291.08	250.00	116.43
3001-3003(1)	UNIT 101(1)	399.97	395.87	300.00	131.96
3001-3003(2)	UNIT 101(1)	399.97	395.87	300.00	131.96
3003-3005(1)	UNIT 101(1)	-380.52	396.73	350.00	113.35
3003-3005(2)	UNIT 101(1)	-380.52	396.73	350.00	113.35
101-151(1)	UNIT 102(1)	1552.40	1552.40	1350.00	114.99

2.52 Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.51. No branches reported a loading of greater than 130% of the branch rating.

Table 2.51: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	189.34	203.59	200.00	101.80
3001-3003(1)	SING OPN LIN 1 101-151(1)	385.94	381.35	300.00	127.12
3001-3003(2)	SING OPN LIN 1 101-151(1)	385.94	381.35	300.00	127.12
3003-3005(1)	SING OPN LIN 1 101-151(1)	-367.73	382.26	350.00	109.22
3003-3005(2)	SING OPN LIN 1 101-151(1)	-367.73	382.26	350.00	109.22
101-151(1)	SING OPN LIN 2 102-151(1)	1552.65	1552.65	1350.00	115.01
154-205(1)	BUS 153	624.80	733.41	660.00	111.12
203-205(1)	BUS 153	-252.44	291.02	250.00	116.41
203-205(2)	BUS 153	-252.44	291.02	250.00	116.41
3001-3003(1)	UNIT 101(1)	385.95	381.37	300.00	127.12
3001-3003(2)	UNIT 101(1)	385.95	381.37	300.00	127.12
3003-3005(1)	UNIT 101(1)	-367.75	382.27	350.00	109.22
3003-3005(2)	UNIT 101(1)	-367.75	382.27	350.00	109.22
101-151(1)	UNIT 102(1)	1552.65	1552.65	1350.00	115.01

2.53 Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.52. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.52: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	194.16	214.94	200.00	107.47
3001-3003(1)	SING OPN LIN 1 101-151(1)	450.75	451.67	300.00	150.56
3001-3003(2)	SING OPN LIN 1 101-151(1)	450.75	451.67	300.00	150.56
3003-3005(1)	SING OPN LIN 1 101-151(1)	-420.68	452.82	350.00	129.38
3003-3005(2)	SING OPN LIN 1 101-151(1)	-420.68	452.82	350.00	129.38
101-151(1)	SING OPN LIN 2 102-151(1)	1618.94	1618.94	1350.00	119.92
3001-3003(1)	UNIT 101(1)	450.81	451.73	300.00	150.58
3001-3003(2)	UNIT 101(1)	450.81	451.73	300.00	150.58
3003-3005(1)	UNIT 101(1)	-420.72	452.88	350.00	129.39
3003-3005(2)	UNIT 101(1)	-420.72	452.88	350.00	129.39
101-151(1)	UNIT 102(1)	1618.92	1618.92	1350.00	119.92

2.54 Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.53. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.53: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	194.45	215.19	200.00	107.59
3001-3003(1)	SING OPN LIN 1 101-151(1)	437.17	437.12	300.00	145.71
3001-3003(2)	SING OPN LIN 1 101-151(1)	437.17	437.12	300.00	145.71
3003-3005(1)	SING OPN LIN 1 101-151(1)	-408.95	438.29	350.00	125.23
3003-3005(2)	SING OPN LIN 1 101-151(1)	-408.95	438.29	350.00	125.23
101-151(1)	SING OPN LIN 2 102-151(1)	1618.85	1618.85	1350.00	119.91
3001-3003(1)	UNIT 101(1)	437.56	437.53	300.00	145.84
3001-3003(2)	UNIT 101(1)	437.56	437.53	300.00	145.84
3003-3005(1)	UNIT 101(1)	-409.28	438.71	350.00	125.35
3003-3005(2)	UNIT 101(1)	-409.28	438.71	350.00	125.35
101-151(1)	UNIT 102(1)	1618.84	1618.84	1350.00	119.91

2.55 Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.54. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.54: Contingencies causing branch flow greater than 100% for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	194.78	215.69	200.00	107.85
3001-3003(1)	SING OPN LIN 1 101-151(1)	423.87	423.04	300.00	141.01
3001-3003(2)	SING OPN LIN 1 101-151(1)	423.87	423.04	300.00	141.01
3003-3005(1)	SING OPN LIN 1 101-151(1)	-397.21	424.25	350.00	121.21
3003-3005(2)	SING OPN LIN 1 101-151(1)	-397.21	424.25	350.00	121.21
101-151(1)	SING OPN LIN 2 102-151(1)	1619.13	1619.13	1350.00	119.94
3001-3003(1)	UNIT 101(1)	424.25	423.45	300.00	141.15
3001-3003(2)	UNIT 101(1)	424.25	423.45	300.00	141.15
3003-3005(1)	UNIT 101(1)	-397.53	424.66	350.00	121.33
3003-3005(2)	UNIT 101(1)	-397.53	424.66	350.00	121.33
101-151(1)	UNIT 102(1)	1619.11	1619.11	1350.00	119.93

2.56 Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.55. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 205.

Table 2.55: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	349.46	341.36	300.00	113.79
3001-3003(2)	SING OPN LIN 1 101-151(1)	349.46	341.36	300.00	113.79
101-151(1)	SING OPN LIN 2 102-151(1)	1409.78	1409.78	1350.00	104.43
154-203(1)	SING OPN LIN 6 152-153(1)	281.94	307.27	250.00	122.91
154-205(1)	SING OPN LIN 6 152-153(1)	708.08	771.70	660.00	116.92
154-203(1)	BUS 153	262.99	279.92	250.00	111.97
154-205(1)	BUS 153	630.93	671.54	660.00	101.75
154-203(1)	BUS 205	357.44	449.67	250.00	179.87
3001-3003(1)	UNIT 101(1)	349.51	341.41	300.00	113.80
3001-3003(2)	UNIT 101(1)	349.51	341.41	300.00	113.80
101-151(1)	UNIT 102(1)	1409.78	1409.78	1350.00	104.43

2.57 Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.56. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 205.

Table 2.56: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	334.92	326.77	300.00	108.92
3001-3003(2)	SING OPN LIN 1 101-151(1)	334.92	326.77	300.00	108.92
101-151(1)	SING OPN LIN 2 102-151(1)	1409.85	1409.85	1350.00	104.43
154-203(1)	SING OPN LIN 6 152-153(1)	279.89	304.45	250.00	121.78
154-205(1)	SING OPN LIN 6 152-153(1)	702.36	763.99	660.00	115.76
154-203(1)	BUS 153	262.57	279.31	250.00	111.73
154-205(1)	BUS 153	629.64	669.80	660.00	101.49
154-203(1)	BUS 205	358.47	453.39	250.00	181.36
3001-3003(1)	UNIT 101(1)	334.96	326.82	300.00	108.94
3001-3003(2)	UNIT 101(1)	334.96	326.82	300.00	108.94
101-151(1)	UNIT 102(1)	1409.85	1409.85	1350.00	104.43

2.58 Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.57. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 205.

Table 2.57: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	322.52	314.41	300.00	104.80
3001-3003(2)	SING OPN LIN 1 101-151(1)	322.52	314.41	300.00	104.80
101-151(1)	SING OPN LIN 2 102-151(1)	1409.95	1409.95	1350.00	104.44
154-203(1)	SING OPN LIN 6 152-153(1)	278.34	302.70	250.00	121.08
154-205(1)	SING OPN LIN 6 152-153(1)	698.68	759.83	660.00	115.13
154-203(1)	BUS 153	262.33	279.13	250.00	111.65
154-205(1)	BUS 153	629.25	669.56	660.00	101.45
154-203(1)	BUS 205	359.62	458.62	250.00	183.45
3001-3003(1)	UNIT 101(1)	322.56	314.45	300.00	104.82
3001-3003(2)	UNIT 101(1)	322.56	314.45	300.00	104.82
101-151(1)	UNIT 102(1)	1409.95	1409.95	1350.00	104.44

2.59 Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.58. A loading of greater than 130% of the branch rating were reported for the branches 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.58: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	388.40	382.49	300.00	127.50
3001-3003(2)	SING OPN LIN 1 101-151(1)	388.40	382.49	300.00	127.50
3003-3005(1)	SING OPN LIN 1 101-151(1)	-372.98	383.08	350.00	109.45
3003-3005(2)	SING OPN LIN 1 101-151(1)	-372.98	383.08	350.00	109.45
101-151(1)	SING OPN LIN 2 102-151(1)	1485.71	1485.71	1350.00	110.05
154-203(1)	SING OPN LIN 6 152-153(1)	295.25	340.23	250.00	136.09
154-205(1)	SING OPN LIN 6 152-153(1)	759.27	874.95	660.00	132.57
205-206(1)	SING OPN LIN 12 201-211(1)	-1288.82	1288.82	1250.00	103.11
205-206(1)	SING OPN LIN 13 201-212(1)	-1288.82	1288.82	1250.00	103.11
154-203(1)	BUS 153	276.63	308.59	250.00	123.44
154-205(1)	BUS 153	678.30	756.68	660.00	114.65
153-154(1)	BUS 203	-320.14	356.97	350.00	101.99
154-203(1)	BUS 3005	216.92	253.08	250.00	101.23
154-205(1)	BUS 3005	598.83	698.68	660.00	105.86
154-3008(1)	BUS 3005	-362.15	445.75	440.00	101.31
3001-3003(1)	UNIT 101(1)	388.50	382.59	300.00	127.53
3001-3003(2)	UNIT 101(1)	388.50	382.59	300.00	127.53
3003-3005(1)	UNIT 101(1)	-373.07	383.19	350.00	109.48
3003-3005(2)	UNIT 101(1)	-373.07	383.19	350.00	109.48
101-151(1)	UNIT 102(1)	1485.71	1485.71	1350.00	110.05
205-206(1)	UNIT 211(1)	-1287.86	1287.86	1250.00	103.03
205-206(1)	UNIT 212(1)	-1287.86	1287.86	1250.00	103.03

2.60 Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.59. A loading of greater than 130% of the branch rating were reported for the branches 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.59: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	373.39	367.05	300.00	122.35
3001-3003(2)	SING OPN LIN 1 101-151(1)	373.39	367.05	300.00	122.35
3003-3005(1)	SING OPN LIN 1 101-151(1)	-359.21	367.65	350.00	105.04
3003-3005(2)	SING OPN LIN 1 101-151(1)	-359.21	367.65	350.00	105.04
101-151(1)	SING OPN LIN 2 102-151(1)	1485.68	1485.68	1350.00	110.05
154-203(1)	SING OPN LIN 6 152-153(1)	293.22	336.96	250.00	134.79
154-205(1)	SING OPN LIN 6 152-153(1)	753.41	865.81	660.00	131.18
205-206(1)	SING OPN LIN 12 201-211(1)	-1288.91	1288.91	1250.00	103.11
205-206(1)	SING OPN LIN 13 201-212(1)	-1288.91	1288.91	1250.00	103.11
154-203(1)	BUS 153	276.21	307.90	250.00	123.16
154-205(1)	BUS 153	676.99	754.66	660.00	114.34
153-154(1)	BUS 203	-321.98	358.91	350.00	102.55
154-203(1)	BUS 3005	215.96	251.16	250.00	100.47
154-205(1)	BUS 3005	595.13	692.16	660.00	104.87
154-3008(1)	BUS 3005	-362.03	443.95	440.00	100.90
3001-3003(1)	UNIT 101(1)	373.49	367.15	300.00	122.38
3001-3003(2)	UNIT 101(1)	373.49	367.15	300.00	122.38
3003-3005(1)	UNIT 101(1)	-359.30	367.75	350.00	105.07
3003-3005(2)	UNIT 101(1)	-359.30	367.75	350.00	105.07
101-151(1)	UNIT 102(1)	1485.68	1485.68	1350.00	110.05
205-206(1)	UNIT 211(1)	-1288.59	1288.59	1250.00	103.09
205-206(1)	UNIT 212(1)	-1288.59	1288.59	1250.00	103.09

2.61 Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.60. A loading of greater than 130% of the branch rating were reported for the branches 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.60: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	360.76	354.18	300.00	118.06
3001-3003(2)	SING OPN LIN 1 101-151(1)	360.76	354.18	300.00	118.06
3003-3005(1)	SING OPN LIN 1 101-151(1)	-347.42	354.82	350.00	101.38
3003-3005(2)	SING OPN LIN 1 101-151(1)	-347.42	354.82	350.00	101.38
101-151(1)	SING OPN LIN 2 102-151(1)	1485.84	1485.84	1350.00	110.06
154-203(1)	SING OPN LIN 6 152-153(1)	291.30	334.56	250.00	133.82
154-205(1)	SING OPN LIN 6 152-153(1)	750.98	862.50	660.00	130.68
205-206(1)	SING OPN LIN 12 201-211(1)	-1288.53	1288.53	1250.00	103.08
205-206(1)	SING OPN LIN 13 201-212(1)	-1288.53	1288.53	1250.00	103.08
154-203(1)	BUS 153	276.00	307.76	250.00	123.10
154-205(1)	BUS 153	676.59	754.45	660.00	114.31
153-154(1)	BUS 203	-323.84	361.36	350.00	103.25
154-203(1)	BUS 3005	215.36	250.35	250.00	100.14
154-205(1)	BUS 3005	593.54	689.98	660.00	104.54
154-3008(1)	BUS 3005	-362.01	443.67	440.00	100.83
3001-3003(1)	UNIT 101(1)	360.85	354.28	300.00	118.09
3001-3003(2)	UNIT 101(1)	360.85	354.28	300.00	118.09
3003-3005(1)	UNIT 101(1)	-347.51	354.92	350.00	101.40
3003-3005(2)	UNIT 101(1)	-347.51	354.92	350.00	101.40
101-151(1)	UNIT 102(1)	1485.84	1485.84	1350.00	110.06
205-206(1)	UNIT 211(1)	-1288.55	1288.55	1250.00	103.08
205-206(1)	UNIT 212(1)	-1288.55	1288.55	1250.00	103.08

2.62 Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.61. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.61: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	423.42	420.90	300.00	140.30
3001-3003(2)	SING OPN LIN 1 101-151(1)	423.42	420.90	300.00	140.30
3003-3005(1)	SING OPN LIN 1 101-151(1)	-400.48	421.83	350.00	120.52
3003-3005(2)	SING OPN LIN 1 101-151(1)	-400.48	421.83	350.00	120.52
101-151(1)	SING OPN LIN 2 102-151(1)	1549.59	1549.59	1350.00	114.78
154-203(1)	SING OPN LIN 6 152-153(1)	306.92	377.77	250.00	151.11
154-205(1)	SING OPN LIN 6 152-153(1)	799.41	983.94	660.00	149.08
203-205(1)	SING OPN LIN 6 152-153(1)	-218.04	262.75	250.00	105.10
203-205(2)	SING OPN LIN 6 152-153(1)	-218.04	262.75	250.00	105.10
205-206(1)	SING OPN LIN 12 201-211(1)	-1351.75	1351.75	1250.00	108.14
205-206(1)	SING OPN LIN 13 201-212(1)	-1351.75	1351.75	1250.00	108.14
154-203(1)	BUS 153	288.88	338.72	250.00	135.49
154-205(1)	BUS 153	715.49	838.95	660.00	127.11
153-154(1)	BUS 203	-332.09	387.85	350.00	110.81
3001-3003(1)	UNIT 101(1)	422.32	419.74	300.00	139.91
3001-3003(2)	UNIT 101(1)	422.32	419.74	300.00	139.91
3003-3005(1)	UNIT 101(1)	-399.52	420.67	350.00	120.19
3003-3005(2)	UNIT 101(1)	-399.52	420.67	350.00	120.19
101-151(1)	UNIT 102(1)	1549.58	1549.58	1350.00	114.78
205-206(1)	UNIT 211(1)	-1351.57	1351.57	1250.00	108.13
205-206(1)	UNIT 212(1)	-1351.57	1351.57	1250.00	108.13

2.63 Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.62. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.62: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	407.93	404.62	300.00	134.87
3001-3003(2)	SING OPN LIN 1 101-151(1)	407.93	404.62	300.00	134.87
3003-3005(1)	SING OPN LIN 1 101-151(1)	-386.74	405.56	350.00	115.87
3003-3005(2)	SING OPN LIN 1 101-151(1)	-386.74	405.56	350.00	115.87
101-151(1)	SING OPN LIN 2 102-151(1)	1549.49	1549.49	1350.00	114.78
154-203(1)	SING OPN LIN 6 152-153(1)	304.91	373.63	250.00	149.45
154-205(1)	SING OPN LIN 6 152-153(1)	793.44	972.25	660.00	147.31
203-205(1)	SING OPN LIN 6 152-153(1)	-216.61	259.94	250.00	103.98
203-205(2)	SING OPN LIN 6 152-153(1)	-216.61	259.94	250.00	103.98
205-206(1)	SING OPN LIN 12 201-211(1)	-1351.54	1351.54	1250.00	108.12
205-206(1)	SING OPN LIN 13 201-212(1)	-1351.54	1351.54	1250.00	108.12
154-203(1)	BUS 153	288.46	337.89	250.00	135.16
154-205(1)	BUS 153	714.16	836.54	660.00	126.75
153-154(1)	BUS 203	-333.81	389.71	350.00	111.34
3001-3003(1)	UNIT 101(1)	408.14	404.84	300.00	134.95
3001-3003(2)	UNIT 101(1)	408.14	404.84	300.00	134.95
3003-3005(1)	UNIT 101(1)	-386.92	405.78	350.00	115.94
3003-3005(2)	UNIT 101(1)	-386.92	405.78	350.00	115.94
101-151(1)	UNIT 102(1)	1549.48	1549.48	1350.00	114.78
205-206(1)	UNIT 211(1)	-1351.37	1351.37	1250.00	108.11
205-206(1)	UNIT 212(1)	-1351.37	1351.37	1250.00	108.11

2.64 Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.63. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 154-203(1), 154-205(1) for single line open contingency SING OPN LIN 6 152-153(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.63: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	395.02	391.21	300.00	130.40
3001-3003(2)	SING OPN LIN 1 101-151(1)	395.02	391.21	300.00	130.40
3003-3005(1)	SING OPN LIN 1 101-151(1)	-375.02	392.19	350.00	112.05
3003-3005(2)	SING OPN LIN 1 101-151(1)	-375.02	392.19	350.00	112.05
101-151(1)	SING OPN LIN 2 102-151(1)	1549.66	1549.66	1350.00	114.79
154-203(1)	SING OPN LIN 6 152-153(1)	303.56	371.85	250.00	148.74
154-205(1)	SING OPN LIN 6 152-153(1)	789.79	967.46	660.00	146.59
203-205(1)	SING OPN LIN 6 152-153(1)	-215.67	258.72	250.00	103.49
203-205(2)	SING OPN LIN 6 152-153(1)	-215.67	258.72	250.00	103.49
205-206(1)	SING OPN LIN 12 201-211(1)	-1351.44	1351.44	1250.00	108.12
205-206(1)	SING OPN LIN 13 201-212(1)	-1351.44	1351.44	1250.00	108.12
154-203(1)	BUS 153	288.29	337.83	250.00	135.13
154-205(1)	BUS 153	713.77	836.41	660.00	126.73
153-154(1)	BUS 203	-336.57	394.02	350.00	112.58
3001-3003(1)	UNIT 101(1)	395.23	391.43	300.00	130.48
3001-3003(2)	UNIT 101(1)	395.23	391.43	300.00	130.48
3003-3005(1)	UNIT 101(1)	-375.21	392.40	350.00	112.11
3003-3005(2)	UNIT 101(1)	-375.21	392.40	350.00	112.11
101-151(1)	UNIT 102(1)	1549.65	1549.65	1350.00	114.79
205-206(1)	UNIT 211(1)	-1351.27	1351.27	1250.00	108.10
205-206(1)	UNIT 212(1)	-1351.27	1351.27	1250.00	108.10

2.65 Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.64. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.64: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	350.98	342.98	300.00	114.33
3001-3003(2)	SING OPN LIN 1 101-151(1)	350.98	342.98	300.00	114.33
101-151(1)	SING OPN LIN 2 102-151(1)	1412.80	1412.80	1350.00	104.65
154-203(1)	SING OPN LIN 6 152-153(1)	303.49	336.05	250.00	134.42
154-205(1)	SING OPN LIN 6 152-153(1)	665.33	736.70	660.00	111.62
154-203(1)	BUS 153	286.36	309.13	250.00	123.65
153-154(1)	BUS 203	-333.56	364.48	350.00	104.14
3001-3003(1)	UNIT 101(1)	351.03	343.03	300.00	114.34
3001-3003(2)	UNIT 101(1)	351.03	343.03	300.00	114.34
101-151(1)	UNIT 102(1)	1412.80	1412.80	1350.00	104.65

2.66 Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.65. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.65: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	332.74	324.67	300.00	108.22
3001-3003(2)	SING OPN LIN 1 101-151(1)	332.74	324.67	300.00	108.22
101-151(1)	SING OPN LIN 2 102-151(1)	1412.91	1412.91	1350.00	104.66
154-203(1)	SING OPN LIN 6 152-153(1)	301.48	333.10	250.00	133.24
154-205(1)	SING OPN LIN 6 152-153(1)	659.59	728.77	660.00	110.42
154-203(1)	BUS 153	285.94	308.50	250.00	123.40
153-154(1)	BUS 203	-335.49	366.52	350.00	104.72
3001-3003(1)	UNIT 101(1)	336.48	328.42	300.00	109.47
3001-3003(2)	UNIT 101(1)	336.48	328.42	300.00	109.47
101-151(1)	UNIT 102(1)	1412.91	1412.91	1350.00	104.66

2.67 Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.66. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.66: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	320.37	312.35	300.00	104.12
3001-3003(2)	SING OPN LIN 1 101-151(1)	320.37	312.35	300.00	104.12
101-151(1)	SING OPN LIN 2 102-151(1)	1413.02	1413.02	1350.00	104.67
154-203(1)	SING OPN LIN 6 152-153(1)	299.91	331.28	250.00	132.51
154-205(1)	SING OPN LIN 6 152-153(1)	656.09	724.71	660.00	109.80
154-203(1)	BUS 153	285.70	308.31	250.00	123.33
153-154(1)	BUS 203	-337.05	368.50	350.00	105.29
3001-3003(1)	UNIT 101(1)	324.08	316.05	300.00	105.35
3001-3003(2)	UNIT 101(1)	324.08	316.05	300.00	105.35
101-151(1)	UNIT 102(1)	1413.02	1413.02	1350.00	104.67

2.68 Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.67. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.67: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	391.28	385.94	300.00	128.65
3001-3003(2)	SING OPN LIN 1 101-151(1)	391.28	385.94	300.00	128.65
3003-3005(1)	SING OPN LIN 1 101-151(1)	-374.58	386.63	350.00	110.47
3003-3005(2)	SING OPN LIN 1 101-151(1)	-374.58	386.63	350.00	110.47
101-151(1)	SING OPN LIN 2 102-151(1)	1490.80	1490.80	1350.00	110.43
154-203(1)	SING OPN LIN 6 152-153(1)	315.39	370.75	250.00	148.30
154-205(1)	SING OPN LIN 6 152-153(1)	717.60	843.57	660.00	127.81
203-205(1)	SING OPN LIN 6 152-153(1)	-232.11	267.82	250.00	107.13
203-205(2)	SING OPN LIN 6 152-153(1)	-232.11	267.82	250.00	107.13
154-203(1)	BUS 153	298.68	338.73	250.00	135.49
154-205(1)	BUS 153	640.69	726.61	660.00	110.09
153-154(1)	BUS 203	-346.96	399.20	350.00	114.06
3001-3003(1)	UNIT 101(1)	391.41	386.08	300.00	128.69
3001-3003(2)	UNIT 101(1)	391.41	386.08	300.00	128.69
3003-3005(1)	UNIT 101(1)	-374.70	386.77	350.00	110.51
3003-3005(2)	UNIT 101(1)	-374.70	386.77	350.00	110.51
101-151(1)	UNIT 102(1)	1490.79	1490.79	1350.00	110.43

2.69 Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.68. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.68: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	376.18	370.37	300.00	123.46
3001-3003(2)	SING OPN LIN 1 101-151(1)	376.18	370.37	300.00	123.46
3003-3005(1)	SING OPN LIN 1 101-151(1)	-360.80	371.07	350.00	106.02
3003-3005(2)	SING OPN LIN 1 101-151(1)	-360.80	371.07	350.00	106.02
101-151(1)	SING OPN LIN 2 102-151(1)	1490.78	1490.78	1350.00	110.43
154-203(1)	SING OPN LIN 6 152-153(1)	313.41	367.25	250.00	146.90
154-205(1)	SING OPN LIN 6 152-153(1)	711.70	833.97	660.00	126.36
203-205(1)	SING OPN LIN 6 152-153(1)	-230.73	265.44	250.00	106.18
203-205(2)	SING OPN LIN 6 152-153(1)	-230.73	265.44	250.00	106.18
154-203(1)	BUS 153	298.27	338.01	250.00	135.20
154-205(1)	BUS 153	639.38	724.55	660.00	109.78
153-154(1)	BUS 203	-348.77	401.16	350.00	114.62
154-203(1)	BUS 3005	231.38	275.92	250.00	110.37
154-205(1)	BUS 3005	559.01	666.60	660.00	101.00
154-3008(1)	BUS 3005	-363.00	458.45	440.00	104.19
3001-3003(1)	UNIT 101(1)	376.31	370.50	300.00	123.50
3001-3003(2)	UNIT 101(1)	376.31	370.50	300.00	123.50
3003-3005(1)	UNIT 101(1)	-360.92	371.21	350.00	106.06
3003-3005(2)	UNIT 101(1)	-360.92	371.21	350.00	106.06
101-151(1)	UNIT 102(1)	1490.78	1490.78	1350.00	110.43

2.70 Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.69. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.69: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	363.50	357.42	300.00	119.14
3001-3003(2)	SING OPN LIN 1 101-151(1)	363.50	357.42	300.00	119.14
3003-3005(1)	SING OPN LIN 1 101-151(1)	-349.02	358.16	350.00	102.33
3003-3005(2)	SING OPN LIN 1 101-151(1)	-349.02	358.16	350.00	102.33
101-151(1)	SING OPN LIN 2 102-151(1)	1490.98	1490.98	1350.00	110.44
154-203(1)	SING OPN LIN 6 152-153(1)	311.93	365.40	250.00	146.16
154-205(1)	SING OPN LIN 6 152-153(1)	708.16	829.53	660.00	125.69
203-205(1)	SING OPN LIN 6 152-153(1)	-229.68	264.14	250.00	105.66
203-205(2)	SING OPN LIN 6 152-153(1)	-229.68	264.14	250.00	105.66
154-203(1)	BUS 153	298.06	337.88	250.00	135.15
154-205(1)	BUS 153	639.07	724.45	660.00	109.77
153-154(1)	BUS 203	-350.19	403.22	350.00	115.21
154-203(1)	BUS 3005	230.79	275.06	250.00	110.02
154-205(1)	BUS 3005	557.65	664.61	660.00	100.70
154-3008(1)	BUS 3005	-362.98	458.12	440.00	104.12
3001-3003(1)	UNIT 101(1)	363.62	357.55	300.00	119.18
3001-3003(2)	UNIT 101(1)	363.62	357.55	300.00	119.18
3003-3005(1)	UNIT 101(1)	-349.14	358.29	350.00	102.37
3003-3005(2)	UNIT 101(1)	-349.14	358.29	350.00	102.37
101-151(1)	UNIT 102(1)	1490.97	1490.97	1350.00	110.44

2.71 Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.70. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.70: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	426.97	425.18	300.00	141.73
3001-3003(2)	SING OPN LIN 1 101-151(1)	426.97	425.18	300.00	141.73
3003-3005(1)	SING OPN LIN 1 101-151(1)	-402.45	426.19	350.00	121.77
3003-3005(2)	SING OPN LIN 1 101-151(1)	-402.45	426.19	350.00	121.77
101-151(1)	SING OPN LIN 2 102-151(1)	1555.40	1555.40	1350.00	115.22
205-206(1)	SING OPN LIN 12 201-211(1)	-1266.29	1266.29	1250.00	101.30
205-206(1)	SING OPN LIN 13 201-212(1)	-1266.29	1266.29	1250.00	101.30
154-203(1)	BUS 153	309.70	370.58	250.00	148.23
154-205(1)	BUS 153	678.60	812.00	660.00	123.03
203-205(1)	BUS 153	-230.07	269.88	250.00	107.95
203-205(2)	BUS 153	-230.07	269.88	250.00	107.95
153-154(1)	BUS 203	-358.30	436.75	350.00	124.79
153-154(2)	BUS 203	-299.30	364.84	350.00	104.24
153-154(3)	BUS 203	-299.30	364.84	350.00	104.24
3001-3003(1)	UNIT 101(1)	427.23	425.46	300.00	141.82
3001-3003(2)	UNIT 101(1)	427.23	425.46	300.00	141.82
3003-3005(1)	UNIT 101(1)	-402.67	426.47	350.00	121.85
3003-3005(2)	UNIT 101(1)	-402.67	426.47	350.00	121.85
101-151(1)	UNIT 102(1)	1555.39	1555.39	1350.00	115.21
205-206(1)	UNIT 211(1)	-1265.78	1265.78	1250.00	101.26
205-206(1)	UNIT 212(1)	-1265.78	1265.78	1250.00	101.26

2.72 Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.71. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.71: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	411.35	408.70	300.00	136.23
3001-3003(2)	SING OPN LIN 1 101-151(1)	411.35	408.70	300.00	136.23
3003-3005(1)	SING OPN LIN 1 101-151(1)	-388.68	409.73	350.00	117.07
3003-3005(2)	SING OPN LIN 1 101-151(1)	-388.68	409.73	350.00	117.07
101-151(1)	SING OPN LIN 2 102-151(1)	1555.33	1555.33	1350.00	115.21
205-206(1)	SING OPN LIN 12 201-211(1)	-1266.14	1266.14	1250.00	101.29
205-206(1)	SING OPN LIN 13 201-212(1)	-1266.14	1266.14	1250.00	101.29
154-203(1)	BUS 153	309.30	369.68	250.00	147.87
154-205(1)	BUS 153	677.26	809.48	660.00	122.65
203-205(1)	BUS 153	-229.79	269.27	250.00	107.71
203-205(2)	BUS 153	-229.79	269.27	250.00	107.71
153-154(1)	BUS 203	-359.97	438.59	350.00	125.31
153-154(2)	BUS 203	-300.69	366.36	350.00	104.67
153-154(3)	BUS 203	-300.69	366.36	350.00	104.67
3001-3003(1)	UNIT 101(1)	411.61	408.97	300.00	136.32
3001-3003(2)	UNIT 101(1)	411.61	408.97	300.00	136.32
3003-3005(1)	UNIT 101(1)	-388.90	410.00	350.00	117.14
3003-3005(2)	UNIT 101(1)	-388.90	410.00	350.00	117.14
101-151(1)	UNIT 102(1)	1555.32	1555.32	1350.00	115.21
205-206(1)	UNIT 211(1)	-1265.65	1265.65	1250.00	101.25
205-206(1)	UNIT 212(1)	-1265.65	1265.65	1250.00	101.25

2.73 Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.72. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.72: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	398.38	395.19	300.00	131.73
3001-3003(2)	SING OPN LIN 1 101-151(1)	398.38	395.19	300.00	131.73
3003-3005(1)	SING OPN LIN 1 101-151(1)	-376.97	396.26	350.00	113.22
3003-3005(2)	SING OPN LIN 1 101-151(1)	-376.97	396.26	350.00	113.22
101-151(1)	SING OPN LIN 2 102-151(1)	1555.54	1555.54	1350.00	115.23
205-206(1)	SING OPN LIN 12 201-211(1)	-1266.06	1266.06	1250.00	101.29
205-206(1)	SING OPN LIN 13 201-212(1)	-1266.06	1266.06	1250.00	101.29
154-203(1)	BUS 153	309.12	369.64	250.00	147.85
154-205(1)	BUS 153	676.96	809.48	660.00	122.65
203-205(1)	BUS 153	-229.67	269.24	250.00	107.69
203-205(2)	BUS 153	-229.67	269.24	250.00	107.69
153-154(1)	BUS 203	-361.26	440.84	350.00	125.96
153-154(2)	BUS 203	-301.76	368.23	350.00	105.21
153-154(3)	BUS 203	-301.76	368.23	350.00	105.21
3001-3003(1)	UNIT 101(1)	398.63	395.45	300.00	131.82
3001-3003(2)	UNIT 101(1)	398.63	395.45	300.00	131.82
3003-3005(1)	UNIT 101(1)	-377.19	396.52	350.00	113.29
3003-3005(2)	UNIT 101(1)	-377.19	396.52	350.00	113.29
101-151(1)	UNIT 102(1)	1555.54	1555.54	1350.00	115.22
205-206(1)	UNIT 211(1)	-1265.58	1265.58	1250.00	101.25
205-206(1)	UNIT 212(1)	-1265.58	1265.58	1250.00	101.25

2.74 Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.73. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.73: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	353.47	346.01	300.00	115.34
3001-3003(2)	SING OPN LIN 1 101-151(1)	353.47	346.01	300.00	115.34
101-151(1)	SING OPN LIN 2 102-151(1)	1418.32	1418.32	1350.00	105.06
154-203(1)	SING OPN LIN 6 152-153(1)	325.07	368.50	250.00	147.40
154-205(1)	SING OPN LIN 6 152-153(1)	622.01	705.10	660.00	106.83
203-205(1)	SING OPN LIN 6 152-153(1)	-248.94	277.71	250.00	111.08
203-205(2)	SING OPN LIN 6 152-153(1)	-248.94	277.71	250.00	111.08
201-204(1)	SING OPN LIN 10 201-202(1)	-1171.34	1356.56	1300.00	104.35
154-203(1)	BUS 153	310.05	341.79	250.00	136.72
203-205(1)	BUS 153	-240.93	261.69	250.00	104.68
203-205(2)	BUS 153	-240.93	261.69	250.00	104.68
153-154(1)	BUS 203	-361.06	409.15	350.00	116.90
154-203(1)	BUS 3005	233.85	265.60	250.00	106.24
3001-3003(1)	UNIT 101(1)	353.54	346.08	300.00	115.36
3001-3003(2)	UNIT 101(1)	353.54	346.08	300.00	115.36
101-151(1)	UNIT 102(1)	1418.32	1418.32	1350.00	105.06

2.75 Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.74. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.74: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	338.71	331.06	300.00	110.35
3001-3003(2)	SING OPN LIN 1 101-151(1)	338.71	331.06	300.00	110.35
101-151(1)	SING OPN LIN 2 102-151(1)	1418.36	1418.36	1350.00	105.06
154-203(1)	SING OPN LIN 6 152-153(1)	323.10	365.34	250.00	146.14
154-205(1)	SING OPN LIN 6 152-153(1)	616.25	696.82	660.00	105.58
203-205(1)	SING OPN LIN 6 152-153(1)	-247.58	275.54	250.00	110.22
203-205(2)	SING OPN LIN 6 152-153(1)	-247.58	275.54	250.00	110.22
201-204(1)	SING OPN LIN 10 201-202(1)	-1171.51	1356.32	1300.00	104.33
154-203(1)	BUS 153	309.65	341.12	250.00	136.45
203-205(1)	BUS 153	-240.65	261.23	250.00	104.49
203-205(2)	BUS 153	-240.65	261.23	250.00	104.49
153-154(1)	BUS 203	-362.94	411.20	350.00	117.49
154-203(1)	BUS 3005	232.91	263.88	250.00	105.55
3001-3003(1)	UNIT 101(1)	338.77	331.13	300.00	110.38
3001-3003(2)	UNIT 101(1)	338.77	331.13	300.00	110.38
101-151(1)	UNIT 102(1)	1418.36	1418.36	1350.00	105.06

2.76 Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, LLS, the branches that are loaded more than 100% of their rating is given in Table 2.75. A loading of greater than 130% of the branch rating was reported for the branch 154-203(1) for bus fault contingency BUS 153 and for single line open contingency SING OPN LIN 6 152-153(1).

Table 2.75: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, LLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	326.20	318.52	300.00	106.17
3001-3003(2)	SING OPN LIN 1 101-151(1)	326.20	318.52	300.00	106.17
101-151(1)	SING OPN LIN 2 102-151(1)	1418.57	1418.57	1350.00	105.08
154-203(1)	SING OPN LIN 6 152-153(1)	321.55	363.46	250.00	145.38
154-205(1)	SING OPN LIN 6 152-153(1)	612.79	692.67	660.00	104.95
203-205(1)	SING OPN LIN 6 152-153(1)	-246.48	274.22	250.00	109.69
203-205(2)	SING OPN LIN 6 152-153(1)	-246.48	274.22	250.00	109.69
201-204(1)	SING OPN LIN 10 201-202(1)	-1172.70	1360.96	1300.00	104.69
154-203(1)	BUS 153	309.40	340.95	250.00	136.38
203-205(1)	BUS 153	-240.47	261.10	250.00	104.44
203-205(2)	BUS 153	-240.47	261.10	250.00	104.44
153-154(1)	BUS 203	-364.44	413.33	350.00	118.09
154-203(1)	BUS 3005	232.24	263.04	250.00	105.22
3001-3003(1)	UNIT 101(1)	326.21	318.52	300.00	106.17
3001-3003(2)	UNIT 101(1)	326.21	318.52	300.00	106.17
101-151(1)	UNIT 102(1)	1418.57	1418.57	1350.00	105.08

2.77 Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.76. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 154-203(1) for bus fault contingency BUS 153; 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.76: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	395.05	390.59	300.00	130.20
3001-3003(2)	SING OPN LIN 1 101-151(1)	395.05	390.59	300.00	130.20
3003-3005(1)	SING OPN LIN 1 101-151(1)	-376.41	391.44	350.00	111.84
3003-3005(2)	SING OPN LIN 1 101-151(1)	-376.41	391.44	350.00	111.84
101-151(1)	SING OPN LIN 2 102-151(1)	1497.42	1497.42	1350.00	110.92
154-203(1)	BUS 153	321.36	374.22	250.00	149.69
154-205(1)	BUS 153	602.82	701.97	660.00	106.36
203-205(1)	BUS 153	-247.33	282.93	250.00	113.17
203-205(2)	BUS 153	-247.33	282.93	250.00	113.17
153-154(1)	BUS 203	-374.00	453.43	350.00	129.55
153-154(2)	BUS 203	-312.30	378.62	350.00	108.18
153-154(3)	BUS 203	-312.30	378.62	350.00	108.18
3001-3003(1)	UNIT 101(1)	395.23	390.78	300.00	130.26
3001-3003(2)	UNIT 101(1)	395.23	390.78	300.00	130.26
3003-3005(1)	UNIT 101(1)	-376.57	391.62	350.00	111.89
3003-3005(2)	UNIT 101(1)	-376.57	391.62	350.00	111.89
101-151(1)	UNIT 102(1)	1497.41	1497.41	1350.00	110.92

2.78 Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.77. A loading of greater than 130% of the branch rating was reported for the branch 153-154(1) for bus fault contingency BUS 203 and for bus fault contingency BUS 153.

Table 2.77: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	379.82	374.83	300.00	124.94
3001-3003(2)	SING OPN LIN 1 101-151(1)	379.82	374.83	300.00	124.94
3003-3005(1)	SING OPN LIN 1 101-151(1)	-362.63	375.69	350.00	107.34
3003-3005(2)	SING OPN LIN 1 101-151(1)	-362.63	375.69	350.00	107.34
101-151(1)	SING OPN LIN 2 102-151(1)	1497.41	1497.41	1350.00	110.92
154-203(1)	BUS 153	320.97	373.42	250.00	149.37
154-205(1)	BUS 153	601.52	699.80	660.00	106.03
203-205(1)	BUS 153	-247.06	282.38	250.00	112.95
203-205(2)	BUS 153	-247.06	282.38	250.00	112.95
153-154(1)	BUS 203	-375.73	455.36	350.00	130.10
153-154(2)	BUS 203	-313.73	380.22	350.00	108.64
153-154(3)	BUS 203	-313.73	380.22	350.00	108.64
3001-3003(1)	UNIT 101(1)	379.99	375.00	300.00	125.00
3001-3003(2)	UNIT 101(1)	379.99	375.00	300.00	125.00
3003-3005(1)	UNIT 101(1)	-362.78	375.87	350.00	107.39
3003-3005(2)	UNIT 101(1)	-362.78	375.87	350.00	107.39
101-151(1)	UNIT 102(1)	1497.41	1497.41	1350.00	110.92

2.79 Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, RLS, the branches that are loaded more than 100% of their rating is given in Table 2.78. A loading of greater than 130% of the branch rating was reported for the branch 153-154(1) for bus fault contingency BUS 203 and for bus fault contingency BUS 153.

Table 2.78: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, RLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	367.07	361.77	300.00	120.59
3001-3003(2)	SING OPN LIN 1 101-151(1)	367.07	361.77	300.00	120.59
3003-3005(1)	SING OPN LIN 1 101-151(1)	-350.87	362.67	350.00	103.62
3003-3005(2)	SING OPN LIN 1 101-151(1)	-350.87	362.67	350.00	103.62
101-151(1)	SING OPN LIN 2 102-151(1)	1497.66	1497.66	1350.00	110.94
154-203(1)	BUS 153	320.76	373.32	250.00	149.33
154-205(1)	BUS 153	601.30	699.83	660.00	106.04
203-205(1)	BUS 153	-246.91	282.30	250.00	112.92
203-205(2)	BUS 153	-246.91	282.30	250.00	112.92
153-154(1)	BUS 203	-377.07	457.74	350.00	130.78
153-154(2)	BUS 203	-314.84	382.20	350.00	109.20
153-154(3)	BUS 203	-314.84	382.20	350.00	109.20
3001-3003(1)	UNIT 101(1)	367.24	361.94	300.00	120.65
3001-3003(2)	UNIT 101(1)	367.24	361.94	300.00	120.65
3003-3005(1)	UNIT 101(1)	-351.02	362.84	350.00	103.67
3003-3005(2)	UNIT 101(1)	-351.02	362.84	350.00	103.67
101-151(1)	UNIT 102(1)	1497.67	1497.67	1350.00	110.94

2.80 Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.79. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.79: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	188.93	208.26	200.00	104.13
3001-3003(1)	SING OPN LIN 1 101-151(1)	431.87	431.25	300.00	143.75
3001-3003(2)	SING OPN LIN 1 101-151(1)	431.87	431.25	300.00	143.75
3003-3005(1)	SING OPN LIN 1 101-151(1)	-404.81	432.40	350.00	123.54
3003-3005(2)	SING OPN LIN 1 101-151(1)	-404.81	432.40	350.00	123.54
101-151(1)	SING OPN LIN 2 102-151(1)	1563.17	1563.17	1350.00	115.79
3001-3003(1)	UNIT 101(1)	432.20	431.61	300.00	143.87
3001-3003(2)	UNIT 101(1)	432.20	431.61	300.00	143.87
3003-3005(1)	UNIT 101(1)	-405.09	432.76	350.00	123.65
3003-3005(2)	UNIT 101(1)	-405.09	432.76	350.00	123.65
101-151(1)	UNIT 102(1)	1563.15	1563.15	1350.00	115.79

2.81 Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.80. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.80: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	189.22	208.50	200.00	104.25
3001-3003(1)	SING OPN LIN 1 101-151(1)	416.06	414.48	300.00	138.16
3001-3003(2)	SING OPN LIN 1 101-151(1)	416.06	414.48	300.00	138.16
3003-3005(1)	SING OPN LIN 1 101-151(1)	-391.03	415.66	350.00	118.76
3003-3005(2)	SING OPN LIN 1 101-151(1)	-391.03	415.66	350.00	118.76
101-151(1)	SING OPN LIN 2 102-151(1)	1563.10	1563.10	1350.00	115.79
3001-3003(1)	UNIT 101(1)	416.38	414.83	300.00	138.28
3001-3003(2)	UNIT 101(1)	416.38	414.83	300.00	138.28
3003-3005(1)	UNIT 101(1)	-391.30	416.00	350.00	118.86
3003-3005(2)	UNIT 101(1)	-391.30	416.00	350.00	118.86
101-151(1)	UNIT 102(1)	1563.09	1563.09	1350.00	115.78

2.82 Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, HLS, the branches that are loaded more than 100% of their rating is given in Table 2.81. A loading of greater than 130% of the branch rating were reported for the branches: 3001-3003(1), 3001-3003(2) for unit fault contingency UNIT 101(1); 3001-3003(1), 3001-3003(2) for single line open contingency SING OPN LIN 1 101-151(1).

Table 2.81: Contingencies causing branch flow greater than 100% for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, HLS

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
154-203(1)	BASE CASE	189.55	209.00	200.00	104.50
3001-3003(1)	SING OPN LIN 1 101-151(1)	402.99	400.80	300.00	133.60
3001-3003(2)	SING OPN LIN 1 101-151(1)	402.99	400.80	300.00	133.60
3003-3005(1)	SING OPN LIN 1 101-151(1)	-379.33	402.02	350.00	114.86
3003-3005(2)	SING OPN LIN 1 101-151(1)	-379.33	402.02	350.00	114.86
101-151(1)	SING OPN LIN 2 102-151(1)	1563.38	1563.38	1350.00	115.81
3001-3003(1)	UNIT 101(1)	403.30	401.14	300.00	133.71
3001-3003(2)	UNIT 101(1)	403.30	401.14	300.00	133.71
3003-3005(1)	UNIT 101(1)	-379.60	402.35	350.00	114.96
3003-3005(2)	UNIT 101(1)	-379.60	402.35	350.00	114.96
101-151(1)	UNIT 102(1)	1563.37	1563.37	1350.00	115.81

2.83 Result Summary

AC contingency calculation carried out on the hypothetical SAVNW system for Topology 6 to observe the branch overload violations showed the contingencies causing most number of overload violations are contingencies SING OPN LIN 1 101-151(1) and UNIT 101(1) reporting 296 violations each, followed by BUS 153 reporting 168 violations and SING OPN LIN 6 152-153(1) reporting 183 violations. The complete counts of contingencies causing overload violations is given in Table 5.8.

The branch reporting the greatest number of violations are the branches 101-151(1), 3001-3003(1) and 3001-3003(2) with 162 violations each, followed by the branch 153-203(1) with 159 violations. The complete counts of each of the branch reporting overloads are given in Table 5.7.

Out of all the studied scenarios, the scenario reporting the greatest number of violations are the scenarios Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS and Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS with 24 violations each. The complete counts of scenarios reporting overload violations is given in Tables 5.5 and 5.6.

Chapter 3

Upper Emergency Bus Voltage Violation

3.1 Introduction

In this chapter the upper voltage limit violations are reported for bus faults, unit faults and single line faults for each of the studied scenario. The violations are reported in tabular format, and in the reported table, Base Voltage is PU base case voltage, Contingency Voltage is PU contingency case voltage, Deviation is difference between contingency case and base case voltage, Range Violation is range violations calculated as Contingency Voltage - maximum range limit (1.1 PU for the upper emergency range).

3.2 Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3005.

Table 3.1: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3005	1.05	1.11	0.06	0.06
212	BUS 3005	1.05	1.11	0.06	0.06

3.3 Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3005.

Table 3.2: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3005	1.05	1.11	0.06	0.06
212	BUS 3005	1.05	1.11	0.06	0.06

3.4 Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3005.

Table 3.3: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3005	1.05	1.11	0.06	0.06
212	BUS 3005	1.05	1.11	0.06	0.06

3.5 Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 4 151-152(1).

Table 3.4: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06
212	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06

3.6 Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 4 151-152(1).

Table 3.5: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06
212	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06

3.7 Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 4 151-152(1).

Table 3.6: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06
212	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06

3.8 Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3003.

Table 3.7: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3003	1.08	1.11	0.03	0.06
212	BUS 3003	1.08	1.11	0.03	0.06

3.9 Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3003.

Table 3.8: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3003	1.08	1.11	0.03	0.06
212	BUS 3003	1.08	1.11	0.03	0.06

3.10 Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3003.

Table 3.9: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3003	1.08	1.11	0.03	0.06
212	BUS 3003	1.08	1.11	0.03	0.06

3.11 Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, LLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3005.

Table 3.10: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3005	1.05	1.11	0.05	0.06
212	BUS 3005	1.05	1.11	0.05	0.06

3.12 Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, LLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3005.

Table 3.11: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3005	1.05	1.11	0.05	0.06
212	BUS 3005	1.05	1.11	0.05	0.06

3.13 Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, LLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3005.

Table 3.12: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3005	1.05	1.10	0.05	0.05
212	BUS 3005	1.05	1.10	0.05	0.05

3.14 Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3003.

Table 3.13: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3003	1.07	1.10	0.03	0.05
212	BUS 3003	1.07	1.10	0.03	0.05

3.15 Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3004.

Table 3.14: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3004	1.07	1.11	0.03	0.06
212	BUS 3004	1.07	1.11	0.03	0.06

3.16 Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3004.

Table 3.15: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3004	1.07	1.11	0.03	0.06
212	BUS 3004	1.07	1.11	0.03	0.06

3.17 Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violations for unit fault UNIT 102(1), for bus fault BUS 3008, and for single line open fault SING OPN LIN 2 102-151(1).

Table 3.16: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	UNIT 102(1)	1.09	1.11	0.02	0.06
212	UNIT 102(1)	1.09	1.11	0.02	0.06
211	BUS 3008	1.09	1.10	0.01	0.05
212	BUS 3008	1.09	1.10	0.01	0.05
211	SING OPN LIN 2 102-151(1)	1.09	1.11	0.02	0.06
212	SING OPN LIN 2 102-151(1)	1.09	1.11	0.02	0.06

3.18 Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violations for unit fault UNIT 102(1), for bus fault BUS 3008, and for single line open fault SING OPN LIN 2 102-151(1).

Table 3.17: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	UNIT 102(1)	1.09	1.11	0.02	0.06
212	UNIT 102(1)	1.09	1.11	0.02	0.06
211	BUS 3008	1.09	1.10	0.01	0.05
212	BUS 3008	1.09	1.10	0.01	0.05
211	SING OPN LIN 2 102-151(1)	1.09	1.11	0.02	0.06
212	SING OPN LIN 2 102-151(1)	1.09	1.11	0.02	0.06

3.19 Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violations for unit fault UNIT 102(1), for bus fault BUS 3008, and for single line open fault SING OPN LIN 2 102-151(1).

Table 3.18: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	UNIT 102(1)	1.09	1.11	0.02	0.06
212	UNIT 102(1)	1.09	1.11	0.02	0.06
211	BUS 3008	1.09	1.10	0.01	0.05
212	BUS 3008	1.09	1.10	0.01	0.05
211	SING OPN LIN 2 102-151(1)	1.09	1.11	0.02	0.06
212	SING OPN LIN 2 102-151(1)	1.09	1.11	0.02	0.06

3.20 Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3005.

Table 3.19: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3005	1.04	1.10	0.06	0.05
212	BUS 3005	1.04	1.10	0.06	0.05

3.21 Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3005.

Table 3.20: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3005	1.04	1.10	0.06	0.05
212	BUS 3005	1.04	1.10	0.06	0.05

3.22 Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3005 and for single line open fault SING OPN LIN 10 201-202(1).

Table 3.21: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3005	1.04	1.10	0.06	0.05
212	BUS 3005	1.04	1.10	0.06	0.05
211	SING OPN LIN 10 201-202(1)	1.04	1.10	0.06	0.05
212	SING OPN LIN 10 201-202(1)	1.04	1.10	0.06	0.05

3.23 Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 4 151-152(1).

Table 3.22: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06
212	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06

3.24 Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 4 151-152(1).

Table 3.23: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06
212	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06

3.25 Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 4 151-152(1).

Table 3.24: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06
212	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06

3.26 Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 4 151-152(1).

Table 3.25: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 4 151-152(1)	1.06	1.10	0.05	0.05
212	SING OPN LIN 4 151-152(1)	1.06	1.10	0.05	0.05

3.27 Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 4 151-152(1).

Table 3.26: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 4 151-152(1)	1.05	1.10	0.05	0.05
212	SING OPN LIN 4 151-152(1)	1.05	1.10	0.05	0.05

3.28 Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 4 151-152(1).

Table 3.27: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 4 151-152(1)	1.06	1.10	0.05	0.05
212	SING OPN LIN 4 151-152(1)	1.06	1.10	0.05	0.05

3.29 Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 203.

Table 3.28: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 203	1.07	1.11	0.04	0.06
212	BUS 203	1.07	1.11	0.04	0.06

3.30 Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 203.

Table 3.29: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 203	1.07	1.11	0.04	0.06
212	BUS 203	1.07	1.11	0.04	0.06

3.31 Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 203.

Table 3.30: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 203	1.07	1.11	0.04	0.06
212	BUS 203	1.07	1.11	0.04	0.06

3.32 Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3004.

Table 3.31: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3004	1.07	1.10	0.03	0.05
212	BUS 3004	1.07	1.10	0.03	0.05

3.33 Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3004.

Table 3.32: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3004	1.07	1.10	0.03	0.05
212	BUS 3004	1.07	1.10	0.03	0.05

3.34 Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3004.

Table 3.33: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3004	1.07	1.10	0.03	0.05
212	BUS 3004	1.07	1.10	0.03	0.05

3.35 Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violations for unit fault UNIT 102(1) and for single line open fault SING OPN LIN 2 102-151(1).

Table 3.34: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	UNIT 102(1)	1.09	1.10	0.01	0.05
212	UNIT 102(1)	1.09	1.10	0.01	0.05
211	SING OPN LIN 2 102-151(1)	1.09	1.10	0.01	0.05
212	SING OPN LIN 2 102-151(1)	1.09	1.10	0.01	0.05

3.36 Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violations for unit fault UNIT 102(1) and for single line open fault SING OPN LIN 2 102-151(1).

Table 3.35: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	UNIT 102(1)	1.09	1.10	0.01	0.05
212	UNIT 102(1)	1.09	1.10	0.01	0.05
211	SING OPN LIN 2 102-151(1)	1.09	1.10	0.01	0.05
212	SING OPN LIN 2 102-151(1)	1.09	1.10	0.01	0.05

3.37 Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violations for unit fault UNIT 102(1) and for single line open fault SING OPN LIN 2 102-151(1).

Table 3.36: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	UNIT 102(1)	1.09	1.10	0.01	0.05
212	UNIT 102(1)	1.09	1.10	0.01	0.05
211	SING OPN LIN 2 102-151(1)	1.09	1.10	0.01	0.05
212	SING OPN LIN 2 102-151(1)	1.09	1.10	0.01	0.05

3.38 Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 4 151-152(1).

Table 3.37: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06
212	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06

3.39 Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 4 151-152(1).

Table 3.38: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06
212	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06

3.40 Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 4 151-152(1).

Table 3.39: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06
212	SING OPN LIN 4 151-152(1)	1.06	1.11	0.05	0.06

3.41 Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3005.

Table 3.40: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3005	1.05	1.11	0.06	0.06
212	BUS 3005	1.05	1.11	0.06	0.06

3.42 Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 3005.

Table 3.41: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 3005	1.05	1.11	0.06	0.06
212	BUS 3005	1.05	1.11	0.06	0.06

3.43 Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 203.

Table 3.42: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 203	1.07	1.10	0.04	0.05
212	BUS 203	1.07	1.10	0.04	0.05

3.44 Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 203.

Table 3.43: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 203	1.07	1.10	0.04	0.05
212	BUS 203	1.07	1.10	0.04	0.05

3.45 Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, HLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 203.

Table 3.44: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 203	1.07	1.11	0.04	0.06
212	BUS 203	1.07	1.11	0.04	0.06

3.46 Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, LLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 10 201-202(1).

Table 3.45: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 10 201-202(1)	1.05	1.11	0.06	0.06
212	SING OPN LIN 10 201-202(1)	1.05	1.11	0.06	0.06

3.47 Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, LLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 10 201-202(1).

Table 3.46: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 10 201-202(1)	1.05	1.11	0.06	0.06
212	SING OPN LIN 10 201-202(1)	1.05	1.11	0.06	0.06

3.48 Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, LLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for single line open fault SING OPN LIN 10 201-202(1).

Table 3.47: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	SING OPN LIN 10 201-202(1)	1.05	1.11	0.06	0.06
212	SING OPN LIN 10 201-202(1)	1.05	1.11	0.06	0.06

3.49 Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 203.

Table 3.48: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 203	1.07	1.11	0.04	0.06
212	BUS 203	1.07	1.11	0.04	0.06

3.50 Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 203.

Table 3.49: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 203	1.06	1.11	0.04	0.06
212	BUS 203	1.06	1.11	0.04	0.06

3.51 Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, RLS, the buses violating the Upper voltage limits are buses 211, 212. The buses 211, 212 reported violation for bus fault BUS 203.

Table 3.50: Contingencies reporting bus voltages greater than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
211	BUS 203	1.07	1.11	0.04	0.06
212	BUS 203	1.07	1.11	0.04	0.06

3.52 Result Summary

For the AC contingency calculation carried out on the hypothetical SAVNW system for Topology 6 to study the upper emergency bus voltage violations, it was seen that only the buses 211 and 212 are reporting the upper voltage range violation across all the studied scenarios and contingencies. It was also noted that for all the grouped scenarios and contingencies, both Bus 211 and 212 reported violations together.

The scenarios for which the greatest number of overvoltage violations were reported for are the scenarios: Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS; Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS; and Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS reporting 56 violations each. The complete counts of each of the scenario reporting upper limit voltage violations are given in Table 5.9.

The contingencies causing the greatest number of overvoltage violations is the bus fault BUS 3004 causing 60 violations, followed by single open line faults SING OPN LIN 5 151-152(2) and SING OPN LIN 4 151-152(1) reporting 48 violations each. The complete counts of each of the contingency causing upper limit violations is given in Table 5.11.

Chapter 4

Lower Emergency Bus Voltage Violation

4.1 Introduction

In this chapter the lower voltage limit violations are reported by bus faults, unit faults and single line faults for each of the studied scenario. The violations are reported in tabular format and in the reported table, Base Voltage is PU base case voltage, Contingency Voltage is PU contingency case voltage, Deviation is difference between contingency case and base case voltage, Range Violation is range violations calculated as Contingency Voltage - minimum range limit (0.9 PU for lower emergency range).

4.2 Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 203, 204, 205, 3008, 153, 3006, 3007, 3009, 3010, 3012 as given in the Table 4.1.

Table 4.1: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 202	1.00	0.87	-0.13	-0.08
154	BUS 202	0.97	0.83	-0.14	-0.12
203	BUS 202	1.00	0.81	-0.19	-0.14
204	BUS 202	0.99	0.87	-0.12	-0.08
205	BUS 202	0.98	0.84	-0.14	-0.11
3008	BUS 202	0.99	0.88	-0.11	-0.07
153	BUS 205	1.01	0.89	-0.12	-0.06
3006	BUS 205	1.01	0.90	-0.11	-0.05
3007	BUS 205	1.00	0.89	-0.11	-0.06
3009	BUS 205	1.01	0.90	-0.11	-0.05
3010	BUS 205	1.01	0.90	-0.11	-0.05
3012	BUS 205	1.01	0.90	-0.11	-0.05
103	SING OPN LIN 11 201-204(1)	1.00	0.87	-0.13	-0.08
154	SING OPN LIN 11 201-204(1)	0.97	0.84	-0.13	-0.11
203	SING OPN LIN 11 201-204(1)	1.00	0.89	-0.11	-0.06
204	SING OPN LIN 11 201-204(1)	0.99	0.84	-0.15	-0.11
205	SING OPN LIN 11 201-204(1)	0.98	0.84	-0.14	-0.11
3008	SING OPN LIN 11 201-204(1)	0.99	0.88	-0.11	-0.07

4.3 Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 203, 204, 205, 3008, 153, 3006, 3007, 3012 as given in the Table 4.2.

Table 4.2: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 202	1.00	0.87	-0.13	-0.08
154	BUS 202	0.97	0.84	-0.14	-0.11
203	BUS 202	1.00	0.81	-0.19	-0.14
204	BUS 202	0.99	0.87	-0.12	-0.08
205	BUS 202	0.98	0.84	-0.14	-0.11
3008	BUS 202	0.99	0.88	-0.11	-0.07
153	BUS 205	1.01	0.89	-0.12	-0.06
3006	BUS 205	1.01	0.89	-0.12	-0.06
3007	BUS 205	1.00	0.88	-0.11	-0.07
3012	BUS 205	1.01	0.89	-0.12	-0.06
103	SING OPN LIN 11 201-204(1)	1.00	0.87	-0.13	-0.08
154	SING OPN LIN 11 201-204(1)	0.97	0.84	-0.13	-0.11
203	SING OPN LIN 11 201-204(1)	1.00	0.89	-0.11	-0.06
204	SING OPN LIN 11 201-204(1)	0.99	0.84	-0.15	-0.11
205	SING OPN LIN 11 201-204(1)	0.98	0.84	-0.14	-0.11
3008	SING OPN LIN 11 201-204(1)	0.99	0.88	-0.11	-0.07

4.4 Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 203, 204, 205, 3008, 153, 3006, 3007, 3009, 3012 as given in the Table 4.3.

Table 4.3: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 202	1.00	0.87	-0.13	-0.08
154	BUS 202	0.97	0.83	-0.14	-0.12
203	BUS 202	1.00	0.81	-0.19	-0.14
204	BUS 202	0.99	0.87	-0.13	-0.08
205	BUS 202	0.98	0.84	-0.14	-0.11
3008	BUS 202	0.99	0.88	-0.11	-0.07
153	BUS 205	1.01	0.88	-0.12	-0.07
3006	BUS 205	1.01	0.89	-0.12	-0.06
3007	BUS 205	1.00	0.88	-0.12	-0.07
3009	BUS 205	1.02	0.90	-0.12	-0.05
3012	BUS 205	1.01	0.88	-0.12	-0.07
103	SING OPN LIN 11 201-204(1)	1.00	0.87	-0.13	-0.08
154	SING OPN LIN 11 201-204(1)	0.97	0.84	-0.14	-0.11
203	SING OPN LIN 11 201-204(1)	1.00	0.88	-0.12	-0.07
204	SING OPN LIN 11 201-204(1)	0.99	0.84	-0.15	-0.11
205	SING OPN LIN 11 201-204(1)	0.98	0.84	-0.14	-0.11
3008	SING OPN LIN 11 201-204(1)	0.99	0.88	-0.11	-0.07

4.5 Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS, the buses reporting lower emergency range violations are the buses 154, 205, 3007, 3008 as given in the Table 4.4.

Table 4.4: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.83	-0.16	-0.12
3008	BUS 3005	0.99	0.86	-0.12	-0.09
154	SING OPN LIN 10 201-202(1)	0.97	0.88	-0.09	-0.07
205	SING OPN LIN 10 201-202(1)	0.98	0.89	-0.09	-0.06

4.6 Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS, the buses reporting lower emergency range violations are the buses 154, 205, 3007, 3008 as given in the Table 4.5.

Table 4.5: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.83	-0.16	-0.12
3008	BUS 3005	0.99	0.87	-0.12	-0.08
154	SING OPN LIN 10 201-202(1)	0.97	0.88	-0.09	-0.07
205	SING OPN LIN 10 201-202(1)	0.98	0.89	-0.09	-0.06

4.7 Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, RLS, the buses reporting lower emergency range violations are the buses 154, 205, 3007, 3008 as given in the Table 4.6.

Table 4.6: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.83	-0.16	-0.12
3008	BUS 3005	0.99	0.87	-0.12	-0.08
154	SING OPN LIN 10 201-202(1)	0.97	0.88	-0.09	-0.07
205	SING OPN LIN 10 201-202(1)	0.98	0.89	-0.09	-0.06

4.8 Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 205, 3006, 3008, 3009, 3010, 3012, 152, 202, 203, 204, 3007, 3018 as given in the Table 4.7.

Table 4.7: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.89	-0.06	-0.06
205	BUS 203	0.96	0.90	-0.06	-0.05
103	BUS 3005	0.98	0.88	-0.10	-0.07
204	BUS 3005	0.97	0.89	-0.08	-0.06
3007	BUS 3005	0.98	0.77	-0.21	-0.18
3008	BUS 3005	0.97	0.80	-0.16	-0.15
3018	BUS 3005	1.03	0.88	-0.15	-0.07
103	SING OPN LIN 6 152-153(1)	0.98	0.89	-0.09	-0.06
153	SING OPN LIN 6 152-153(1)	0.99	0.86	-0.12	-0.09
154	SING OPN LIN 6 152-153(1)	0.95	0.86	-0.09	-0.09
205	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
3006	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3008	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3009	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.11	-0.06
3010	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.11	-0.06
3012	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.11	-0.06
152	SING OPN LIN 10 201-202(1)	1.01	0.89	-0.12	-0.06
202	SING OPN LIN 10 201-202(1)	1.00	0.86	-0.14	-0.09
203	SING OPN LIN 10 201-202(1)	0.98	0.84	-0.14	-0.11
204	SING OPN LIN 10 201-202(1)	0.97	0.85	-0.12	-0.10
3007	SING OPN LIN 10 201-202(1)	0.98	0.87	-0.10	-0.08

4.9 Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 205, 3006, 3008, 3009, 3010, 3012, 152, 202, 203, 204, 3007, 3018 as given in the Table 4.8.

Table 4.8: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.89	-0.06	-0.06
205	BUS 203	0.96	0.90	-0.06	-0.05
103	BUS 3005	0.98	0.88	-0.10	-0.07
204	BUS 3005	0.97	0.90	-0.08	-0.05
3007	BUS 3005	0.98	0.77	-0.21	-0.18
3008	BUS 3005	0.97	0.81	-0.16	-0.14
3018	BUS 3005	1.03	0.88	-0.15	-0.07
103	SING OPN LIN 6 152-153(1)	0.98	0.89	-0.09	-0.06
153	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
154	SING OPN LIN 6 152-153(1)	0.95	0.86	-0.09	-0.09
205	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
3006	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.11	-0.06
3008	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.07	-0.06
3009	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
3010	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.11	-0.05
3012	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.11	-0.06
152	SING OPN LIN 10 201-202(1)	1.01	0.89	-0.12	-0.06
202	SING OPN LIN 10 201-202(1)	1.00	0.86	-0.14	-0.09
203	SING OPN LIN 10 201-202(1)	0.98	0.84	-0.14	-0.11
204	SING OPN LIN 10 201-202(1)	0.97	0.85	-0.12	-0.10
3007	SING OPN LIN 10 201-202(1)	0.98	0.87	-0.10	-0.08

4.10 Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 205, 3006, 3008, 3009, 3010, 3012, 204, 3007, 3018 as given in the Table 4.9.

Table 4.9: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.89	-0.06	-0.06
205	BUS 203	0.96	0.90	-0.06	-0.05
103	BUS 3005	0.98	0.88	-0.10	-0.07
204	BUS 3005	0.97	0.90	-0.08	-0.05
3007	BUS 3005	0.98	0.77	-0.21	-0.18
3008	BUS 3005	0.97	0.81	-0.16	-0.14
3018	BUS 3005	1.03	0.88	-0.15	-0.07
103	SING OPN LIN 6 152-153(1)	0.98	0.89	-0.08	-0.06
153	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
154	SING OPN LIN 6 152-153(1)	0.95	0.86	-0.09	-0.09
205	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
3006	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.10	-0.06
3008	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.07	-0.06
3009	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
3010	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.11	-0.05
3012	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07

4.11 Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, LLS, the buses reporting lower emergency range violations are the buses 3007 as given in the Table 4.10.

Table 4.10: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.87	-0.12	-0.08

4.12 Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, LLS, the buses reporting lower emergency range violations are the buses 3007 as given in the Table 4.11.

Table 4.11: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.88	-0.12	-0.07

4.13 Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, LLS, the buses reporting lower emergency range violations are the buses 3007 as given in the Table 4.12.

Table 4.12: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.88	-0.12	-0.07

4.14 Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, RLS, the buses reporting lower emergency range violations are the buses 153, 154, 205, 103, 202, 203, 204, 3007, 3008 as given in the Table 4.13.

Table 4.13: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 203	0.96	0.90	-0.07	-0.05
205	BUS 3005	0.97	0.90	-0.08	-0.05
3007	BUS 3005	0.99	0.81	-0.17	-0.14
3008	BUS 3005	0.98	0.84	-0.14	-0.11
153	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
154	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
205	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
103	SING OPN LIN 10 201-202(1)	0.99	0.88	-0.11	-0.07
202	SING OPN LIN 10 201-202(1)	1.01	0.89	-0.12	-0.06
203	SING OPN LIN 10 201-202(1)	0.99	0.87	-0.12	-0.08
204	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.11	-0.07
3007	SING OPN LIN 10 201-202(1)	0.99	0.90	-0.09	-0.05
3008	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07

4.15 Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, RLS, the buses reporting lower emergency range violations are the buses 153, 154, 103, 202, 203, 204, 205, 3008, 3007 as given in the Table 4.14.

Table 4.14: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 203	0.96	0.90	-0.07	-0.05
3007	BUS 3005	0.99	0.81	-0.17	-0.14
3008	BUS 3005	0.98	0.85	-0.13	-0.10
153	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
154	SING OPN LIN 6 152-153(1)	0.96	0.89	-0.08	-0.06
103	SING OPN LIN 10 201-202(1)	0.99	0.88	-0.11	-0.07
202	SING OPN LIN 10 201-202(1)	1.01	0.89	-0.12	-0.06
203	SING OPN LIN 10 201-202(1)	0.99	0.87	-0.12	-0.08
204	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.11	-0.07
205	SING OPN LIN 10 201-202(1)	0.97	0.86	-0.12	-0.09
3008	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07

4.16 Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, RLS, the buses reporting lower emergency range violations are the buses 153, 154, 103, 202, 203, 204, 205, 3007, 3008, 3012 as given in the Table 4.15.

Table 4.15: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 203	0.96	0.90	-0.07	-0.05
3007	BUS 3005	0.99	0.81	-0.17	-0.14
3008	BUS 3005	0.98	0.85	-0.13	-0.10
153	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
154	SING OPN LIN 6 152-153(1)	0.96	0.89	-0.08	-0.06
103	SING OPN LIN 10 201-202(1)	0.99	0.88	-0.11	-0.07
202	SING OPN LIN 10 201-202(1)	1.01	0.89	-0.12	-0.06
203	SING OPN LIN 10 201-202(1)	0.99	0.87	-0.12	-0.08
204	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.11	-0.07
205	SING OPN LIN 10 201-202(1)	0.97	0.86	-0.12	-0.09
3007	SING OPN LIN 10 201-202(1)	0.99	0.90	-0.09	-0.05
3008	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07
3012	SING OPN LIN 10 201-202(1)	1.00	0.90	-0.10	-0.05

4.17 Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, HLS, the buses reporting lower emergency range violations are the buses 154, 103, 153, 204, 205, 3006, 3007, 3008, 3009, 3010, 3012 as given in the Table 4.16.

Table 4.16: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.94	0.87	-0.07	-0.08
205	BUS 153	0.95	0.89	-0.06	-0.06
103	BUS 203	0.97	0.89	-0.08	-0.06
204	BUS 203	0.97	0.89	-0.08	-0.06
3008	BUS 203	0.96	0.90	-0.06	-0.05
154	SING OPN LIN 4 151-152(1)	0.94	0.90	-0.04	-0.05
103	SING OPN LIN 6 152-153(1)	0.97	0.87	-0.10	-0.08
153	SING OPN LIN 6 152-153(1)	0.98	0.85	-0.13	-0.10
204	SING OPN LIN 6 152-153(1)	0.97	0.88	-0.08	-0.07
205	SING OPN LIN 6 152-153(1)	0.95	0.86	-0.09	-0.09
3006	SING OPN LIN 6 152-153(1)	0.98	0.87	-0.12	-0.08
3007	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
3008	SING OPN LIN 6 152-153(1)	0.96	0.87	-0.09	-0.08
3009	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
3010	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
3012	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08

4.18 Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, HLS, the buses reporting lower emergency range violations are the buses 154, 103, 153, 204, 205, 3006, 3007, 3008, 3009, 3010, 3012 as given in the Table 4.17.

Table 4.17: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.94	0.87	-0.07	-0.08
205	BUS 153	0.95	0.89	-0.06	-0.06
103	BUS 203	0.97	0.89	-0.08	-0.06
204	BUS 203	0.97	0.89	-0.08	-0.06
3008	BUS 203	0.96	0.90	-0.06	-0.05
154	SING OPN LIN 4 151-152(1)	0.94	0.90	-0.04	-0.05
103	SING OPN LIN 6 152-153(1)	0.97	0.88	-0.09	-0.07
153	SING OPN LIN 6 152-153(1)	0.98	0.85	-0.13	-0.10
204	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
205	SING OPN LIN 6 152-153(1)	0.95	0.86	-0.09	-0.09
3006	SING OPN LIN 6 152-153(1)	0.98	0.87	-0.11	-0.08
3007	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
3008	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
3009	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3010	SING OPN LIN 6 152-153(1)	1.00	0.88	-0.11	-0.07
3012	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08

4.19 Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, HLS, the buses reporting lower emergency range violations are the buses 154, 103, 153, 204, 205, 3006, 3007, 3008, 3009, 3010, 3012 as given in the Table 4.18.

Table 4.18: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.94	0.87	-0.07	-0.08
205	BUS 153	0.95	0.89	-0.06	-0.06
103	BUS 203	0.97	0.89	-0.08	-0.06
204	BUS 203	0.97	0.89	-0.08	-0.06
3008	BUS 203	0.96	0.90	-0.06	-0.05
154	SING OPN LIN 4 151-152(1)	0.94	0.90	-0.04	-0.05
103	SING OPN LIN 6 152-153(1)	0.97	0.88	-0.09	-0.07
153	SING OPN LIN 6 152-153(1)	0.98	0.85	-0.13	-0.10
204	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
205	SING OPN LIN 6 152-153(1)	0.95	0.86	-0.09	-0.09
3006	SING OPN LIN 6 152-153(1)	0.98	0.87	-0.11	-0.08
3007	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
3008	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
3009	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.12	-0.07
3010	SING OPN LIN 6 152-153(1)	1.00	0.88	-0.12	-0.07
3012	SING OPN LIN 6 152-153(1)	0.98	0.86	-0.12	-0.09

4.20 Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, LLS, the buses reporting lower emergency range violations are the buses 154, 203, 204, 205, 3007, 3008 as given in the Table 4.19.

Table 4.19: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.85	-0.14	-0.10
3008	BUS 3005	0.99	0.88	-0.11	-0.07
154	SING OPN LIN 10 201-202(1)	0.97	0.88	-0.10	-0.07
203	SING OPN LIN 10 201-202(1)	1.00	0.90	-0.10	-0.05
204	SING OPN LIN 10 201-202(1)	0.99	0.90	-0.09	-0.05
205	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07

4.21 Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, LLS, the buses reporting lower emergency range violations are the buses 154, 203, 204, 205, 3007, 3008 as given in the Table 4.20.

Table 4.20: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.85	-0.14	-0.10
3008	BUS 3005	0.99	0.88	-0.10	-0.07
154	SING OPN LIN 10 201-202(1)	0.97	0.88	-0.10	-0.07
203	SING OPN LIN 10 201-202(1)	1.00	0.90	-0.10	-0.05
204	SING OPN LIN 10 201-202(1)	0.99	0.90	-0.09	-0.05
205	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07

4.22 Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, LLS, the buses reporting lower emergency range violations are the buses 154, 203, 204, 205, 3007, 3008 as given in the Table 4.21.

Table 4.21: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.85	-0.14	-0.10
3008	BUS 3005	0.99	0.88	-0.10	-0.07
154	SING OPN LIN 10 201-202(1)	0.97	0.87	-0.10	-0.08
203	SING OPN LIN 10 201-202(1)	1.00	0.89	-0.10	-0.06
204	SING OPN LIN 10 201-202(1)	0.99	0.90	-0.09	-0.05
205	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07

4.23 Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 204, 205, 3006, 3008, 3009, 3010, 3012, 3007, 3018 as given in the Table 4.22.

Table 4.22: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.89	-0.06	-0.06
103	BUS 203	0.98	0.89	-0.09	-0.06
204	BUS 203	0.97	0.88	-0.09	-0.07
205	BUS 203	0.96	0.87	-0.10	-0.08
3008	BUS 203	0.97	0.90	-0.07	-0.05
3007	BUS 3005	0.98	0.78	-0.20	-0.17
3018	BUS 3005	1.03	0.89	-0.15	-0.06
103	SING OPN LIN 6 152-153(1)	0.98	0.89	-0.09	-0.06
153	SING OPN LIN 6 152-153(1)	0.99	0.86	-0.12	-0.09
154	SING OPN LIN 6 152-153(1)	0.95	0.86	-0.09	-0.09
204	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.08	-0.05
205	SING OPN LIN 6 152-153(1)	0.96	0.87	-0.09	-0.08
3006	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3008	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3009	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.11	-0.06
3010	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.11	-0.06
3012	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.11	-0.06

4.24 Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, RLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 204, 205, 3006, 3008, 3009, 3010, 3012, 3007, 3018 as given in the Table 4.23.

Table 4.23: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.89	-0.06	-0.06
103	BUS 203	0.98	0.89	-0.09	-0.06
204	BUS 203	0.97	0.88	-0.09	-0.07
205	BUS 203	0.96	0.87	-0.10	-0.08
3008	BUS 203	0.97	0.90	-0.07	-0.05
3007	BUS 3005	0.98	0.78	-0.20	-0.17
3018	BUS 3005	1.03	0.89	-0.14	-0.06
103	SING OPN LIN 6 152-153(1)	0.98	0.89	-0.09	-0.06
153	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
154	SING OPN LIN 6 152-153(1)	0.95	0.86	-0.09	-0.09
204	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
205	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
3006	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.11	-0.06
3008	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3009	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
3010	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.11	-0.05
3012	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.11	-0.06

4.25 Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 204, 205, 3006, 3008, 3009, 3010, 3012, 3007, 3018 as given in the Table 4.24.

Table 4.24: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.89	-0.06	-0.06
103	BUS 203	0.98	0.89	-0.09	-0.06
204	BUS 203	0.97	0.88	-0.09	-0.07
205	BUS 203	0.96	0.86	-0.10	-0.09
3008	BUS 203	0.97	0.90	-0.07	-0.05
3007	BUS 3005	0.98	0.78	-0.20	-0.17
3018	BUS 3005	1.03	0.89	-0.14	-0.06
103	SING OPN LIN 6 152-153(1)	0.98	0.89	-0.09	-0.06
153	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
154	SING OPN LIN 6 152-153(1)	0.95	0.86	-0.09	-0.09
204	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
205	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
3006	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.10	-0.06
3008	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3009	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
3010	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.11	-0.05
3012	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07

4.26 Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3008 as given in the Table 4.25.

Table 4.25: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 153	0.96	0.88	-0.08	-0.07
154	BUS 153	0.93	0.85	-0.08	-0.10
204	BUS 153	0.95	0.89	-0.07	-0.06
205	BUS 153	0.94	0.86	-0.08	-0.09
3008	BUS 153	0.95	0.89	-0.06	-0.06

4.27 Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3008 as given in the Table 4.26.

Table 4.26: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 153	0.96	0.88	-0.08	-0.07
154	BUS 153	0.93	0.85	-0.08	-0.10
204	BUS 153	0.95	0.89	-0.07	-0.06
205	BUS 153	0.94	0.86	-0.08	-0.09
3008	BUS 153	0.95	0.90	-0.05	-0.05

4.28 Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3008 as given in the Table 4.27.

Table 4.27: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 153	0.96	0.88	-0.08	-0.07
154	BUS 153	0.93	0.85	-0.08	-0.10
204	BUS 153	0.95	0.89	-0.07	-0.06
205	BUS 153	0.94	0.86	-0.08	-0.09
3008	BUS 153	0.95	0.89	-0.06	-0.06

4.29 Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3008, 203, 3007, 206 as given in the Table 4.28.

Table 4.28: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	UNIT 206(1)	1.00	0.87	-0.13	-0.08
154	UNIT 206(1)	0.97	0.84	-0.13	-0.11
203	UNIT 206(1)	1.00	0.90	-0.10	-0.05
204	UNIT 206(1)	0.99	0.88	-0.12	-0.07
205	UNIT 206(1)	0.98	0.84	-0.14	-0.11
206	UNIT 206(1)	1.03	0.84	-0.19	-0.11
3007	UNIT 206(1)	1.00	0.89	-0.10	-0.06
3008	UNIT 206(1)	0.99	0.87	-0.12	-0.08
103	BUS 202	1.00	0.88	-0.12	-0.07
154	BUS 202	0.97	0.85	-0.12	-0.10
203	BUS 202	1.00	0.83	-0.17	-0.12
204	BUS 202	0.99	0.88	-0.11	-0.07
205	BUS 202	0.98	0.86	-0.12	-0.09
3008	BUS 202	0.99	0.89	-0.10	-0.06
3007	BUS 205	1.00	0.90	-0.10	-0.05
103	SING OPN LIN 11 201-204(1)	1.00	0.89	-0.11	-0.06
154	SING OPN LIN 11 201-204(1)	0.97	0.86	-0.12	-0.09
204	SING OPN LIN 11 201-204(1)	0.99	0.86	-0.13	-0.09
205	SING OPN LIN 11 201-204(1)	0.98	0.86	-0.12	-0.09
3008	SING OPN LIN 11 201-204(1)	0.99	0.89	-0.10	-0.06
203	SING OPN LIN 20 205-206(1)	1.00	0.90	-0.10	-0.05
3007	SING OPN LIN 20 205-206(1)	1.00	0.89	-0.10	-0.06

4.30 Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3008, 203, 3007, 206 as given in the Table 4.29.

Table 4.29: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	UNIT 206(1)	1.00	0.87	-0.13	-0.08
154	UNIT 206(1)	0.97	0.84	-0.13	-0.11
203	UNIT 206(1)	1.00	0.90	-0.10	-0.05
204	UNIT 206(1)	0.99	0.88	-0.11	-0.07
205	UNIT 206(1)	0.98	0.84	-0.14	-0.11
206	UNIT 206(1)	1.03	0.84	-0.19	-0.11
3007	UNIT 206(1)	1.00	0.90	-0.10	-0.05
3008	UNIT 206(1)	0.99	0.88	-0.11	-0.07
103	BUS 202	1.00	0.88	-0.12	-0.07
154	BUS 202	0.97	0.85	-0.12	-0.10
203	BUS 202	1.00	0.83	-0.17	-0.12
204	BUS 202	0.99	0.88	-0.11	-0.07
205	BUS 202	0.98	0.86	-0.12	-0.09
3008	BUS 202	0.99	0.89	-0.10	-0.06
3007	BUS 205	1.00	0.89	-0.10	-0.06
103	SING OPN LIN 11 201-204(1)	1.00	0.89	-0.11	-0.06
154	SING OPN LIN 11 201-204(1)	0.97	0.86	-0.12	-0.09
204	SING OPN LIN 11 201-204(1)	0.99	0.86	-0.13	-0.09
205	SING OPN LIN 11 201-204(1)	0.98	0.86	-0.12	-0.09
3008	SING OPN LIN 11 201-204(1)	0.99	0.89	-0.09	-0.06
203	SING OPN LIN 20 205-206(1)	1.00	0.90	-0.10	-0.05
3007	SING OPN LIN 20 205-206(1)	1.00	0.90	-0.10	-0.05

4.31 Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 203, 204, 205, 3008, 3007, 153, 3012, 206 as given in the Table 4.30.

Table 4.30: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	UNIT 206(1)	1.00	0.87	-0.13	-0.08
154	UNIT 206(1)	0.97	0.84	-0.13	-0.11
203	UNIT 206(1)	1.00	0.90	-0.10	-0.05
204	UNIT 206(1)	0.99	0.88	-0.11	-0.07
205	UNIT 206(1)	0.98	0.84	-0.14	-0.11
206	UNIT 206(1)	1.03	0.84	-0.19	-0.11
3007	UNIT 206(1)	1.00	0.90	-0.10	-0.05
3008	UNIT 206(1)	0.99	0.88	-0.11	-0.07
103	BUS 202	1.00	0.88	-0.12	-0.07
154	BUS 202	0.97	0.85	-0.12	-0.10
203	BUS 202	1.00	0.83	-0.17	-0.12
204	BUS 202	0.99	0.88	-0.11	-0.07
205	BUS 202	0.98	0.86	-0.12	-0.09
3008	BUS 202	0.99	0.89	-0.10	-0.06
153	BUS 205	1.01	0.90	-0.11	-0.05
3007	BUS 205	1.00	0.89	-0.11	-0.06
3012	BUS 205	1.01	0.90	-0.11	-0.05
103	SING OPN LIN 11 201-204(1)	1.00	0.89	-0.11	-0.06
154	SING OPN LIN 11 201-204(1)	0.97	0.86	-0.12	-0.09
203	SING OPN LIN 11 201-204(1)	1.00	0.90	-0.10	-0.05
204	SING OPN LIN 11 201-204(1)	0.99	0.86	-0.13	-0.09
205	SING OPN LIN 11 201-204(1)	0.98	0.86	-0.12	-0.09
3008	SING OPN LIN 11 201-204(1)	0.99	0.89	-0.10	-0.06
3007	SING OPN LIN 20 205-206(1)	1.00	0.90	-0.10	-0.05

4.32 Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, RLS, the buses reporting lower emergency range violations are the buses 154, 103, 152, 153, 203, 3005, 3006, 3007, 3008, 3009, 3010, 3012, 3018 as given in the Table 4.31.

Table 4.31: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 205	1.00	0.81	-0.19	-0.14
152	BUS 205	1.03	0.90	-0.13	-0.05
153	BUS 205	1.00	0.86	-0.14	-0.09
154	BUS 205	0.97	0.77	-0.20	-0.18
203	BUS 205	1.00	0.87	-0.13	-0.08
3005	BUS 205	1.01	0.88	-0.12	-0.07
3006	BUS 205	1.01	0.87	-0.14	-0.08
3007	BUS 205	0.99	0.85	-0.14	-0.10
3008	BUS 205	0.99	0.82	-0.16	-0.13
3009	BUS 205	1.01	0.87	-0.14	-0.08
3010	BUS 205	1.01	0.87	-0.14	-0.08
3012	BUS 205	1.01	0.87	-0.14	-0.08
3018	BUS 205	1.05	0.90	-0.15	-0.05
154	SING OPN LIN 10 201-202(1)	0.97	0.89	-0.08	-0.06

4.33 Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, RLS, the buses reporting lower emergency range violations are the buses 154, 103, 152, 153, 203, 3005, 3006, 3007, 3008, 3009, 3010, 3012, 3018 as given in the Table 4.32.

Table 4.32: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 205	1.00	0.80	-0.20	-0.15
152	BUS 205	1.03	0.89	-0.13	-0.06
153	BUS 205	1.00	0.86	-0.15	-0.09
154	BUS 205	0.97	0.77	-0.20	-0.18
203	BUS 205	1.00	0.87	-0.13	-0.08
3005	BUS 205	1.01	0.88	-0.13	-0.07
3006	BUS 205	1.01	0.86	-0.14	-0.09
3007	BUS 205	0.99	0.85	-0.14	-0.10
3008	BUS 205	0.99	0.82	-0.17	-0.13
3009	BUS 205	1.01	0.87	-0.14	-0.08
3010	BUS 205	1.02	0.87	-0.14	-0.08
3012	BUS 205	1.01	0.86	-0.14	-0.09
3018	BUS 205	1.05	0.89	-0.16	-0.06
154	SING OPN LIN 10 201-202(1)	0.97	0.89	-0.08	-0.06

4.34 Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, RLS, the buses reporting lower emergency range violations are the buses 154, 103, 152, 153, 202, 203, 3004, 3005, 3006, 3007, 3008, 3009, 3010, 3012, 3018 as given in the Table 4.33.

Table 4.33: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 205	1.00	0.79	-0.21	-0.16
152	BUS 205	1.03	0.89	-0.14	-0.06
153	BUS 205	1.00	0.85	-0.15	-0.10
154	BUS 205	0.97	0.76	-0.21	-0.19
202	BUS 205	1.02	0.90	-0.12	-0.05
203	BUS 205	1.00	0.86	-0.14	-0.09
3004	BUS 205	1.03	0.90	-0.13	-0.05
3005	BUS 205	1.01	0.87	-0.13	-0.08
3006	BUS 205	1.01	0.86	-0.15	-0.09
3007	BUS 205	0.99	0.84	-0.15	-0.11
3008	BUS 205	0.99	0.81	-0.17	-0.14
3009	BUS 205	1.02	0.86	-0.15	-0.09
3010	BUS 205	1.02	0.87	-0.15	-0.08
3012	BUS 205	1.00	0.85	-0.15	-0.10
3018	BUS 205	1.05	0.88	-0.16	-0.07
154	SING OPN LIN 10 201-202(1)	0.97	0.89	-0.08	-0.06

4.35 Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 205, 3006, 3008, 3009, 3010, 3012, 202, 203, 204, 3007, 3018 as given in the Table 4.34.

Table 4.34: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.90	-0.06	-0.05
103	BUS 3005	0.98	0.89	-0.09	-0.06
205	BUS 3005	0.96	0.87	-0.09	-0.08
3007	BUS 3005	0.98	0.78	-0.20	-0.17
3008	BUS 3005	0.97	0.81	-0.16	-0.14
3018	BUS 3005	1.03	0.89	-0.15	-0.06
103	SING OPN LIN 6 152-153(1)	0.98	0.90	-0.08	-0.05
153	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
154	SING OPN LIN 6 152-153(1)	0.95	0.87	-0.09	-0.08
205	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
3006	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.10	-0.06
3008	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
3009	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.10	-0.06
3010	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.10	-0.06
3012	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.10	-0.06
202	SING OPN LIN 10 201-202(1)	1.01	0.88	-0.13	-0.07
203	SING OPN LIN 10 201-202(1)	0.98	0.86	-0.13	-0.09
204	SING OPN LIN 10 201-202(1)	0.98	0.87	-0.11	-0.08
3007	SING OPN LIN 10 201-202(1)	0.98	0.89	-0.09	-0.06

4.36 Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, HLS, the buses reporting lower emergency range violations are the buses 153, 154, 205, 3006, 3012, 103, 202, 203, 204, 3007, 3008, 3018 as given in the Table 4.35.

Table 4.35: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.90	-0.06	-0.05
103	BUS 3005	0.98	0.89	-0.09	-0.06
205	BUS 3005	0.96	0.88	-0.09	-0.07
3007	BUS 3005	0.98	0.78	-0.20	-0.17
3008	BUS 3005	0.97	0.82	-0.15	-0.13
3018	BUS 3005	1.03	0.89	-0.14	-0.06
153	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.12	-0.07
154	SING OPN LIN 6 152-153(1)	0.95	0.87	-0.08	-0.08
205	SING OPN LIN 6 152-153(1)	0.96	0.89	-0.08	-0.06
3006	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.10	-0.06
3012	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.10	-0.06
103	SING OPN LIN 10 201-202(1)	0.98	0.86	-0.12	-0.09
202	SING OPN LIN 10 201-202(1)	1.01	0.88	-0.13	-0.07
203	SING OPN LIN 10 201-202(1)	0.98	0.86	-0.13	-0.09
204	SING OPN LIN 10 201-202(1)	0.98	0.87	-0.11	-0.08
3007	SING OPN LIN 10 201-202(1)	0.98	0.89	-0.09	-0.06
3008	SING OPN LIN 10 201-202(1)	0.97	0.87	-0.10	-0.08

4.37 Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, HLS, the buses reporting lower emergency range violations are the buses 153, 154, 205, 3006, 3012, 103, 202, 203, 204, 3007, 3008, 3009, 3018 as given in the Table 4.36.

Table 4.36: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.90	-0.06	-0.05
103	BUS 3005	0.98	0.89	-0.09	-0.06
205	BUS 3005	0.96	0.88	-0.09	-0.07
3007	BUS 3005	0.98	0.78	-0.20	-0.17
3008	BUS 3005	0.97	0.82	-0.15	-0.13
3018	BUS 3005	1.03	0.89	-0.14	-0.06
153	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
154	SING OPN LIN 6 152-153(1)	0.95	0.87	-0.08	-0.08
205	SING OPN LIN 6 152-153(1)	0.96	0.89	-0.08	-0.06
3006	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.10	-0.06
3012	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.10	-0.06
103	SING OPN LIN 10 201-202(1)	0.98	0.86	-0.12	-0.09
202	SING OPN LIN 10 201-202(1)	1.01	0.88	-0.13	-0.07
203	SING OPN LIN 10 201-202(1)	0.98	0.86	-0.13	-0.09
204	SING OPN LIN 10 201-202(1)	0.98	0.87	-0.11	-0.08
3007	SING OPN LIN 10 201-202(1)	0.98	0.89	-0.09	-0.06
3008	SING OPN LIN 10 201-202(1)	0.97	0.87	-0.11	-0.08
3009	SING OPN LIN 10 201-202(1)	1.00	0.90	-0.10	-0.05

4.38 Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 152, 153, 154, 202, 203, 3004, 3005, 3006, 3007, 3008, 3009, 3010, 3012, 3018 as given in the Table 4.37.

Table 4.37: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 205	1.00	0.80	-0.20	-0.15
152	BUS 205	1.03	0.88	-0.14	-0.07
153	BUS 205	1.01	0.85	-0.16	-0.10
154	BUS 205	0.97	0.76	-0.21	-0.19
202	BUS 205	1.02	0.89	-0.12	-0.06
203	BUS 205	1.00	0.86	-0.14	-0.09
3004	BUS 205	1.03	0.89	-0.14	-0.06
3005	BUS 205	1.01	0.87	-0.14	-0.08
3006	BUS 205	1.01	0.85	-0.15	-0.10
3007	BUS 205	1.00	0.84	-0.15	-0.11
3008	BUS 205	0.99	0.81	-0.18	-0.14
3009	BUS 205	1.01	0.86	-0.15	-0.09
3010	BUS 205	1.01	0.86	-0.15	-0.09
3012	BUS 205	1.01	0.86	-0.15	-0.09
3018	BUS 205	1.05	0.89	-0.17	-0.06

4.39 Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, LLS, the buses reporting lower emergency range violations are the buses 3007 as given in the Table 4.38.

Table 4.38: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	1.00	0.88	-0.11	-0.07

4.40 Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3008, 3007, 206 as given in the Table 4.39.

Table 4.39: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	UNIT 206(1)	1.00	0.88	-0.12	-0.07
154	UNIT 206(1)	0.97	0.85	-0.13	-0.10
204	UNIT 206(1)	0.99	0.88	-0.11	-0.07
205	UNIT 206(1)	0.98	0.85	-0.13	-0.10
206	UNIT 206(1)	1.04	0.85	-0.19	-0.10
3008	UNIT 206(1)	0.99	0.88	-0.10	-0.07
3007	BUS 3005	1.00	0.88	-0.11	-0.07
103	SING OPN LIN 20 205-206(1)	1.00	0.88	-0.12	-0.07
154	SING OPN LIN 20 205-206(1)	0.97	0.85	-0.13	-0.10
204	SING OPN LIN 20 205-206(1)	0.99	0.88	-0.11	-0.07
205	SING OPN LIN 20 205-206(1)	0.98	0.85	-0.13	-0.10
3008	SING OPN LIN 20 205-206(1)	0.99	0.88	-0.10	-0.07

4.41 Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, RLS, the buses reporting lower emergency range violations are the buses 153, 154, 103, 203, 204, 205, 3008, 3007 as given in the Table 4.40.

Table 4.40: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 3005	0.97	0.89	-0.08	-0.06
3007	BUS 3005	0.99	0.82	-0.17	-0.13
3008	BUS 3005	0.98	0.85	-0.13	-0.10
153	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
154	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
103	SING OPN LIN 10 201-202(1)	1.00	0.89	-0.10	-0.06
203	SING OPN LIN 10 201-202(1)	0.99	0.88	-0.11	-0.07
204	SING OPN LIN 10 201-202(1)	0.99	0.89	-0.10	-0.06
205	SING OPN LIN 10 201-202(1)	0.98	0.87	-0.11	-0.08
3008	SING OPN LIN 10 201-202(1)	0.98	0.89	-0.09	-0.06

4.42 Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, RLS, the buses reporting lower emergency range violations are the buses 153, 154, 103, 203, 204, 205, 3008, 3007 as given in the Table 4.41.

Table 4.41: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 3005	0.97	0.89	-0.07	-0.06
3007	BUS 3005	0.99	0.82	-0.17	-0.13
3008	BUS 3005	0.98	0.86	-0.13	-0.09
153	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
154	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.07	-0.06
103	SING OPN LIN 10 201-202(1)	1.00	0.89	-0.10	-0.06
203	SING OPN LIN 10 201-202(1)	0.99	0.89	-0.11	-0.06
204	SING OPN LIN 10 201-202(1)	0.99	0.89	-0.10	-0.06
205	SING OPN LIN 10 201-202(1)	0.98	0.87	-0.11	-0.08
3008	SING OPN LIN 10 201-202(1)	0.98	0.89	-0.09	-0.06

4.43 Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, RLS, the buses reporting lower emergency range violations are the buses 153, 154, 103, 203, 204, 205, 3008, 3007 as given in the Table 4.42.

Table 4.42: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 3005	0.97	0.89	-0.07	-0.06
3007	BUS 3005	0.99	0.82	-0.17	-0.13
3008	BUS 3005	0.98	0.86	-0.13	-0.09
153	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
154	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.07	-0.06
103	SING OPN LIN 10 201-202(1)	0.99	0.89	-0.10	-0.06
203	SING OPN LIN 10 201-202(1)	0.99	0.88	-0.11	-0.07
204	SING OPN LIN 10 201-202(1)	0.99	0.89	-0.10	-0.06
205	SING OPN LIN 10 201-202(1)	0.98	0.87	-0.11	-0.08
3008	SING OPN LIN 10 201-202(1)	0.98	0.89	-0.09	-0.06

4.44 Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 204, 205, 3006, 3008, 3009, 3010, 3012 as given in the Table 4.43.

Table 4.43: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.88	-0.06	-0.07
205	BUS 153	0.96	0.90	-0.06	-0.05
103	BUS 203	0.97	0.90	-0.08	-0.05
204	BUS 203	0.97	0.90	-0.07	-0.05
103	SING OPN LIN 6 152-153(1)	0.97	0.88	-0.09	-0.07
153	SING OPN LIN 6 152-153(1)	0.98	0.86	-0.13	-0.09
154	SING OPN LIN 6 152-153(1)	0.95	0.85	-0.10	-0.10
204	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
205	SING OPN LIN 6 152-153(1)	0.96	0.87	-0.09	-0.08
3006	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3008	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
3009	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3010	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3012	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07

4.45 Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 204, 205, 3006, 3008, 3009, 3010, 3012, 3007, 3018 as given in the Table 4.44.

Table 4.44: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.88	-0.06	-0.07
205	BUS 153	0.96	0.90	-0.06	-0.05
103	BUS 203	0.97	0.90	-0.08	-0.05
204	BUS 203	0.97	0.90	-0.07	-0.05
3007	BUS 3005	0.97	0.76	-0.22	-0.19
3008	BUS 3005	0.96	0.79	-0.17	-0.16
3018	BUS 3005	1.03	0.87	-0.16	-0.08
103	SING OPN LIN 6 152-153(1)	0.97	0.88	-0.09	-0.07
153	SING OPN LIN 6 152-153(1)	0.99	0.86	-0.13	-0.09
154	SING OPN LIN 6 152-153(1)	0.95	0.85	-0.09	-0.10
204	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
205	SING OPN LIN 6 152-153(1)	0.96	0.87	-0.09	-0.08
3006	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3008	SING OPN LIN 6 152-153(1)	0.96	0.89	-0.08	-0.06
3009	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
3010	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
3012	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07

4.46 Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 204, 205, 3006, 3008, 3009, 3010, 3012, 3007, 3018 as given in the Table 4.45.

Table 4.45: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.88	-0.06	-0.07
205	BUS 153	0.96	0.90	-0.06	-0.05
103	BUS 203	0.97	0.90	-0.08	-0.05
204	BUS 203	0.97	0.90	-0.07	-0.05
3007	BUS 3005	0.97	0.76	-0.22	-0.19
3008	BUS 3005	0.96	0.79	-0.17	-0.16
3018	BUS 3005	1.03	0.87	-0.16	-0.08
103	SING OPN LIN 6 152-153(1)	0.97	0.88	-0.09	-0.07
153	SING OPN LIN 6 152-153(1)	0.98	0.86	-0.13	-0.09
154	SING OPN LIN 6 152-153(1)	0.95	0.85	-0.09	-0.10
204	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.07	-0.06
205	SING OPN LIN 6 152-153(1)	0.96	0.87	-0.09	-0.08
3006	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3008	SING OPN LIN 6 152-153(1)	0.96	0.89	-0.08	-0.06
3009	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
3010	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
3012	SING OPN LIN 6 152-153(1)	0.98	0.87	-0.11	-0.08

4.47 Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, LLS, the buses reporting lower emergency range violations are the buses 154, 205, 3007, 3008 as given in the Table 4.46.

Table 4.46: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.86	-0.13	-0.09
3008	BUS 3005	0.99	0.89	-0.10	-0.06
154	SING OPN LIN 10 201-202(1)	0.97	0.89	-0.08	-0.06
205	SING OPN LIN 10 201-202(1)	0.98	0.90	-0.08	-0.05

4.48 Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, LLS, the buses reporting lower emergency range violations are the buses 154, 205, 3007, 3008 as given in the Table 4.47.

Table 4.47: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.86	-0.13	-0.09
3008	BUS 3005	0.99	0.89	-0.10	-0.06
154	SING OPN LIN 10 201-202(1)	0.97	0.89	-0.08	-0.06
205	SING OPN LIN 10 201-202(1)	0.98	0.90	-0.08	-0.05

4.49 Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, LLS, the buses reporting lower emergency range violations are the buses 154, 205, 3007, 3008 as given in the Table 4.48.

Table 4.48: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.86	-0.13	-0.09
3008	BUS 3005	0.99	0.89	-0.10	-0.06
154	SING OPN LIN 10 201-202(1)	0.97	0.89	-0.08	-0.06
205	SING OPN LIN 10 201-202(1)	0.98	0.90	-0.08	-0.05

4.50 Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, RLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 205, 3006, 3008, 3009, 3010, 3012, 204, 3007, 3018 as given in the Table 4.49.

Table 4.49: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.96	0.90	-0.06	-0.05
103	BUS 203	0.98	0.90	-0.09	-0.05
204	BUS 203	0.98	0.89	-0.08	-0.06
205	BUS 203	0.97	0.88	-0.09	-0.07
3007	BUS 3005	0.98	0.79	-0.19	-0.16
3008	BUS 3005	0.97	0.83	-0.15	-0.12
3018	BUS 3005	1.04	0.90	-0.14	-0.05
103	SING OPN LIN 6 152-153(1)	0.98	0.90	-0.09	-0.05
153	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
154	SING OPN LIN 6 152-153(1)	0.96	0.87	-0.09	-0.08
205	SING OPN LIN 6 152-153(1)	0.97	0.88	-0.08	-0.07
3006	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.10	-0.06
3008	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
3009	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.10	-0.06
3010	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.10	-0.06
3012	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.10	-0.06

4.51 Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, RLS, the buses reporting lower emergency range violations are the buses 153, 154, 205, 3006, 3012, 103, 204, 3007, 3008, 3018 as given in the Table 4.50.

Table 4.50: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.96	0.90	-0.06	-0.05
103	BUS 203	0.98	0.90	-0.09	-0.05
204	BUS 203	0.98	0.89	-0.08	-0.06
205	BUS 203	0.97	0.88	-0.09	-0.07
3007	BUS 3005	0.98	0.79	-0.19	-0.16
3008	BUS 3005	0.97	0.83	-0.15	-0.12
3018	BUS 3005	1.04	0.90	-0.14	-0.05
153	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.12	-0.07
154	SING OPN LIN 6 152-153(1)	0.96	0.87	-0.09	-0.08
205	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3006	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.10	-0.06
3012	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.10	-0.06

4.52 Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, RLS, the buses reporting lower emergency range violations are the buses 153, 154, 205, 3006, 3012, 103, 204, 3007, 3008, 3018 as given in the Table 4.51.

Table 4.51: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.96	0.90	-0.06	-0.05
103	BUS 203	0.98	0.90	-0.09	-0.05
204	BUS 203	0.98	0.89	-0.08	-0.06
205	BUS 203	0.97	0.87	-0.09	-0.08
3007	BUS 3005	0.98	0.79	-0.19	-0.16
3008	BUS 3005	0.97	0.83	-0.14	-0.12
3018	BUS 3005	1.04	0.90	-0.14	-0.05
153	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.12	-0.07
154	SING OPN LIN 6 152-153(1)	0.96	0.87	-0.08	-0.08
205	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3006	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.10	-0.06
3012	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.10	-0.06

4.53 Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3007, 3008 as given in the Table 4.52.

Table 4.52: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 153	0.96	0.89	-0.07	-0.06
154	BUS 153	0.93	0.86	-0.08	-0.09
204	BUS 153	0.96	0.90	-0.06	-0.05
205	BUS 153	0.94	0.87	-0.07	-0.08
3007	BUS 203	0.96	0.90	-0.07	-0.05
3008	BUS 203	0.95	0.87	-0.09	-0.08

4.54 Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3007, 3008 as given in the Table 4.53.

Table 4.53: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 153	0.96	0.89	-0.07	-0.06
154	BUS 153	0.93	0.86	-0.08	-0.09
204	BUS 153	0.96	0.90	-0.06	-0.05
205	BUS 153	0.94	0.87	-0.07	-0.08
3007	BUS 203	0.96	0.90	-0.07	-0.05
3008	BUS 203	0.95	0.87	-0.09	-0.08

4.55 Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3007, 3008 as given in the Table 4.54.

Table 4.54: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 153	0.96	0.89	-0.07	-0.06
154	BUS 153	0.93	0.86	-0.08	-0.09
204	BUS 153	0.96	0.90	-0.06	-0.05
205	BUS 153	0.94	0.87	-0.07	-0.08
3007	BUS 203	0.96	0.90	-0.07	-0.05
3008	BUS 203	0.95	0.87	-0.09	-0.08

4.56 Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 203, 3007, 3008, 206 as given in the Table 4.55.

Table 4.55: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	UNIT 206(1)	1.00	0.87	-0.13	-0.08
154	UNIT 206(1)	0.97	0.84	-0.13	-0.11
203	UNIT 206(1)	1.00	0.90	-0.10	-0.05
204	UNIT 206(1)	0.99	0.88	-0.11	-0.07
205	UNIT 206(1)	0.98	0.84	-0.14	-0.11
206	UNIT 206(1)	1.03	0.84	-0.19	-0.11
3007	UNIT 206(1)	1.00	0.90	-0.10	-0.05
3008	UNIT 206(1)	0.99	0.88	-0.11	-0.07
103	BUS 202	1.00	0.89	-0.11	-0.06
154	BUS 202	0.97	0.86	-0.11	-0.09
203	BUS 202	1.00	0.84	-0.16	-0.11
204	BUS 202	0.99	0.89	-0.10	-0.06
205	BUS 202	0.98	0.87	-0.11	-0.08
3008	BUS 205	0.99	0.88	-0.11	-0.07
103	SING OPN LIN 11 201-204(1)	1.00	0.90	-0.10	-0.05
154	SING OPN LIN 11 201-204(1)	0.97	0.87	-0.10	-0.08
204	SING OPN LIN 11 201-204(1)	0.99	0.88	-0.12	-0.07
205	SING OPN LIN 11 201-204(1)	0.98	0.88	-0.10	-0.07
203	SING OPN LIN 20 205-206(1)	1.00	0.90	-0.10	-0.05
3007	SING OPN LIN 20 205-206(1)	1.00	0.90	-0.10	-0.05
3008	SING OPN LIN 20 205-206(1)	0.99	0.88	-0.11	-0.07

4.57 Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3008, 203, 3007, 206 as given in the Table 4.56.

Table 4.56: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	UNIT 206(1)	1.00	0.87	-0.13	-0.08
154	UNIT 206(1)	0.97	0.85	-0.13	-0.10
204	UNIT 206(1)	0.99	0.88	-0.11	-0.07
205	UNIT 206(1)	0.98	0.85	-0.13	-0.10
206	UNIT 206(1)	1.03	0.85	-0.18	-0.10
3008	UNIT 206(1)	0.99	0.88	-0.11	-0.07
103	BUS 202	1.00	0.89	-0.11	-0.06
154	BUS 202	0.97	0.86	-0.11	-0.09
203	BUS 202	1.00	0.84	-0.16	-0.11
204	BUS 202	0.99	0.89	-0.10	-0.06
205	BUS 202	0.98	0.87	-0.11	-0.08
3007	BUS 205	1.00	0.90	-0.10	-0.05
3008	BUS 205	0.99	0.88	-0.11	-0.07
103	SING OPN LIN 11 201-204(1)	1.00	0.90	-0.10	-0.05
154	SING OPN LIN 11 201-204(1)	0.97	0.87	-0.10	-0.08
204	SING OPN LIN 11 201-204(1)	0.99	0.88	-0.12	-0.07
205	SING OPN LIN 11 201-204(1)	0.98	0.88	-0.10	-0.07
3008	SING OPN LIN 20 205-206(1)	0.99	0.88	-0.11	-0.07

4.58 Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3008, 203, 3007, 206 as given in the Table 4.57.

Table 4.57: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	UNIT 206(1)	1.00	0.87	-0.12	-0.08
154	UNIT 206(1)	0.97	0.85	-0.13	-0.10
204	UNIT 206(1)	0.99	0.88	-0.11	-0.07
205	UNIT 206(1)	0.98	0.85	-0.13	-0.10
206	UNIT 206(1)	1.03	0.85	-0.18	-0.10
3008	UNIT 206(1)	0.99	0.88	-0.11	-0.07
103	BUS 202	1.00	0.89	-0.11	-0.06
154	BUS 202	0.97	0.86	-0.11	-0.09
203	BUS 202	1.00	0.84	-0.16	-0.11
204	BUS 202	0.99	0.89	-0.10	-0.06
205	BUS 202	0.98	0.87	-0.11	-0.08
3007	BUS 205	1.00	0.90	-0.10	-0.05
3008	BUS 205	0.99	0.87	-0.12	-0.08
103	SING OPN LIN 11 201-204(1)	1.00	0.90	-0.10	-0.05
154	SING OPN LIN 11 201-204(1)	0.97	0.87	-0.10	-0.08
204	SING OPN LIN 11 201-204(1)	0.99	0.87	-0.12	-0.08
205	SING OPN LIN 11 201-204(1)	0.98	0.87	-0.11	-0.08
3008	SING OPN LIN 20 205-206(1)	0.99	0.88	-0.11	-0.07

4.59 Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, RLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 203, 3005, 3006, 3007, 3008, 3009, 3010, 3012 as given in the Table 4.58.

Table 4.58: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 205	1.00	0.81	-0.18	-0.14
153	BUS 205	1.01	0.87	-0.13	-0.08
154	BUS 205	0.97	0.79	-0.18	-0.16
203	BUS 205	1.00	0.88	-0.12	-0.07
3005	BUS 205	1.01	0.89	-0.11	-0.06
3006	BUS 205	1.01	0.88	-0.13	-0.07
3007	BUS 205	0.99	0.87	-0.13	-0.08
3008	BUS 205	0.99	0.84	-0.15	-0.11
3009	BUS 205	1.01	0.88	-0.13	-0.07
3010	BUS 205	1.01	0.88	-0.13	-0.07
3012	BUS 205	1.01	0.88	-0.13	-0.07

4.60 Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, RLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 203, 3005, 3006, 3007, 3008, 3009, 3010, 3012 as given in the Table 4.59.

Table 4.59: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 205	1.00	0.81	-0.19	-0.14
153	BUS 205	1.01	0.87	-0.14	-0.08
154	BUS 205	0.97	0.78	-0.19	-0.17
203	BUS 205	1.00	0.88	-0.12	-0.07
3005	BUS 205	1.01	0.89	-0.12	-0.06
3006	BUS 205	1.01	0.87	-0.13	-0.08
3007	BUS 205	0.99	0.86	-0.13	-0.09
3008	BUS 205	0.99	0.83	-0.16	-0.12
3009	BUS 205	1.02	0.88	-0.13	-0.07
3010	BUS 205	1.02	0.89	-0.13	-0.06
3012	BUS 205	1.01	0.87	-0.13	-0.08

4.61 Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, RLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 203, 3005, 3006, 3007, 3008, 3009, 3010, 3012, 3018 as given in the Table 4.60.

Table 4.60: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 205	1.00	0.80	-0.19	-0.15
153	BUS 205	1.01	0.86	-0.14	-0.09
154	BUS 205	0.97	0.78	-0.20	-0.17
203	BUS 205	1.00	0.87	-0.12	-0.08
3005	BUS 205	1.01	0.88	-0.12	-0.07
3006	BUS 205	1.01	0.87	-0.14	-0.08
3007	BUS 205	0.99	0.86	-0.14	-0.09
3008	BUS 205	0.99	0.83	-0.16	-0.12
3009	BUS 205	1.02	0.88	-0.14	-0.07
3010	BUS 205	1.02	0.88	-0.14	-0.07
3012	BUS 205	1.00	0.86	-0.14	-0.09
3018	BUS 205	1.05	0.90	-0.15	-0.05

4.62 Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, HLS, the buses reporting lower emergency range violations are the buses 153, 154, 205, 3006, 3009, 3010, 3012, 103, 202, 203, 204, 3007, 3008, 3018 as given in the Table 4.61.

Table 4.61: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 3005	0.98	0.89	-0.09	-0.06
154	BUS 3005	0.96	0.87	-0.09	-0.08
205	BUS 3005	0.97	0.88	-0.08	-0.07
3007	BUS 3005	0.98	0.79	-0.19	-0.16
3008	BUS 3005	0.97	0.82	-0.15	-0.13
3018	BUS 3005	1.04	0.89	-0.14	-0.06
153	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.12	-0.07
154	SING OPN LIN 6 152-153(1)	0.96	0.87	-0.08	-0.08
205	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3006	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
3009	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
3010	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
3012	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
103	SING OPN LIN 10 201-202(1)	0.98	0.87	-0.11	-0.08
202	SING OPN LIN 10 201-202(1)	1.01	0.89	-0.12	-0.06
203	SING OPN LIN 10 201-202(1)	0.99	0.87	-0.12	-0.08
204	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07
3007	SING OPN LIN 10 201-202(1)	0.98	0.90	-0.08	-0.05
3008	SING OPN LIN 10 201-202(1)	0.97	0.88	-0.10	-0.07

4.63 Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, HLS, the buses reporting lower emergency range violations are the buses 153, 154, 205, 3006, 3012, 103, 202, 203, 204, 3007, 3008, 3018 as given in the Table 4.62.

Table 4.62: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 3005	0.98	0.89	-0.09	-0.06
154	BUS 3005	0.96	0.87	-0.09	-0.08
205	BUS 3005	0.97	0.88	-0.08	-0.07
3007	BUS 3005	0.98	0.79	-0.19	-0.16
3008	BUS 3005	0.97	0.82	-0.15	-0.13
3018	BUS 3005	1.04	0.90	-0.14	-0.05
153	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
154	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
205	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.07	-0.06
3006	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
3012	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
103	SING OPN LIN 10 201-202(1)	0.98	0.87	-0.11	-0.08
202	SING OPN LIN 10 201-202(1)	1.01	0.89	-0.12	-0.06
203	SING OPN LIN 10 201-202(1)	0.99	0.87	-0.12	-0.08
204	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07
3007	SING OPN LIN 10 201-202(1)	0.98	0.90	-0.08	-0.05
3008	SING OPN LIN 10 201-202(1)	0.97	0.88	-0.10	-0.07

4.64 Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, HLS, the buses reporting lower emergency range violations are the buses 153, 154, 205, 3006, 3012, 103, 202, 203, 204, 3007, 3008, 3018 as given in the Table 4.63.

Table 4.63: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 3005	0.98	0.89	-0.09	-0.06
154	BUS 3005	0.96	0.87	-0.09	-0.08
205	BUS 3005	0.97	0.88	-0.08	-0.07
3007	BUS 3005	0.98	0.79	-0.19	-0.16
3008	BUS 3005	0.97	0.83	-0.15	-0.12
3018	BUS 3005	1.04	0.90	-0.14	-0.05
153	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
154	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
205	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.07	-0.06
3006	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
3012	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.10	-0.06
103	SING OPN LIN 10 201-202(1)	0.98	0.87	-0.11	-0.08
202	SING OPN LIN 10 201-202(1)	1.01	0.89	-0.12	-0.06
203	SING OPN LIN 10 201-202(1)	0.99	0.87	-0.12	-0.08
204	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07
3007	SING OPN LIN 10 201-202(1)	0.98	0.90	-0.08	-0.05
3008	SING OPN LIN 10 201-202(1)	0.97	0.88	-0.10	-0.07

4.65 Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3008, 152, 153, 203, 3005, 3006, 3007, 3009, 3010, 3012, 3018, 206 as given in the Table 4.64.

Table 4.64: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	UNIT 206(1)	1.00	0.87	-0.12	-0.08
154	UNIT 206(1)	0.97	0.85	-0.13	-0.10
204	UNIT 206(1)	0.99	0.88	-0.11	-0.07
205	UNIT 206(1)	0.98	0.85	-0.13	-0.10
206	UNIT 206(1)	1.04	0.85	-0.19	-0.10
3008	UNIT 206(1)	0.99	0.89	-0.10	-0.06
103	BUS 205	1.00	0.81	-0.19	-0.14
152	BUS 205	1.03	0.90	-0.13	-0.05
153	BUS 205	1.01	0.86	-0.14	-0.09
154	BUS 205	0.97	0.78	-0.19	-0.17
203	BUS 205	1.00	0.87	-0.13	-0.08
3005	BUS 205	1.01	0.88	-0.13	-0.07
3006	BUS 205	1.01	0.87	-0.14	-0.08
3007	BUS 205	1.00	0.86	-0.14	-0.09
3008	BUS 205	0.99	0.83	-0.16	-0.12
3009	BUS 205	1.01	0.87	-0.14	-0.08
3010	BUS 205	1.01	0.87	-0.14	-0.08
3012	BUS 205	1.01	0.87	-0.14	-0.08
3018	BUS 205	1.05	0.90	-0.15	-0.05
103	SING OPN LIN 20 205-206(1)	1.00	0.87	-0.12	-0.08
154	SING OPN LIN 20 205-206(1)	0.97	0.85	-0.13	-0.10
204	SING OPN LIN 20 205-206(1)	0.99	0.88	-0.11	-0.07
205	SING OPN LIN 20 205-206(1)	0.98	0.85	-0.13	-0.10
3008	SING OPN LIN 20 205-206(1)	0.99	0.89	-0.10	-0.06

4.66 Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3008, 152, 153, 202, 203, 3004, 3005, 3006, 3007, 3009, 3010, 3012, 3018, 206 as given in the Table 4.65.

Table 4.65: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	UNIT 206(1)	1.00	0.88	-0.12	-0.07
154	UNIT 206(1)	0.97	0.85	-0.12	-0.10
204	UNIT 206(1)	0.99	0.88	-0.11	-0.07
205	UNIT 206(1)	0.98	0.85	-0.13	-0.10
206	UNIT 206(1)	1.04	0.85	-0.19	-0.10
3008	UNIT 206(1)	0.99	0.89	-0.10	-0.06
103	BUS 205	1.00	0.80	-0.20	-0.15
152	BUS 205	1.03	0.89	-0.14	-0.06
153	BUS 205	1.01	0.86	-0.15	-0.09
154	BUS 205	0.97	0.77	-0.20	-0.18
202	BUS 205	1.02	0.90	-0.12	-0.05
203	BUS 205	1.00	0.87	-0.13	-0.08
3004	BUS 205	1.03	0.90	-0.13	-0.05
3005	BUS 205	1.01	0.88	-0.13	-0.07
3006	BUS 205	1.01	0.86	-0.15	-0.09
3007	BUS 205	1.00	0.85	-0.15	-0.10
3008	BUS 205	0.99	0.82	-0.17	-0.13
3009	BUS 205	1.02	0.87	-0.15	-0.08
3010	BUS 205	1.02	0.87	-0.15	-0.08
3012	BUS 205	1.01	0.86	-0.15	-0.09
3018	BUS 205	1.05	0.89	-0.16	-0.06
103	SING OPN LIN 20 205-206(1)	1.00	0.88	-0.12	-0.07
154	SING OPN LIN 20 205-206(1)	0.97	0.85	-0.12	-0.10
204	SING OPN LIN 20 205-206(1)	0.99	0.88	-0.11	-0.07
205	SING OPN LIN 20 205-206(1)	0.98	0.85	-0.13	-0.10
3008	SING OPN LIN 20 205-206(1)	0.99	0.89	-0.10	-0.06

4.67 Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, LLS, the buses reporting lower emergency range violations are the buses 103, 154, 204, 205, 3008, 152, 153, 202, 203, 3004, 3005, 3006, 3007, 3009, 3010, 3012, 3018, 206 as given in the Table 4.66.

Table 4.66: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	UNIT 206(1)	1.00	0.88	-0.12	-0.07
154	UNIT 206(1)	0.97	0.85	-0.12	-0.10
204	UNIT 206(1)	0.99	0.88	-0.11	-0.07
205	UNIT 206(1)	0.98	0.85	-0.13	-0.10
206	UNIT 206(1)	1.04	0.85	-0.19	-0.10
3008	UNIT 206(1)	0.99	0.89	-0.10	-0.06
103	BUS 205	1.00	0.79	-0.21	-0.16
152	BUS 205	1.03	0.89	-0.14	-0.06
153	BUS 205	1.01	0.85	-0.16	-0.10
154	BUS 205	0.97	0.76	-0.21	-0.19
202	BUS 205	1.02	0.89	-0.12	-0.06
203	BUS 205	1.00	0.86	-0.14	-0.09
3004	BUS 205	1.03	0.89	-0.14	-0.06
3005	BUS 205	1.01	0.87	-0.14	-0.08
3006	BUS 205	1.01	0.85	-0.16	-0.10
3007	BUS 205	1.00	0.84	-0.16	-0.11
3008	BUS 205	0.99	0.81	-0.18	-0.14
3009	BUS 205	1.02	0.86	-0.16	-0.09
3010	BUS 205	1.02	0.86	-0.16	-0.09
3012	BUS 205	1.01	0.85	-0.16	-0.10
3018	BUS 205	1.05	0.88	-0.17	-0.07
103	SING OPN LIN 20 205-206(1)	1.00	0.88	-0.12	-0.07
154	SING OPN LIN 20 205-206(1)	0.97	0.85	-0.12	-0.10
204	SING OPN LIN 20 205-206(1)	0.99	0.88	-0.11	-0.07
205	SING OPN LIN 20 205-206(1)	0.98	0.85	-0.13	-0.10
3008	SING OPN LIN 20 205-206(1)	0.99	0.89	-0.10	-0.06

4.68 Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, RLS, the buses reporting lower emergency range violations are the buses 153, 154, 103, 203, 204, 205, 3007, 3008 as given in the Table 4.67.

Table 4.67: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 3005	0.97	0.90	-0.07	-0.05
3007	BUS 3005	0.99	0.83	-0.16	-0.12
3008	BUS 3005	0.99	0.86	-0.13	-0.09
153	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
154	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
103	SING OPN LIN 10 201-202(1)	1.00	0.90	-0.10	-0.05
203	SING OPN LIN 10 201-202(1)	1.00	0.89	-0.10	-0.06
204	SING OPN LIN 10 201-202(1)	0.99	0.90	-0.09	-0.05
205	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07

4.69 Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, RLS, the buses reporting lower emergency range violations are the buses 154, 103, 203, 205, 3007, 3008 as given in the Table 4.68.

Table 4.68: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 3005	0.97	0.90	-0.07	-0.05
3007	BUS 3005	0.99	0.83	-0.16	-0.12
3008	BUS 3005	0.99	0.86	-0.12	-0.09
154	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
103	SING OPN LIN 10 201-202(1)	1.00	0.90	-0.10	-0.05
203	SING OPN LIN 10 201-202(1)	1.00	0.90	-0.10	-0.05
205	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07

4.70 Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, RLS, the buses reporting lower emergency range violations are the buses 154, 103, 203, 204, 205, 3007, 3008 as given in the Table 4.69.

Table 4.69: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 3005	0.97	0.90	-0.07	-0.05
3007	BUS 3005	0.99	0.83	-0.16	-0.12
3008	BUS 3005	0.98	0.86	-0.12	-0.09
154	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
103	SING OPN LIN 10 201-202(1)	0.99	0.90	-0.10	-0.05
203	SING OPN LIN 10 201-202(1)	1.00	0.89	-0.10	-0.06
204	SING OPN LIN 10 201-202(1)	0.99	0.90	-0.09	-0.05
205	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07

4.71 Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 204, 205, 3006, 3008, 3009, 3010, 3012, 3007, 3018 as given in the Table 4.70.

Table 4.70: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.89	-0.06	-0.06
205	BUS 203	0.96	0.88	-0.08	-0.07
103	BUS 3005	0.97	0.87	-0.10	-0.08
204	BUS 3005	0.97	0.89	-0.08	-0.06
3007	BUS 3005	0.98	0.76	-0.21	-0.19
3008	BUS 3005	0.97	0.80	-0.17	-0.15
3018	BUS 3005	1.03	0.87	-0.16	-0.08
103	SING OPN LIN 6 152-153(1)	0.97	0.88	-0.09	-0.07
153	SING OPN LIN 6 152-153(1)	0.99	0.86	-0.13	-0.09
154	SING OPN LIN 6 152-153(1)	0.95	0.86	-0.09	-0.09
204	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
205	SING OPN LIN 6 152-153(1)	0.96	0.87	-0.09	-0.08
3006	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3008	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3009	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3010	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3012	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07

4.72 Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 204, 205, 3006, 3008, 3009, 3010, 3012, 3007, 3018 as given in the Table 4.71.

Table 4.71: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.89	-0.06	-0.06
205	BUS 203	0.96	0.88	-0.08	-0.07
103	BUS 3005	0.97	0.87	-0.10	-0.08
204	BUS 3005	0.97	0.89	-0.08	-0.06
3007	BUS 3005	0.98	0.77	-0.21	-0.18
3008	BUS 3005	0.97	0.80	-0.16	-0.15
3018	BUS 3005	1.03	0.88	-0.15	-0.07
103	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.09	-0.06
153	SING OPN LIN 6 152-153(1)	0.99	0.86	-0.12	-0.09
154	SING OPN LIN 6 152-153(1)	0.95	0.86	-0.09	-0.09
204	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
205	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
3006	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3008	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3009	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
3010	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
3012	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07

4.73 Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, HLS, the buses reporting lower emergency range violations are the buses 103, 153, 154, 204, 205, 3006, 3008, 3009, 3010, 3012, 3007, 3018 as given in the Table 4.72.

Table 4.72: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 153	0.95	0.89	-0.06	-0.06
205	BUS 203	0.96	0.88	-0.08	-0.07
103	BUS 3005	0.97	0.87	-0.10	-0.08
204	BUS 3005	0.97	0.89	-0.08	-0.06
3007	BUS 3005	0.98	0.77	-0.21	-0.18
3008	BUS 3005	0.97	0.80	-0.16	-0.15
3018	BUS 3005	1.03	0.88	-0.15	-0.07
103	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.09	-0.06
153	SING OPN LIN 6 152-153(1)	0.99	0.86	-0.12	-0.09
154	SING OPN LIN 6 152-153(1)	0.95	0.86	-0.09	-0.09
204	SING OPN LIN 6 152-153(1)	0.97	0.90	-0.07	-0.05
205	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
3006	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
3008	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.07	-0.06
3009	SING OPN LIN 6 152-153(1)	1.00	0.89	-0.11	-0.06
3010	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.11	-0.05
3012	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07

4.74 Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, LLS, the buses reporting lower emergency range violations are the buses 154, 3007, 3008 as given in the Table 4.73.

Table 4.73: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.86	-0.13	-0.09
3008	BUS 3005	0.99	0.89	-0.09	-0.06
154	SING OPN LIN 10 201-202(1)	0.97	0.90	-0.07	-0.05

4.75 Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, LLS, the buses reporting lower emergency range violations are the buses 154, 3007, 3008 as given in the Table 4.74.

Table 4.74: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.87	-0.13	-0.08
3008	BUS 3005	0.99	0.90	-0.09	-0.05
154	SING OPN LIN 10 201-202(1)	0.97	0.90	-0.07	-0.05

4.76 Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, LLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, LLS, the buses reporting lower emergency range violations are the buses 154, 3007, 3008 as given in the Table 4.75.

Table 4.75: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, LLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
3007	BUS 3005	0.99	0.87	-0.13	-0.08
3008	BUS 3005	0.99	0.90	-0.09	-0.05
154	SING OPN LIN 10 201-202(1)	0.97	0.90	-0.08	-0.05

4.77 Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, RLS, the buses reporting lower emergency range violations are the buses 153, 154, 205, 3006, 3009, 3010, 3012, 103, 152, 202, 203, 204, 3007, 3008 as given in the Table 4.76.

Table 4.76: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 203	0.96	0.88	-0.08	-0.07
205	BUS 203	0.97	0.88	-0.09	-0.07
103	BUS 3005	0.98	0.90	-0.09	-0.05
3007	BUS 3005	0.98	0.80	-0.18	-0.15
3008	BUS 3005	0.98	0.83	-0.14	-0.12
153	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.12	-0.07
154	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
205	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3006	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
3009	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
3010	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
3012	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
103	SING OPN LIN 10 201-202(1)	0.98	0.85	-0.14	-0.10
152	SING OPN LIN 10 201-202(1)	1.02	0.90	-0.12	-0.05
202	SING OPN LIN 10 201-202(1)	1.01	0.87	-0.14	-0.08
203	SING OPN LIN 10 201-202(1)	0.99	0.85	-0.14	-0.10
204	SING OPN LIN 10 201-202(1)	0.98	0.85	-0.13	-0.10
3007	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07
3008	SING OPN LIN 10 201-202(1)	0.98	0.86	-0.12	-0.09

4.78 Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, RLS, the buses reporting lower emergency range violations are the buses 153, 154, 205, 3006, 3012, 103, 152, 202, 203, 204, 3007, 3008, 3009, 3010 as given in the Table 4.77.

Table 4.77: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 203	0.96	0.88	-0.08	-0.07
205	BUS 203	0.97	0.88	-0.09	-0.07
3007	BUS 3005	0.98	0.80	-0.18	-0.15
3008	BUS 3005	0.98	0.84	-0.14	-0.11
153	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
154	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
205	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3006	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
3012	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
103	SING OPN LIN 10 201-202(1)	0.98	0.85	-0.14	-0.10
152	SING OPN LIN 10 201-202(1)	1.02	0.90	-0.12	-0.05
202	SING OPN LIN 10 201-202(1)	1.01	0.87	-0.14	-0.08
203	SING OPN LIN 10 201-202(1)	0.99	0.85	-0.14	-0.10
204	SING OPN LIN 10 201-202(1)	0.98	0.85	-0.13	-0.10
3007	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07
3008	SING OPN LIN 10 201-202(1)	0.98	0.86	-0.12	-0.09
3009	SING OPN LIN 10 201-202(1)	1.01	0.89	-0.11	-0.06
3010	SING OPN LIN 10 201-202(1)	1.01	0.89	-0.11	-0.06

4.79 Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, RLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, RLS, the buses reporting lower emergency range violations are the buses 153, 154, 205, 3006, 3012, 103, 152, 202, 203, 204, 3007, 3008, 3009, 3010 as given in the Table 4.78.

Table 4.78: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, RLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 203	0.96	0.88	-0.08	-0.07
204	BUS 203	0.98	0.90	-0.08	-0.05
205	BUS 203	0.97	0.88	-0.09	-0.07
3007	BUS 3005	0.98	0.80	-0.18	-0.15
3008	BUS 3005	0.98	0.84	-0.14	-0.11
153	SING OPN LIN 6 152-153(1)	0.99	0.88	-0.11	-0.07
154	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
205	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3006	SING OPN LIN 6 152-153(1)	1.00	0.90	-0.10	-0.05
3012	SING OPN LIN 6 152-153(1)	0.99	0.89	-0.10	-0.06
103	SING OPN LIN 10 201-202(1)	0.98	0.85	-0.14	-0.10
152	SING OPN LIN 10 201-202(1)	1.02	0.89	-0.12	-0.06
202	SING OPN LIN 10 201-202(1)	1.01	0.86	-0.14	-0.09
203	SING OPN LIN 10 201-202(1)	0.99	0.84	-0.14	-0.11
204	SING OPN LIN 10 201-202(1)	0.98	0.85	-0.13	-0.10
3007	SING OPN LIN 10 201-202(1)	0.98	0.88	-0.10	-0.07
3008	SING OPN LIN 10 201-202(1)	0.98	0.86	-0.12	-0.09
3009	SING OPN LIN 10 201-202(1)	1.01	0.89	-0.12	-0.06
3010	SING OPN LIN 10 201-202(1)	1.01	0.89	-0.12	-0.06

4.80 Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, HLS, the buses reporting lower emergency range violations are the buses 154, 103, 205, 204, 3008 as given in the Table 4.79.

Table 4.79: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 153	0.96	0.89	-0.07	-0.06
154	BUS 153	0.94	0.86	-0.07	-0.09
205	BUS 153	0.95	0.88	-0.07	-0.07
204	BUS 203	0.96	0.86	-0.10	-0.09
3008	BUS 203	0.96	0.88	-0.08	-0.07
154	SING OPN LIN 4 151-152(1)	0.94	0.89	-0.05	-0.06

4.81 Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, HLS, the buses reporting lower emergency range violations are the buses 154, 103, 153, 204, 205, 3006, 3007, 3008, 3009, 3010, 3012 as given in the Table 4.80.

Table 4.80: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 153	0.96	0.89	-0.07	-0.06
154	BUS 153	0.94	0.87	-0.07	-0.08
205	BUS 153	0.95	0.88	-0.07	-0.07
204	BUS 203	0.96	0.86	-0.10	-0.09
3008	BUS 203	0.96	0.88	-0.08	-0.07
154	SING OPN LIN 4 151-152(1)	0.94	0.89	-0.05	-0.06
103	SING OPN LIN 6 152-153(1)	0.96	0.86	-0.10	-0.09
153	SING OPN LIN 6 152-153(1)	0.98	0.84	-0.14	-0.11
204	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.09	-0.07
205	SING OPN LIN 6 152-153(1)	0.95	0.85	-0.10	-0.10
3006	SING OPN LIN 6 152-153(1)	0.98	0.86	-0.12	-0.09
3007	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3008	SING OPN LIN 6 152-153(1)	0.96	0.87	-0.09	-0.08
3009	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
3010	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
3012	SING OPN LIN 6 152-153(1)	0.98	0.86	-0.12	-0.09

4.82 Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, HLS

For the studied scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, HLS, the buses reporting lower emergency range violations are the buses 154, 103, 153, 204, 205, 3006, 3007, 3008, 3009, 3010, 3012 as given in the Table 4.81.

Table 4.81: Contingencies reporting bus voltages lower than emergency voltage of 1.1 PU for scenario Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, HLS

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
103	BUS 153	0.96	0.89	-0.07	-0.06
154	BUS 153	0.94	0.87	-0.07	-0.08
205	BUS 153	0.95	0.88	-0.07	-0.07
204	BUS 203	0.96	0.86	-0.10	-0.09
3008	BUS 203	0.96	0.88	-0.08	-0.07
154	SING OPN LIN 4 151-152(1)	0.94	0.89	-0.05	-0.06
103	SING OPN LIN 6 152-153(1)	0.96	0.86	-0.10	-0.09
153	SING OPN LIN 6 152-153(1)	0.98	0.84	-0.14	-0.11
204	SING OPN LIN 6 152-153(1)	0.96	0.88	-0.08	-0.07
205	SING OPN LIN 6 152-153(1)	0.95	0.85	-0.10	-0.10
3006	SING OPN LIN 6 152-153(1)	0.98	0.86	-0.12	-0.09
3007	SING OPN LIN 6 152-153(1)	0.97	0.89	-0.08	-0.06
3008	SING OPN LIN 6 152-153(1)	0.96	0.87	-0.09	-0.08
3009	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
3010	SING OPN LIN 6 152-153(1)	0.99	0.87	-0.12	-0.08
3012	SING OPN LIN 6 152-153(1)	0.98	0.85	-0.12	-0.10

4.83 Result Summary

For the AC contingency calculation carried out on the hypothetical SAVNW system for Topology 6 to study the lower emergency bus voltage violations, the bus reporting the highest number of lower voltage limit violation is the bus 154 with 944 violations followed by bus 205 with 614 violations. The Table 5.13 in the Appendix gives the count of times a bus violates the lower voltage limit $< 0.9PU$.

The contingency causing the greatest number of lower voltage limit violation is the single line open contingency SING OPN LIN 6 152-153(1) with 576 violations followed by the bus faults BUS 203 with 364 and BUS 3005 with 312 violations. The Table 5.12 in the Appendix gives the contingencies and no of times a contingency causes a lower voltage limit violation $< 0.9PU$.

The scenario reporting the greatest number of lower voltage limit violation is the scenario Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS reporting 92 lower limit voltage violation followed by Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS with 90, and Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS with 86 violations. Table 5.14 and 5.15 in the Appendix gives the number of times a studied scenario reported a lower voltage limit violation $< 0.9PU$.

Chapter 5

Conculsion

The contingency analysis was carried out on the hypothetical SAVNW study system for the Year 5, Topology 6 for 81 study scenarios for bus, single line open, unit contingencies. Out of all the studied contingencies, the contingencies for which the system did not converge are tabulated in Table 5.1 with the corresponding number of scenarios.

It can be seen from the Table 5.1, for the contingencies corresponding to bus faults BUS 152, BUS 151, BUS 204, BUS 201, and BUS 202, single open line contingencies SING OPN LIN 20 205-206(1), SING OPN LIN 11 201-204(1), and unit fault UNIT 206(1) load flow did not converge for any of the 81 studied scenarios. The Bus fault BUS 205 did not converge for 73 of the studied scenarios followed by single open line fault SING OPN LIN 10 201-202(1) for 54 scenarios.

Table 5.1: Count of scenarios for which tested contingency failed to converge

Contingency	Count of Scenarios
BUS 152	81
BUS 151	81
SING OPN LIN 20 205-206(1)	81
BUS 204	81
SING OPN LIN 11 201-204(1)	81
UNIT 206(1)	81
BUS 201	81
BUS 202	81
BUS 205	73
SING OPN LIN 10 201-202(1)	54
BUS 3005	43
SING OPN LIN 6 152-153(1)	31
SING OPN LIN 4 151-152(1)	30
SING OPN LIN 5 151-152(2)	30
BUS 203	18
BUS 153	18
BUS 3004	9
BUS 3003	9

For the converged contingency scenarios, Chapter 2 studied the branch overload violations, Chapter 3 studied the upper limit voltage violation, and Chapter 4 studied the lower limit voltage violations.

In the Chapter 2 branches reporting loading greater than 100% were tabulated for each scenario and from the table, branches reporting loading greater than 130% were also noted down. It was seen that the branch with most number of overload violations reported are the branches 101-151(1), 3001-3003(1) and 3001-3003(2) with 162 violations each, followed by the branch 153-203(1) with 159 violations.

In the Chapter 3 buses reporting voltage greater than 1.1 PU (violation of upper limit) were studied and it was seen that the generator buses 211 and 212 reported the upper limit violation of 254 times each. In the Chapter 4 buses reporting voltage less than 0.9 PU (violation of lower limit) were tabulated, there were multiple buses reporting lower limit violations and bus 154 with 944 violations, followed by bus 205 with 614, bus 3008 with 352, and bus 103 with 244 reported the top number of violations.

The system scenarios for which the faults caused violation needs to be studied for long term planning for system reinforcements, and for alternate dispatch arrangements and demand side measures for operational planning to ensure system operates within the specified tolerance levels when such a contingency occurs.

Appendix

5.1 Non-converged Contingencies for Studied Scenarios

Table 5.2: Tabulation of contingencies that failed to converge for the studied scenario

[illegible]

Table 5.3: Tabulation of contingencies that failed to converge for the studied scenario

[illegible]

Table 5.4: Tabulation of contingencies that failed to converge for the studied scenario

[illegible]

5.2 Loading Violation > 100% Counts

5.2.1 Overload Count - Scenario

Table 5.5: Overload Count - Scenario

Scenario	Overload Counts
Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS	24
Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS	24
Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, LLS	23
Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, LLS	23
Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, RLS	22
Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, LLS	22
Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, RLS	22
Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, RLS	22
Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, RLS	22
Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, RLS	22
Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, RLS	22
Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, RLS	22
Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, HLS	21
Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, HLS	21
Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, HLS	21
Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, HLS	21
Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, LLS	21
Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, HLS	21
Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, LLS	21
Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, HLS	21
Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, HLS	21
Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, HLS	21
Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, RLS	21
Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, RLS	21
Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, RLS	21
Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, RLS	20
Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, RLS	20
Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, RLS	19
Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, HLS	19
Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, RLS	19
Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, RLS	19
Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, HLS	19
Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, HLS	19
Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, LLS	18
Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, LLS	18
Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, LLS	18
Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, HLS	17
Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, HLS	17
Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, LLS	17
Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, RLS	17
Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, HLS	17
Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, LLS	17
Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, LLS	17
Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, HLS	17
Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, RLS	17
Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, RLS	17
Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, RLS	17
Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, LLS	16
Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, LLS	16
Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, LLS	16
Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, LLS	16
Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, HLS	15
Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, LLS	15
Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, LLS	15
Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, HLS	15
Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, HLS	15
Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, RLS	14
Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, RLS	14
Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, RLS	14
Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, LLS	13
Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, LLS	12

Table 5.6: Overload Count - Scenario - Continued

Scenario	Overload Counts
Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS	11
Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS	11
Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, RLS	11
Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS	11
Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS	11
Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, HLS	11
Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, HLS	11
Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, HLS	11
Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, LLS	11
Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS	11
Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS	11
Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, LLS	11
Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, LLS	11
Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, LLS	11
Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, LLS	11
Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS	11
Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, HLS	11
Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, LLS	10
Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, LLS	10
Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, LLS	10

5.2.2 Overload Count - Branch

Table 5.7: Overload Count - Branch

Branch	Overload Counts
3001-3003(1)	162
3001-3003(2)	162
101-151(1)	162
154-203(1)	159
3003-3005(1)	134
3003-3005(2)	134
154-205(1)	115
205-206(1)	108
153-154(1)	61
203-205(1)	57
203-205(2)	57
153-154(2)	18
153-154(3)	18
154-3008(1)	11
201-204(1)	3

5.2.3 Overload Count - Contingency

Table 5.8: Overload Count - Contingency

Contingency	Overload Counts
SING OPN LIN 1 101-151(1)	296
UNIT 101(1)	296
BUS 153	168
SING OPN LIN 6 152-153(1)	160
BUS 203	92
SING OPN LIN 2 102-151(1)	81
UNIT 102(1)	81
BUS 3005	47
SING OPN LIN 12 201-211(1)	27
SING OPN LIN 13 201-212(1)	27
UNIT 211(1)	27
UNIT 212(1)	27
BASE CASE	18
BUS 205	11
SING OPN LIN 10 201-202(1)	3

5.3 Upper Voltage Limit Violation $> 1.1PU$ Counts

5.3.1 Upper Voltage Limit Counts - Scenario

Table 5.9: Upper Voltage Limit Counts - Scenario

Scenario	Limit Violation Counts
Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS	56
Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS	56
Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, HLS	56
Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, HLS	12
Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, HLS	12
Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, HLS	12
Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, HLS	12
Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, RLS	12
Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, HLS	12
Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, HLS	12
Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, HLS	12
Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, HLS	12
Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, RLS	12
Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, RLS	12
Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, RLS	12
Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, HLS	8
Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, RLS	8
Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS	8
Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS	8
Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, RLS	8
Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, HLS	8
Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, RLS	8
Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, HLS	8
Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, HLS	8
Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS	8
Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, HLS	8
Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, HLS	8
Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, HLS	8
Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS	8
Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, LLS	4
Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, RLS	4
Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, LLS	4
Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, LLS	4
Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, RLS	4
Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, RLS	4
Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, RLS	4
Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS	4
Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, RLS	4
Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, RLS	4
Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, RLS	4
Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS	4
Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, RLS	4
Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, RLS	4
Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS	4
Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, RLS	4
Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, LLS	4
Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, LLS	4
Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, LLS	4
Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS	4
Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, RLS	4
Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, RLS	4

5.3.2 Upper Voltage Limit Counts - Bus

Table 5.10: Upper Voltage Limit Counts - Bus

Bus	Limit Violation Counts
211	254
212	254

5.3.3 Upper Voltage Limit Counts - Contingency

Table 5.11: Upper Voltage Limit Counts - Contingency

Contingency	Limit Violation Counts
BUS 3004	60
SING OPN LIN 5 151-152(2)	48
SING OPN LIN 4 151-152(1)	48
BUS 3005	44
BUS 203	36
SING OPN LIN 6 152-153(1)	36
BUS 3003	28
UNIT 102(1)	24
SING OPN LIN 2 102-151(1)	24
SING OPN LIN 10 201-202(1)	16
SING OPN LIN 27 3003-3005(2)	12
UNIT 3018(1)	12
BUS 3008	12
SING OPN LIN 35 3008-3018(1)	12
SING OPN LIN 19 204-205(2)	12
SING OPN LIN 26 3003-3005(1)	12
SING OPN LIN 18 204-205(1)	12
SING OPN LIN 14 202-203(1)	12
SING OPN LIN 9 153-154(3)	12
SING OPN LIN 8 153-154(2)	12
SING OPN LIN 7 153-154(1)	12
SING OPN LIN 15 202-203(2)	12

5.4 Lower Voltage Limit Violation $< 0.9PU$ Counts

5.4.1 Lower Voltage Limit Counts - Contingency

Table 5.12: Lower Voltage Limit Counts - Scenario

Contingency	Limit Violation Counts
SING OPN LIN 6 152-153(1)	576
BUS 203	364
BUS 3005	312
SING OPN LIN 10 201-202(1)	306
BUS 153	242
BUS 205	198
BUS 3003	158
BUS 3004	158
UNIT 3018(1)	66
SING OPN LIN 35 3008-3018(1)	54
BUS 3008	50
SING OPN LIN 18 204-205(1)	46
SING OPN LIN 19 204-205(2)	46
SING OPN LIN 7 153-154(1)	42
SING OPN LIN 14 202-203(1)	42
SING OPN LIN 15 202-203(2)	42
SING OPN LIN 5 151-152(2)	36
SING OPN LIN 4 151-152(1)	36
SING OPN LIN 1 101-151(1)	32
SING OPN LIN 9 153-154(3)	30
SING OPN LIN 8 153-154(2)	30
UNIT 101(1)	30
UNIT 103(1)	26
SING OPN LIN 27 3003-3005(2)	24
SING OPN LIN 26 3003-3005(1)	24
SING OPN LIN 16 203-205(1)	18
SING OPN LIN 17 203-205(2)	18
SING OPN LIN 32 3005-3008(1)	18
SING OPN LIN 3 103-154(1)	16
SING OPN LIN 28 3004-3005(1)	12
SING OPN LIN 2 102-151(1)	10
UNIT 102(1)	10
SING OPN LIN 34 3007-3008(1)	8
SING OPN LIN 21 3001-3002(1)	6
SING OPN LIN 25 3002-3004(1)	6
SING OPN LIN 30 3005-3007(1)	6
SING OPN LIN 31 3005-3007(2)	6
BUS 3002	6
BUS 3006	6
SING OPN LIN 29 3005-3006(1)	2

5.4.2 Lower Voltage Limit Counts - Bus

Table 5.13: Lower Voltage Limit Counts - Bus

Bus	Limit Violation Counts
154	944
205	614
3008	352
103	244
3007	186
204	182
153	104
3012	94
3006	92
3009	78
3010	70
203	54
3018	48
202	24
3005	16
152	12
3004	4

5.4.3 Lower Voltage Limit Counts - Scenario

Table 5.14: Lower Voltage Limit Counts - Scenario

Scenario	Limit Violation Counts
Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, HLS	92
Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, HLS	90
Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, HLS	86
Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, RLS	66
Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, RLS	66
Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, HLS	64
Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, RLS	64
Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, HLS	64
Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, HLS	60
Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, HLS	60
Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, RLS	60
Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, HLS	58
Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, HLS	58
Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, HLS	56
Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, HLS	56
Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, HLS	52
Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, HLS	52
Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, RLS	50
Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, RLS	50
Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, HLS	48
Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, HLS	48
Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, HLS	48
Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, RLS	46
Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, RLS	46
Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, HLS	44
Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, LLS	44
Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, HLS	44
Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, HLS	44
Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, RLS	44
Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, RLS	44
Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, RLS	44
Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, LLS	44
Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, RLS	42
Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, RLS	42
Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, HLS	42
Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, HLS	42

Table 5.15: Lower Voltage Limit Counts - Scenario

Scenario	Limit Violation Counts
Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, LLS	40
Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, RLS	40
Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, RLS	40
Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, LLS	40
Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, HLS	38
Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, HLS	38
Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, HLS	38
Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, RLS	38
Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, RLS	38
Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, LLS	36
Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, HLS	36
Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, LLS	36
Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, LLS	36
Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, LLS	36
Solar = 0 MW, Hydro = 400 MW, Wind = 0 MW, HLS	34
Solar = 0 MW, Hydro = 400 MW, Wind = 40 MW, HLS	34
Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, RLS	32
Solar = 50 MW, Hydro = 400 MW, Wind = 75 MW, LLS	30
Solar = 50 MW, Hydro = 400 MW, Wind = 40 MW, LLS	28
Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, LLS	28
Solar = 100 MW, Hydro = 400 MW, Wind = 0 MW, LLS	26
Solar = 100 MW, Hydro = 400 MW, Wind = 40 MW, LLS	26
Solar = 100 MW, Hydro = 400 MW, Wind = 75 MW, LLS	26
Solar = 50 MW, Hydro = 400 MW, Wind = 0 MW, LLS	26
Solar = 100 MW, Hydro = 600 MW, Wind = 40 MW, RLS	24
Solar = 0 MW, Hydro = 500 MW, Wind = 75 MW, LLS	24
Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, LLS	24
Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, LLS	24
Solar = 100 MW, Hydro = 600 MW, Wind = 0 MW, RLS	24
Solar = 0 MW, Hydro = 500 MW, Wind = 40 MW, LLS	22
Solar = 0 MW, Hydro = 500 MW, Wind = 0 MW, LLS	22
Solar = 100 MW, Hydro = 600 MW, Wind = 75 MW, RLS	22
Solar = 50 MW, Hydro = 600 MW, Wind = 40 MW, RLS	16
Solar = 50 MW, Hydro = 600 MW, Wind = 0 MW, RLS	16
Solar = 50 MW, Hydro = 500 MW, Wind = 0 MW, LLS	16
Solar = 50 MW, Hydro = 600 MW, Wind = 75 MW, RLS	16
Solar = 0 MW, Hydro = 600 MW, Wind = 40 MW, RLS	12
Solar = 0 MW, Hydro = 600 MW, Wind = 0 MW, RLS	12
Solar = 50 MW, Hydro = 500 MW, Wind = 40 MW, LLS	12
Solar = 50 MW, Hydro = 500 MW, Wind = 75 MW, LLS	12
Solar = 0 MW, Hydro = 600 MW, Wind = 75 MW, RLS	10
Solar = 100 MW, Hydro = 500 MW, Wind = 40 MW, LLS	8
Solar = 100 MW, Hydro = 500 MW, Wind = 0 MW, LLS	8
Solar = 100 MW, Hydro = 500 MW, Wind = 75 MW, LLS	8
Solar = 0 MW, Hydro = 400 MW, Wind = 75 MW, LLS	6